

68,416
LEs
FPGA

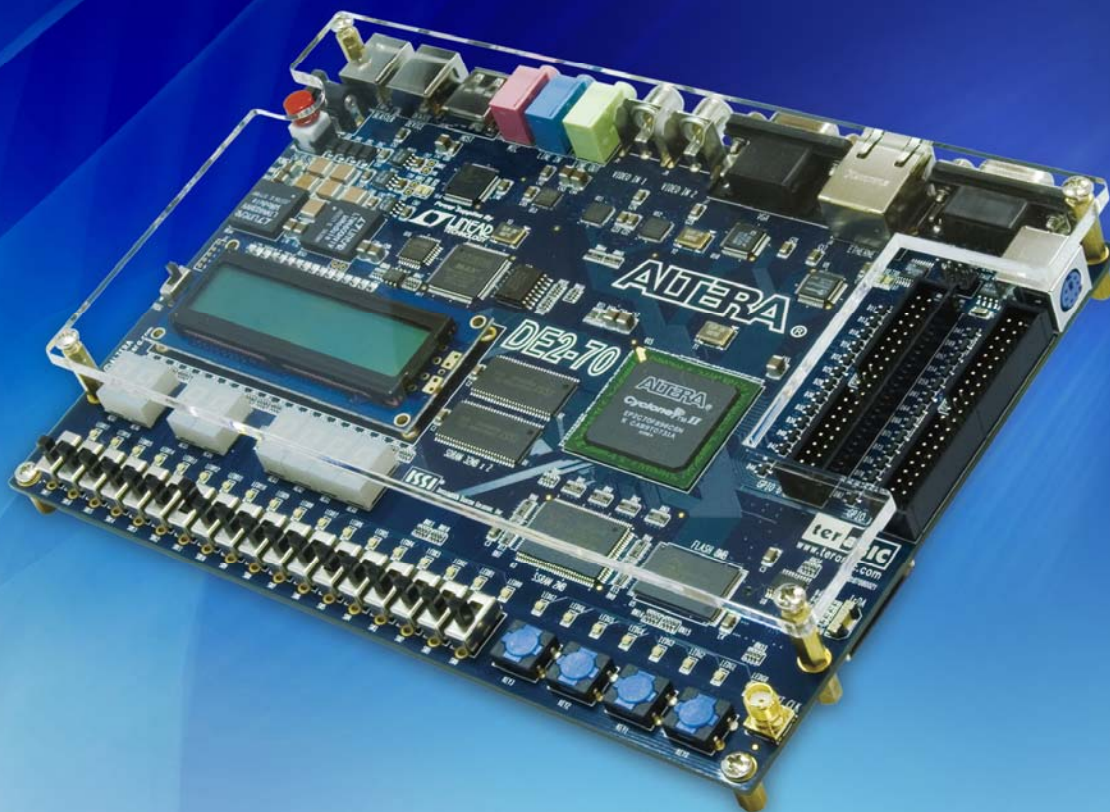
32M
Byte
**X2
SDRAM**

2M
Byte
SSRAM

8M
Byte
FLASH

**X2
TV
Decoder**

terasic
www.terasic.com



ALTERA
**UNIVERSITY
PROGRAM**

Learning through Innovation

DE2-70

Development and Education Board

User Manual



CONTENTS

Chapter 1 DE2-70 Package	1
1.1 Package Contents	1
1.2 The DE2-70 Board Assembly	2
1.3 Getting Help	3
Chapter 2 Altera DE2-70 Board	4
2.1 Layout and Components	4
2.2 Block Diagram of the DE2-70 Board	5
2.3 Power-up the DE2-70 Board	9
Chapter 3 DE2-70 Control Panel	11
3.1 Control Panel Setup	11
3.2 Controlling the LEDs, 7-Segment Displays and LCD Display	13
3.3 Switches and Buttons	15
3.4 SDRAM/SSRAM/Flash Controller and Programmer	16
3.5 USB Monitoring	18
3.6 PS2 Device	19
3.7 SD CARD	20
3.8 Audio Playing and Recording	21
3.9 Overall Structure of the DE2-70 Control Panel	23
Chapter 4 DE2-70 Video Utility	25
4.1 Video Utility Setup	25
4.2 VGA Display	26
4.3 Video Capture	27
4.4 Overall Structure of the DE2-70 Video Utility	28
Chapter 5 Using the DE2-70 Board	30
5.1 Configuring the Cyclone II FPGA	30
5.2 Using the LEDs and Switches	32
5.3 Using the 7-segment Displays	36
5.4 Clock Circuitry	38
5.5 Using the LCD Module	40
5.6 Using the Expansion Header	41
5.7 Using VGA	45
5.8 Using the 24-bit Audio CODEC	48
5.9 RS-232 Serial Port	49
5.10 PS/2 Serial Port	49
5.11 Fast Ethernet Network Controller	50

5.12	TV Decoder.....	52
5.13	Implementing a TV Encoder.....	54
5.14	Using USB Host and Device.....	55
5.15	Using IrDA.....	56
5.16	Using SDRAM/SRAM/Flash.....	57
Chapter 6 Examples of Advanced Demonstrations		66
6.1	DE2-70 Factory Configuration	66
6.2	TV Box Demonstration.....	67
6.3	TV Box Picture in Picture (PIP) Demonstration.....	69
6.4	USB Paintbrush.....	72
6.5	USB Device.....	74
6.6	A Karaoke Machine	76
6.7	Ethernet Packet Sending/Receiving.....	78
6.8	SD Card Music Player.....	80
6.9	Music Synthesizer Demonstration	83
6.10	Audio Recording and Playing.....	87
Chapter 7 Appendix		90
7.1	Revision History	90
7.2	Copyright Statement	90

Chapter 1

DE2-70 Package

The DE2-70 package contains all components needed to use the DE2-70 board in conjunction with a computer that runs the Microsoft Windows software.

1.1 Package Contents

Figure 1.1 shows a photograph of the DE2-70 package.



Figure 1.1. The DE2-70 package contents.

The DE2-70 package includes:

- The DE2-70 board
- USB Cable for FPGA programming and control
- DE2-70 System CD containing the DE2-70 documentation and supporting materials, including the User Manual, the Control Panel utility, reference designs and demonstrations, device datasheets, tutorials, and a set of laboratory exercises
- CD-ROMs containing Altera's Quartus® II Web Edition and the Nios® II Embedded Design Suit Evaluation Edition software.
- Bag of six rubber (silicon) covers for the DE2-70 board stands. The bag also contains some extender pins, which can be used to facilitate easier probing with testing equipment of the board's I/O expansion headers
- Clear plastic cover for the board
- 12V DC wall-mount power supply

1.2 The DE2-70 Board Assembly

To assemble the included stands for the DE2-70 board:

- Assemble a rubber (silicon) cover, as shown in Figure 1.2, for each of the six copper stands on the DE2-70 board
- The clear plastic cover provides extra protection, and is mounted over the top of the board by using additional stands and screws

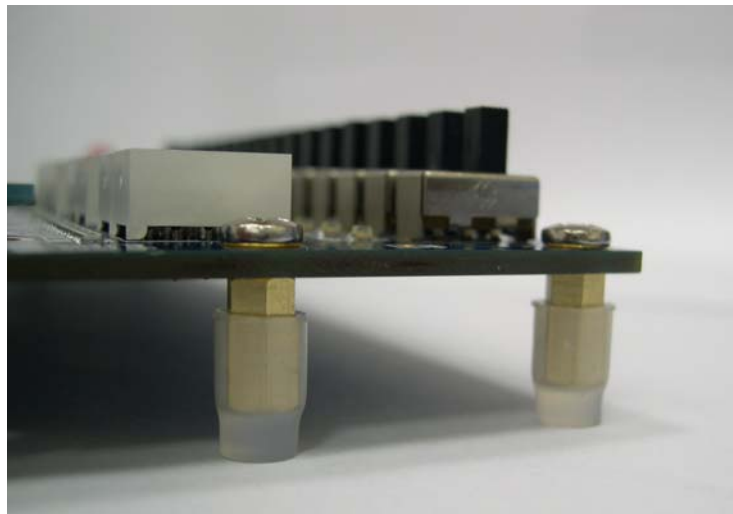


Figure 1.2. The feet for the DE2-70 board.

1.3 Getting Help

Here are the addresses where you can get help if you encounter problems:

- Altera Corporation
101 Innovation Drive
San Jose, California, 95134 USA
Email: university@altera.com
- Terasic Technologies
No. 356, Sec. 1, Fusing E. Rd.
Jhubei City, HsinChu County, Taiwan, 302
Email: support@terasic.com
Web: DE2-70.terasic.com

Chapter 2

Altera DE2-70 Board

This chapter presents the features and design characteristics of the DE2-70 board.

2.1 Layout and Components

A photograph of the DE2-70 board is shown in Figure 2.1. It depicts the layout of the board and indicates the location of the connectors and key components.

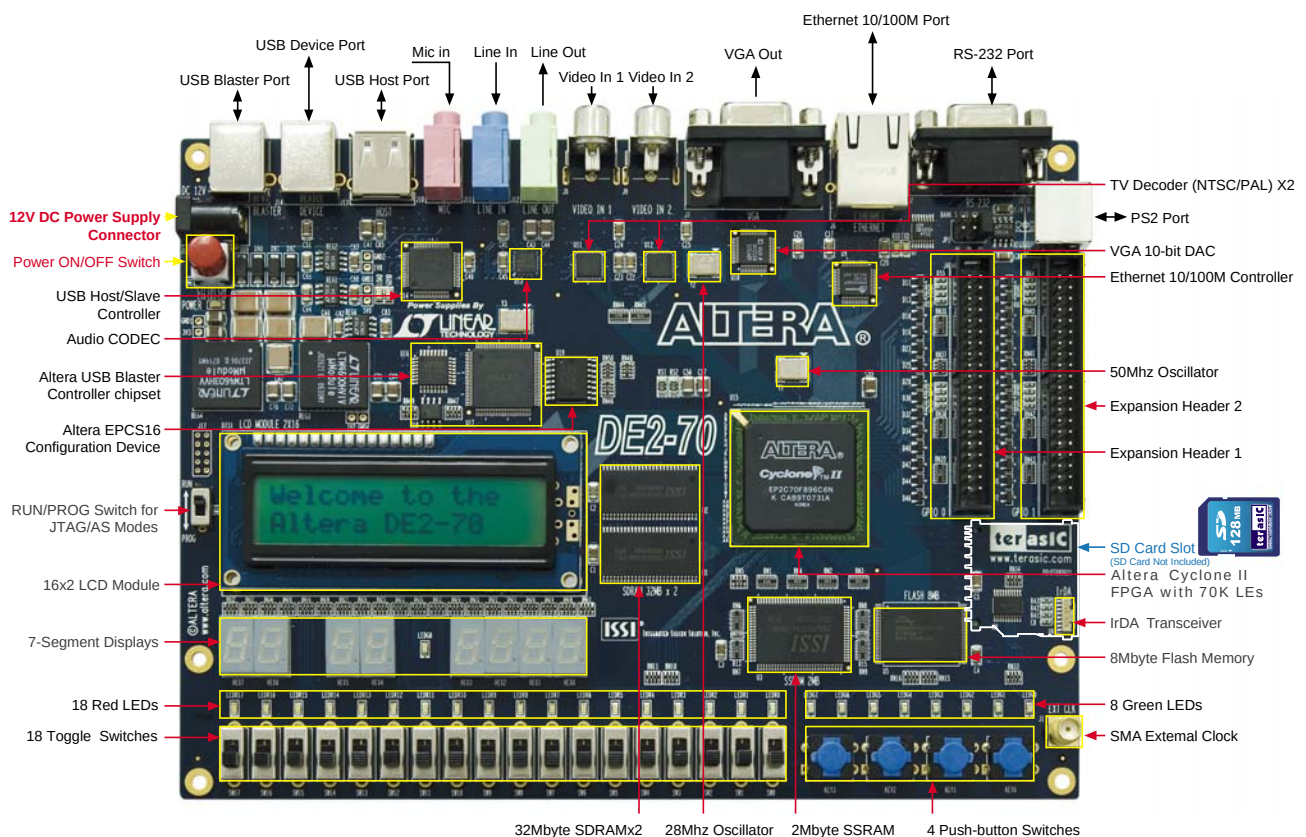


Figure 2.1. The DE2-70 board.

The DE2-70 board has many features that allow the user to implement a wide range of designed circuits, from simple circuits to various multimedia projects.

The following hardware is provided on the DE2-70 board:

- Altera Cyclone® II 2C70 FPGA device
- Altera Serial Configuration device - EPCS16
- USB Blaster (on board) for programming and user API control; both JTAG and Active Serial

(AS) programming modes are supported

- 2-Mbyte SSRAM
- Two 32-Mbyte SDRAM
- 8-Mbyte Flash memory
- SD Card socket
- 4 pushbutton switches
- 18 toggle switches
- 18 red user LEDs
- 9 green user LEDs
- 50-MHz oscillator and 28.63-MHz oscillator for clock sources
- 24-bit CD-quality audio CODEC with line-in, line-out, and microphone-in jacks
- VGA DAC (10-bit high-speed triple DACs) with VGA-out connector
- 2 TV Decoder (NTSC/PAL/SECAM) and TV-in connector
- 10/100 Ethernet Controller with a connector
- USB Host/Slave Controller with USB type A and type B connectors
- RS-232 transceiver and 9-pin connector
- PS/2 mouse/keyboard connector
- IrDA transceiver
- 1 SMA connector
- Two 40-pin Expansion Headers with diode protection

In addition to these hardware features, the DE2-70 board has software support for standard I/O interfaces and a control panel facility for accessing various components. Also, software is provided for a number of demonstrations that illustrate the advanced capabilities of the DE2-70 board.

In order to use the DE2-70 board, the user has to be familiar with the Quartus II software. The necessary knowledge can be acquired by reading the tutorials *Getting Started with Altera's DE2-70 Board* and *Quartus II Introduction* (which exists in three versions based on the design entry method used, namely Verilog, VHDL or schematic entry). These tutorials are provided in the directory *DE2_70_tutorials* on the **DE2-70 System CD-ROM** that accompanies the DE2-70 board and can also be found on Altera's DE2-70 web pages.

2.2 Block Diagram of the DE2-70 Board

Figure 2.2 gives the block diagram of the DE2-70 board. To provide maximum flexibility for the user, all connections are made through the Cyclone II FPGA device. Thus, the user can configure the FPGA to implement any system design.

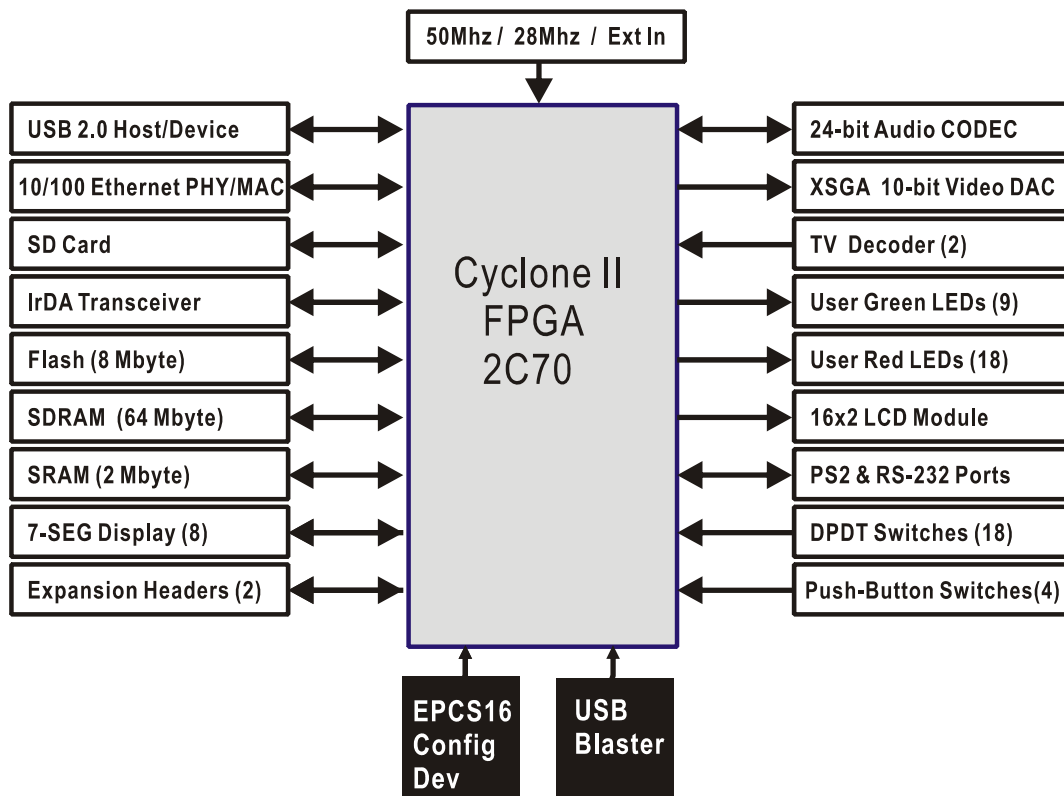


Figure 2.2. Block diagram of the DE2-70 board.

Following is more detailed information about the blocks in Figure 2.2:

Cyclone II 2C70 FPGA

- 68,416 LEs
- 250 M4K RAM blocks
- 1,152,000 total RAM bits
- 150 embedded multipliers
- 4 PLLs
- 622 user I/O pins
- FineLine BGA 896-pin package

Serial Configuration device and USB Blaster circuit

- Altera's EPCS16 Serial Configuration device
- On-board USB Blaster for programming and user API control
- JTAG and AS programming modes are supported

SSRAM

- 2-Mbyte standard synchronous SRAM
- Organized as 512K x 36 bits
- Accessible as memory for the Nios II processor and by the DE2-70 Control Panel

SDRAM

- Two 32-Mbyte Single Data Rate Synchronous Dynamic RAM memory chips
- Organized as 4M x 16 bits x 4 banks
- Accessible as memory for the Nios II processor and by the DE2-70 Control Panel

Flash memory

- 8-Mbyte NOR Flash memory
- Support both byte and word mode access
- Accessible as memory for the Nios II processor and by the DE2-70 Control Panel

SD card socket

- Provides SPI and 1-bit SD mode for SD Card access
- Accessible as memory for the Nios II processor with the DE2-70 SD Card Driver

Pushbutton switches

- 4 pushbutton switches
- Debounced by a Schmitt trigger circuit
- Normally high; generates one active-low pulse when the switch is pressed

Toggle switches

- 18 toggle switches for user inputs
- A switch causes logic 0 when in the DOWN (closest to the edge of the DE2-70 board) position and logic 1 when in the UP position

Clock inputs

- 50-MHz oscillator
- 28.63-MHz oscillator
- SMA external clock input

Audio CODEC

- Wolfson WM8731 24-bit sigma-delta audio CODEC
- Line-level input, line-level output, and microphone input jacks
- Sampling frequency: 8 to 96 KHz
- Applications for MP3 players and recorders, PDAs, smart phones, voice recorders, etc.

VGA output

- Uses the ADV7123 140-MHz triple 10-bit high-speed video DAC
- With 15-pin high-density D-sub connector
- Supports up to 1600 x 1200 at 100-Hz refresh rate
- Can be used with the Cyclone II FPGA to implement a high-performance TV Encoder

NTSC/PAL/ SECAM TV decoder circuit

- Uses two ADV7180 Multi-format SDTV Video Decoders
- Supports worldwide NTSC/PAL/SECAM color demodulation
- One 10-bit ADC, 4X over-sampling for CVBS
- Supports Composite Video (CVBS) RCA jack input
- Supports digital output formats : 8-bit ITU-R BT.656 YCrCb 4:2:2 output + HS, VS, and FIELD
- Applications: DVD recorders, LCD TV, Set-top boxes, Digital TV, Portable video devices, and TV PIP (picture in picture) display.

10/100 Ethernet controller

- Integrated MAC and PHY with a general processor interface
- Supports 100Base-T and 10Base-T applications
- Supports full-duplex operation at 10 Mb/s and 100 Mb/s, with auto-MDIX
- Fully compliant with the IEEE 802.3u Specification
- Supports IP/TCP/UDP checksum generation and checking
- Supports back-pressure mode for half-duplex mode flow control

USB Host/Slave controller

- Complies fully with Universal Serial Bus Specification Rev. 2.0
- Supports data transfer at full-speed and low-speed
- Supports both USB host and device
- Two USB ports (one type A for a host and one type B for a device)
- Provides a high-speed parallel interface to most available processors; supports Nios II with a Terasic driver
- Supports Programmed I/O (PIO) and Direct Memory Access (DMA)

Serial ports

- One RS-232 port
- One PS/2 port
- DB-9 serial connector for the RS-232 port
- PS/2 connector for connecting a PS2 mouse or keyboard to the DE2-70 board

IrDA transceiver

- Contains a 115.2-kb/s infrared transceiver
- 32 mA LED drive current
- Integrated EMI shield
- IEC825-1 Class 1 eye safe
- Edge detection input

Two 40-pin expansion headers

- 72 Cyclone II I/O pins, as well as 8 power and ground lines, are brought out to two 40-pin expansion connectors
- 40-pin header is designed to accept a standard 40-pin ribbon cable used for IDE hard drives
- Diode and resistor protection is provided

2.3 Power-up the DE2-70 Board

The DE2-70 board comes with a preloaded configuration bit stream to demonstrate some features of the board. This bit stream also allows users to see quickly if the board is working properly. To power-up the board perform the following steps:

1. Connect the provided USB cable from the host computer to the USB Blaster connector on the DE2-70 board. For communication between the host and the DE2-70 board, it is necessary to install the Altera USB Blaster driver software. If this driver is not already installed on the host computer, it can be installed as explained in the tutorial *Getting Started with Altera's DE2-70 Board*. This tutorial is available in the directory *DE2_70_tutorials* on the **DE2-70 System CD-ROM**.
2. Connect the 12V adapter to the DE2-70 board
3. Connect a VGA monitor to the VGA port on the DE2-70 board
4. Connect your headset to the Line-out audio port on the DE2-70 board
5. Turn the RUN/PROG switch on the left edge of the DE2-70 board to RUN position; the PROG position is used only for the AS Mode programming
6. Turn the power on by pressing the ON/OFF switch on the DE2-70 board

At this point you should observe the following:

- All user LEDs are flashing
- All 7-segment displays are cycling through the numbers 0 to F
- The LCD display shows **Welcome to the Altera DE2-70**
- The VGA monitor displays the image shown in Figure 2.3.
- Set the toggle switch SW17 to the DOWN position; you should hear a 1-kHz sound
- Set the toggle switch SW17 to the UP position and connect the output of an audio player to the Line-in connector on the DE2-70 board; on your headset you should hear the music played from the audio player (MP3, PC, iPod, or the like)
- You can also connect a microphone to the Microphone-in connector on the DE2-70 board; your voice will be mixed with the music played from the audio player



Figure 2.3. The default VGA output pattern.

Chapter 3

DE2-70 Control Panel

The DE2-70 board comes with a Control Panel facility that allows users to access various components on the board from a host computer. The host computer communicates with the board through an USB connection. The facility can be used to verify the functionality of components on the board or be used as a debug tool while developing RTL code.

This chapter first presents some basic functions of the Control Panel, then describes its structure in block diagram form, and finally describes its capabilities.

3.1 Control Panel Setup

The Control Panel Software Utility is located in the “*DE2_70_control_pane/SW*” folder in the **DE2-70 System CD-ROM**. To install it, just copy the whole folder to your host computer. Launch the control panel by executing the “*DE2_70_Control_Panel.exe*”.

Specific control codes should be downloaded to your FPGA board before the control panel can request it to perform required tasks. The control codes include one *.sof* file and one *.elf* file. To download the codes, just click the “**Download Code**” button on the program. The program will call Quartus II and Nios II tools to download the control codes to the FPGA board through USB-Blaster[USB-0] connection. The *.sof* file is downloaded to FPGA. The *.elf* file is downloaded to either SDRAM-U2 or SSRAM, according to the user option.

To activate the Control Panel, perform the following steps:

1. Make sure Quartus II and NIOS II are installed successfully on your PC.
2. Connect the supplied USB cable to the USB Blaster port, connect the 12V power supply, and turn the power switch ON
3. Set the RUN/PROG switch to the RUN position
4. Start the executable *DE2_70_control_panel.exe* on the host computer. The Control Panel user interface shown in Figure 3.1 will appear.
5. Select the target memory, **SDRAM-U2** or **SSRAM**, on the control panel. Note. The *.elf* file will be downloaded to the target memory and the memory will be read-only in later memory access operation.
6. Click **Download Code** button. **Note, the Control Panel will occupy the USB port until you**

close that port; you cannot use Quartus II to download a configuration file into the FPGA until you close the USB port.

7. The Control Panel is now ready for use; experiment by setting the value of some LEDs display and observing the result on the DE2-70 board.

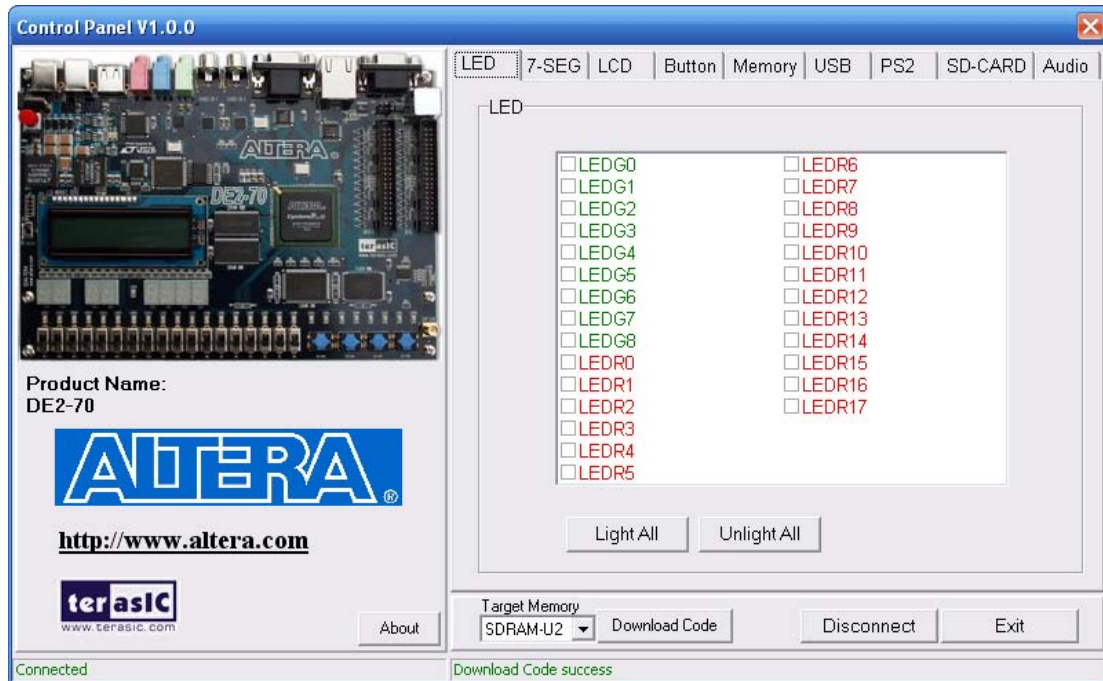


Figure 3.1. The DE2-70 Control Panel.

The concept of the DE2-70 Control Panel is illustrated in Figure 3.2. The “Control Codes” that performs the control functions is implemented in the FPGA board. It communicates with the Control Panel window, which is active on the host computer, via the USB Blaster link. The graphical interface is used to issue commands to the control codes. It handles all requests and performs data transfers between the computer and the DE2-70 board.

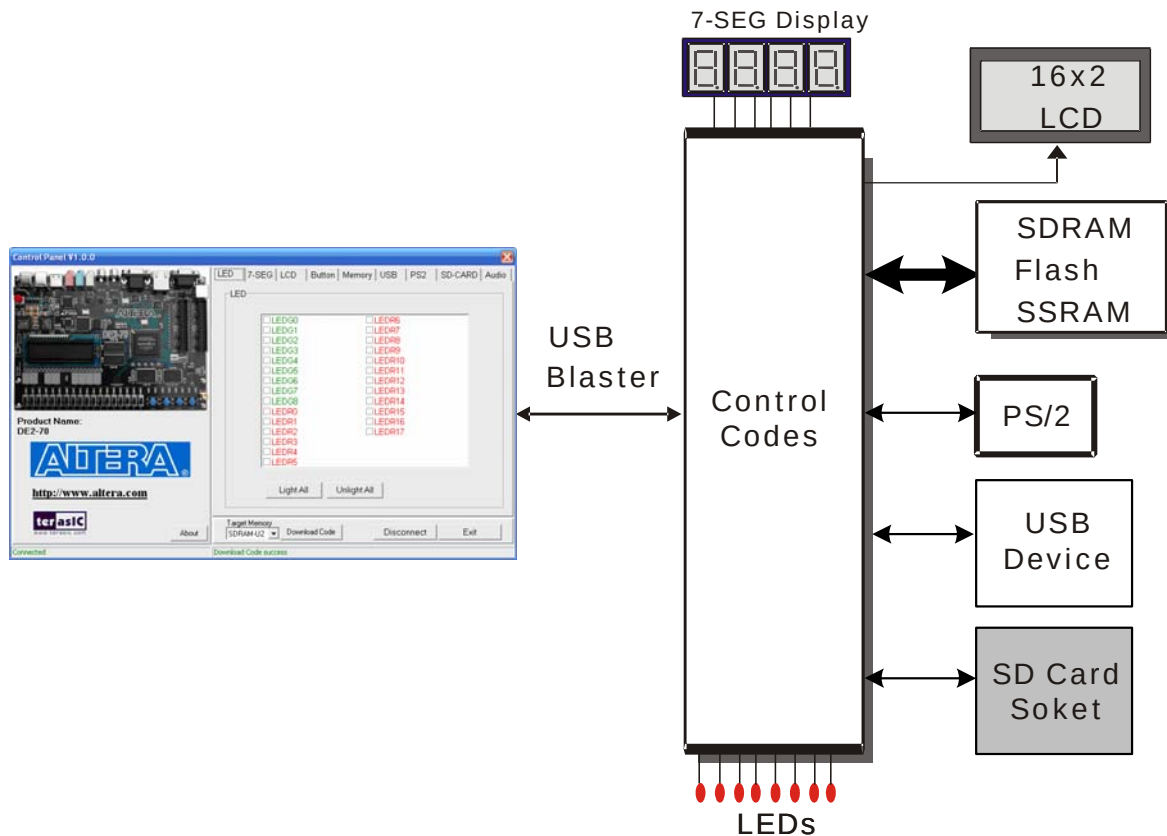


Figure 3.2. The DE2-70 Control Panel concept.

The DE2-70 Control Panel can be used to light up LEDs, change the values displayed on 7-segment and LCD displays, monitor buttons/switches status, read/write the SDRAM, SSRAM and Flash Memory, monitor the status of an USB mouse, read data from a PS/2 keyboard, and read SD-CARD specification information. The feature of reading/writing a word or an entire file from/to the Flash Memory allows the user to develop multimedia applications (Flash Audio Player, Flash Picture Viewer) without worrying about how to build a Memory Programmer.

3.2 Controlling the LEDs, 7-Segment Displays and LCD Display

A simple function of the Control Panel is to allow setting the values displayed on LEDs, 7-segment displays, and the LCD character display.

Choosing the **LED** tab leads to the window in Figure 3.3. Here, you can directly turn the individual LEDs on or off by selecting them or click “Light All” or “Unlight All”.

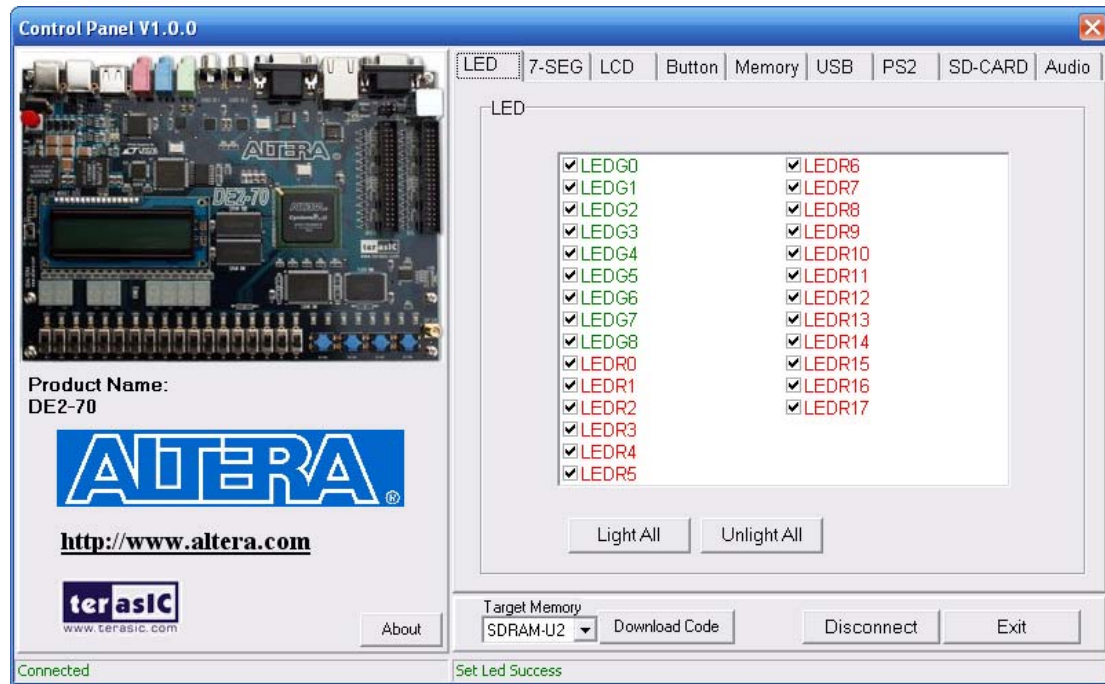


Figure 3.3. Controlling LEDs.

Choosing the **7-SEG** tab leads to the window in Figure 3.4. In the tab sheet, directly use the **Up-Down** control and **Dot** Check box to specified desired patterns, the 7-SEG patterns on the board will be updated immediately.

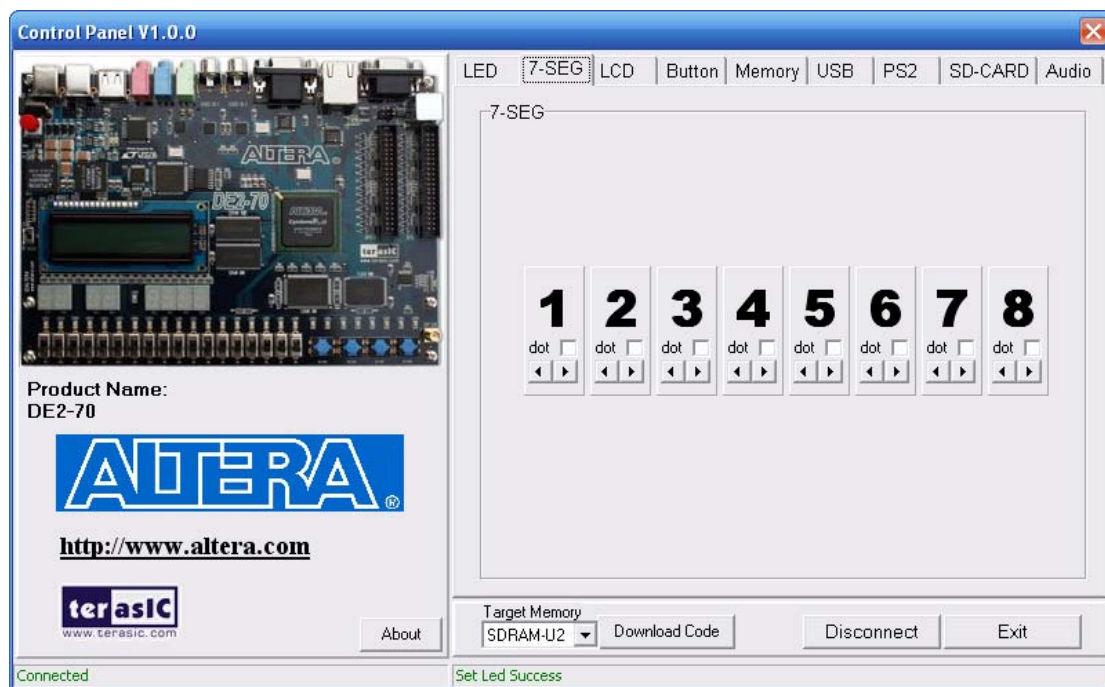


Figure 3.4. Controlling 7-SEG display.

Choosing the **LCD** tab leads to the window in Figure 3.5. Text can be written to the LCD display by typing it in the LCD box and pressing the **Set** button.

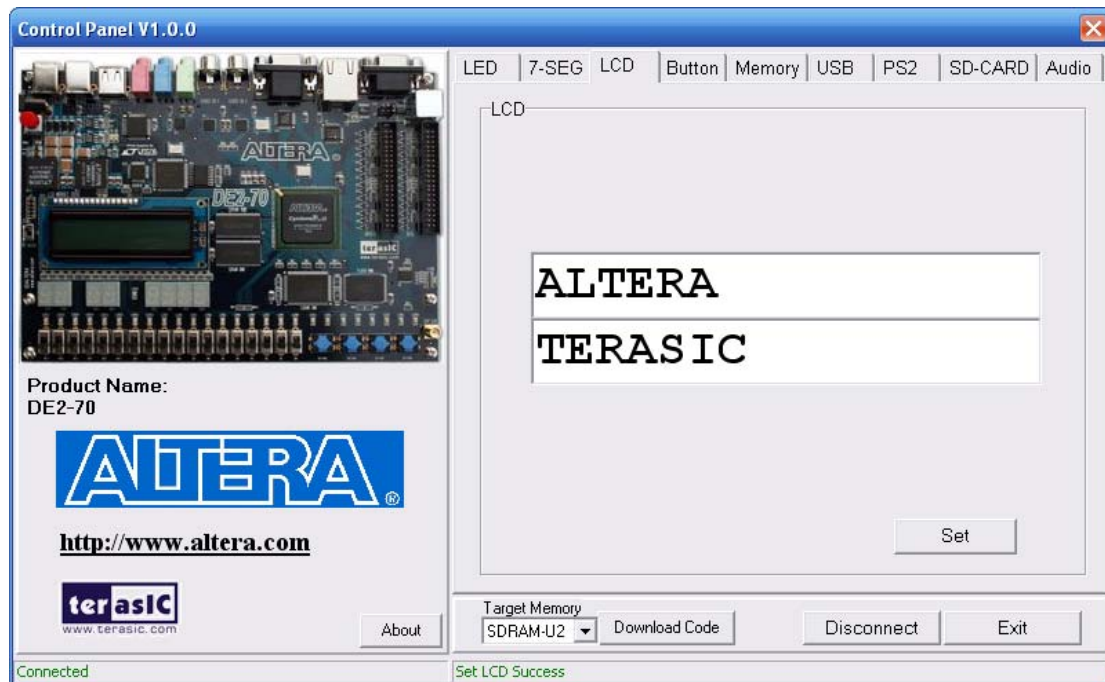


Figure 3.5. Controlling LEDs and the LCD display.

The ability to set arbitrary values into simple display devices is not needed in typical design activities. However, it gives the user a simple mechanism for verifying that these devices are functioning correctly in case a malfunction is suspected. Thus, it can be used for troubleshooting purposes.

3.3 Switches and Buttons

Choosing the **Button** tab leads to the window in Figure 3.6. The function is designed to monitor the status of switches and buttons in real time and show the status in a graphical user interface. It can be used to verify the functionality of the switches and buttons.

Press the **Start** button to start button/switch status monitoring process, and button caption is changed from **Start** to **Stop**. In the monitoring process, the status of buttons and switches on the board is shown in the GUI window and updated in real time. Press **Stop** to end the monitoring process.

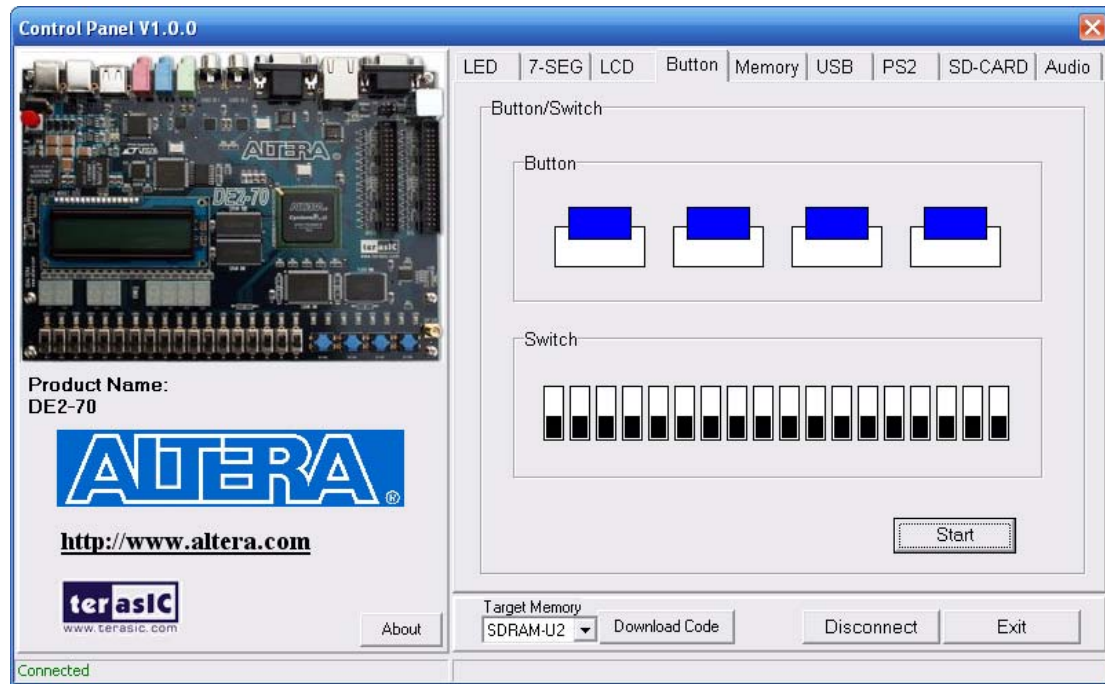


Figure 3.6. Monitoring switches and buttons.

The ability to check the status of button and switch is not needed in typical design activities. However, it provides users a simple mechanism for verifying if the buttons and switches are functioning correctly. Thus, it can be used for troubleshooting purposes.

3.4 SDRAM/SSRAM/Flash Controller and Programmer

The Control Panel can be used to write/read data to/from the SDRAM, SSRAM, and FLASH chips on the DE2-70 board. We will describe how the SDRAM-U1 may be accessed; the same approach is used to access the SDRAM-U2, SRAM, and FLASH. Click on the **Memory** tab and select “SDRAM-U1” to reach the window in Figure 3.7. Please note the target memory chosen for storing .elf file is read-only. Also, please erase the flash before writing data to it.

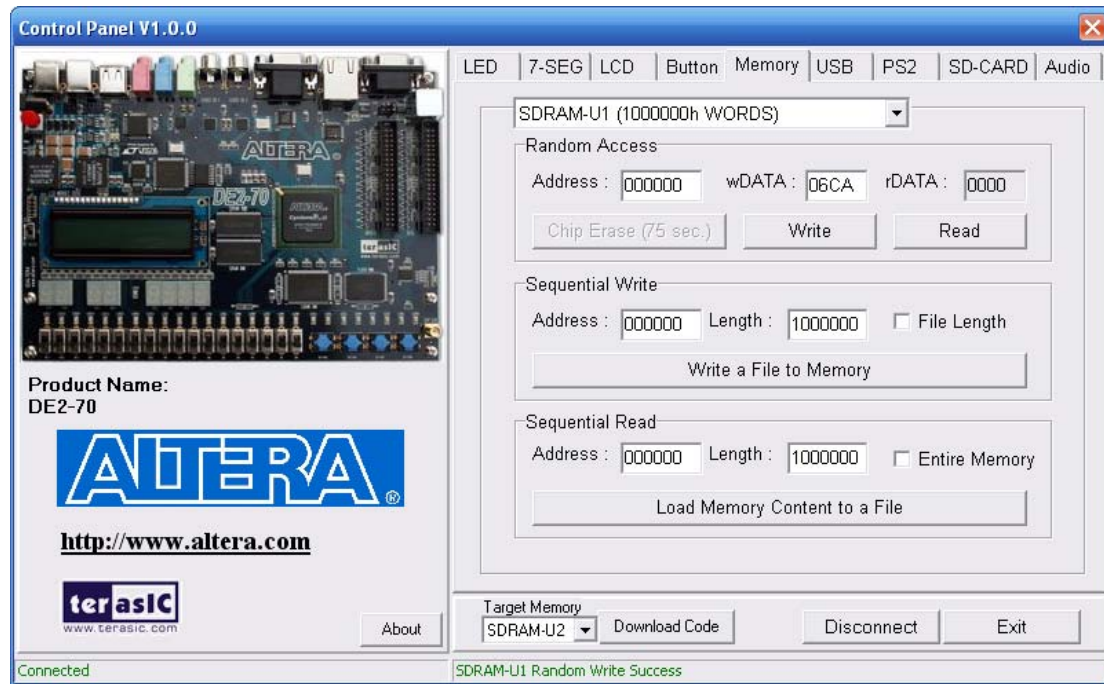


Figure 3.7. Accessing the SDRAM-U1.

A 16-bit word can be written into the SDRAM by entering the address of the desired location, specifying the data to be written, and pressing the **Write** button. Contents of the location can be read by pressing the **Read** button. Figure 3.7 depicts the result of writing the hexadecimal value 06CA into location 200, followed by reading the same location.

The Sequential Write function of the Control Panel is used to write the contents of a file into the SDRAM as follows:

1. Specify the starting address in the **Address** box.
2. Specify the number of bytes to be written in the **Length** box. If the entire file is to be loaded, then a checkmark may be placed in the **File Length** box instead of giving the number of bytes.
3. To initiate the writing of data, click on the **Write a File to Memory** button.
4. When the Control Panel responds with the standard Windows dialog box asking for the source file, specify the desired file in the usual manner.

The Control Panel also supports loading files with a *.hex* extension. Files with a *.hex* extension are ASCII text files that specify memory values using ASCII characters to represent hexadecimal values. For example, a file containing the line

0123456789ABCDEF

defines four 8-bit values: 01, 23, 45, 67, 89, AB, CD, EF. These values will be loaded consecutively

into the memory.

The Sequential Read function is used to read the contents of the SDRAM-U1 and place them into a file as follows:

1. Specify the starting address in the **Address** box.
2. Specify the number of bytes to be copied into the file in the **Length** box. If the entire contents of the SDRAM-U1 are to be copied (which involves all 32 Mbytes), then place a checkmark in the **Entire Memory** box.
3. Press **Load Memory Content to a File** button.
4. When the Control Panel responds with the standard Windows dialog box asking for the destination file, specify the desired file in the usual manner.

Users can use the similar way to access the SSRAM and Flash. Please note that users need to erase the flash before writing data to it.

3.5 USB Monitoring

The Control Panel provides users a USB monitoring tool which monitors the real-time status of a USB mouse connected to the DE2-70 board. The movement of the mouse and the status of the three buttons will be shown in the graphical and text interface. The mouse movement is translated as a position (x,y) with range from (0,0)~(1023,767). This function can be used to verify the functionality of the USB Host.

Follow the steps below to exercise the USB Mouse Monitoring tool:

1. Choosing the **USB** tab leads to the window in Figure 3.8.
2. Plug an USB mouse to the USB HOST port on the DE2-70 board.
3. Press the **Start** button to start the USB mouse monitoring process, and button caption is changed from **Start** to **Stop**. In the monitoring process, the status of the USB mouse is updated and shown in the Control Panel's GUI window in real-time. Press **Stop** to terminate the monitoring process.

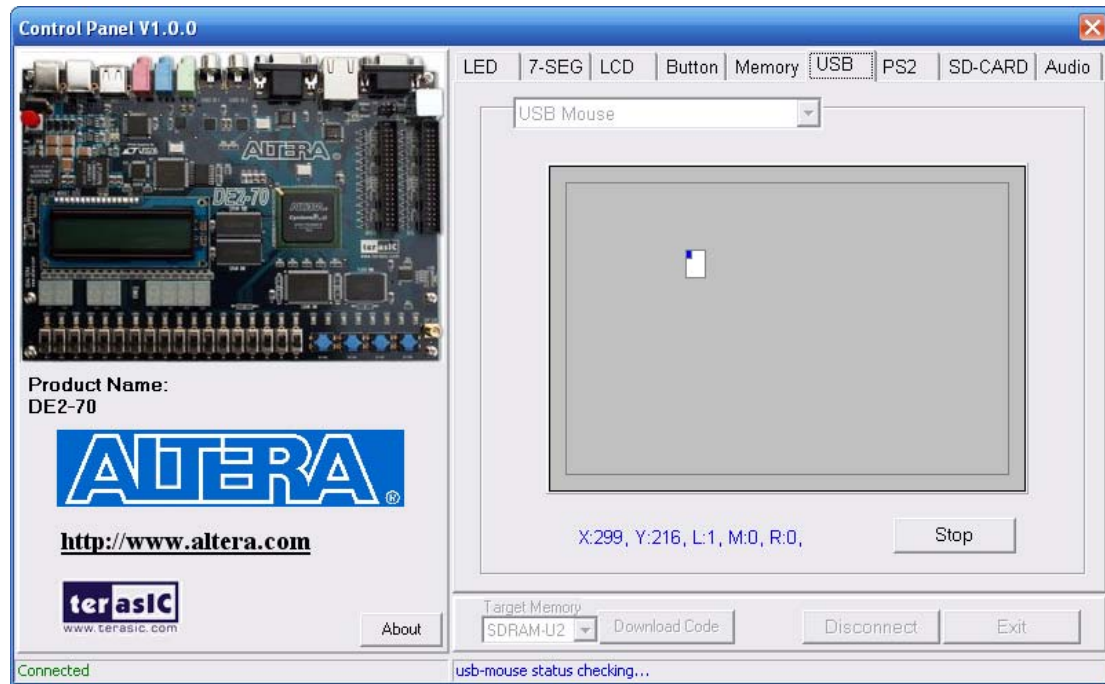


Figure 3.8. USB Mouse Monitoring Tool.

3.6 PS2 Device

The Control Panel provides users a tool to receive the inputs from a PS2 keyboard in real time. The received scan-codes are translated to ASCII code and displayed in the control window. Only visible ASCII codes are displayed. For control key, only “Carriage Return/ENTER” key is implemented. This function can be used to verify the functionality of the PS2 Interface. Please follow the steps below to exercise the PS2 device:

1. Choosing the **PS2** tab leads to the window in Figure 3.9.
2. Plug a PS2 Keyboard to the FPGA board. Then,
3. Press the **Start** button to start PS2Keyboard input receiving process; Button caption is changed from **Start** to **Stop**.
4. In the receiving process, users can start to press the attached keyboard. The input data will be displayed in the control window in real time. Press **Stop** to terminate the monitoring process.

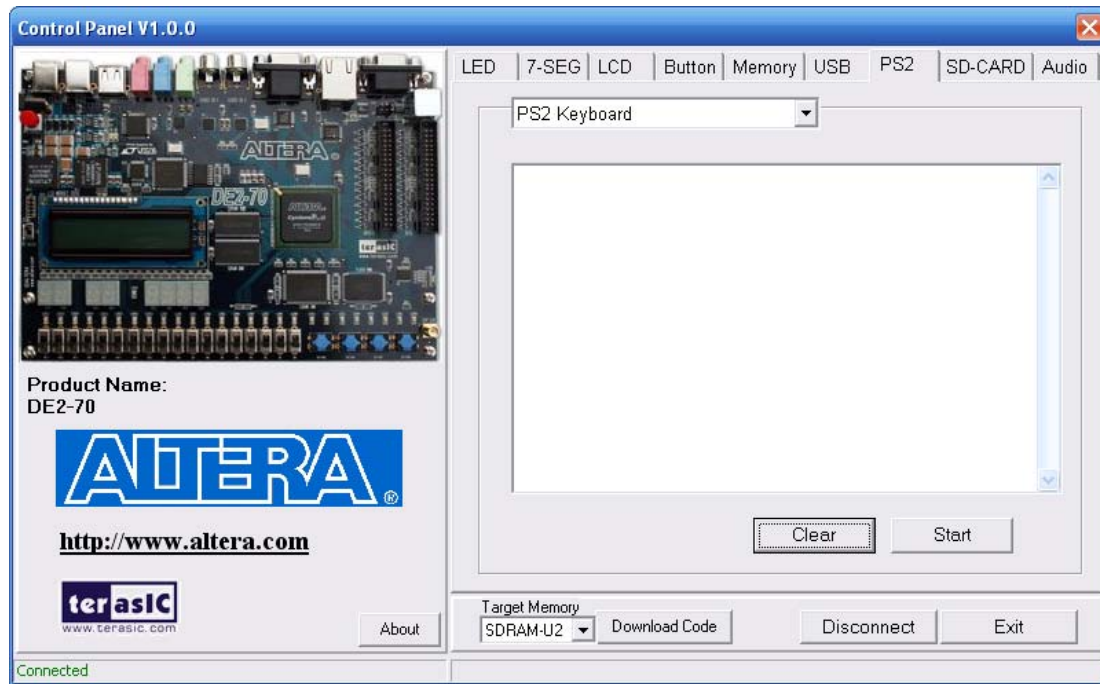


Figure 3.9. Reading the PS2 Keyboard.

3.7 SD CARD

The function is designed to read the identification and specification of the SD card. The 1-bit SD MODE is used to access the SD card. This function can be used to verify the functionality of SD-CARD Interface. Follow the steps below to exercise the SD card:

1. Choosing the **SD-CARD** tab leads to the window in Figure 3.10. First,
2. Insert a SD card to the DE2-70 board, then press the **Read** button to read the SD card. The SD card's identification and specification will be displayed in the control window.

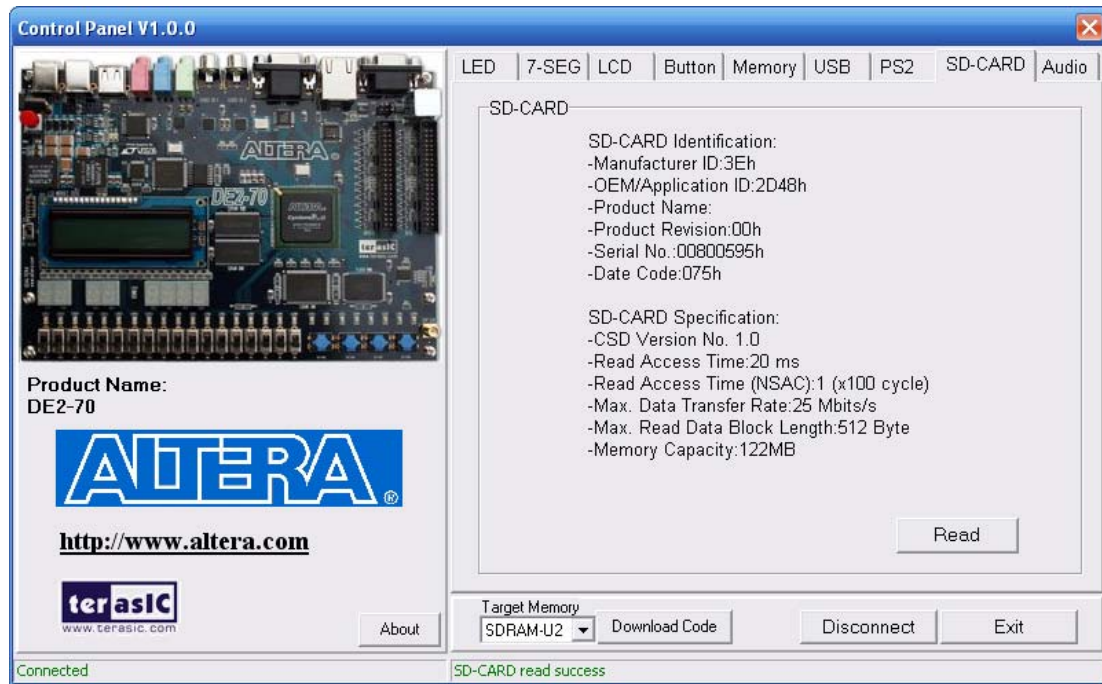


Figure 3.10. Reading the SD card Identification and Specification.

3.8 Audio Playing and Recording

This interesting audio tool is designed to control the audio chip on the DE2-70 board for audio playing and recording. It can play audio stored in a given WAVE file, record audio, and save the audio signal as a wave file. The WAVE file must be uncompressed, stereo (2 channels per sample), and 16-bits per channel. Its sample rate must be either 96K, 48K, 44.1K, 32K, or 8K. Follow the steps below to exercise this tool.

1. Choosing the **Audio** tab leads to the window in Figure 3.11.
2. To play audio, plug a headset or speaker to the LINE-OUT port on the board.
3. Select the “Play Audio” item in the com-box, as shown in Figure 3.11.
4. Click “Open Wave” to select a WAVE file. The waveform of the specified wave file will be displayed in the waveform window. The sampling rate of the wave file also is displayed in the Sample Rate Combo-Box. You can drag the scrollbar to browse the waveform. In the waveform window, the blue line represents left-channel signal and green line represents right-channel signal.
5. Click “Start Play” to start audio play. The program will download the waveform to SDRAM-U1, configure the audio chip for audio playing, and then start the audio playing process. You will hear the audio sound from the headset or speaker. To stop the audio playing, simply click “Stop Play”.

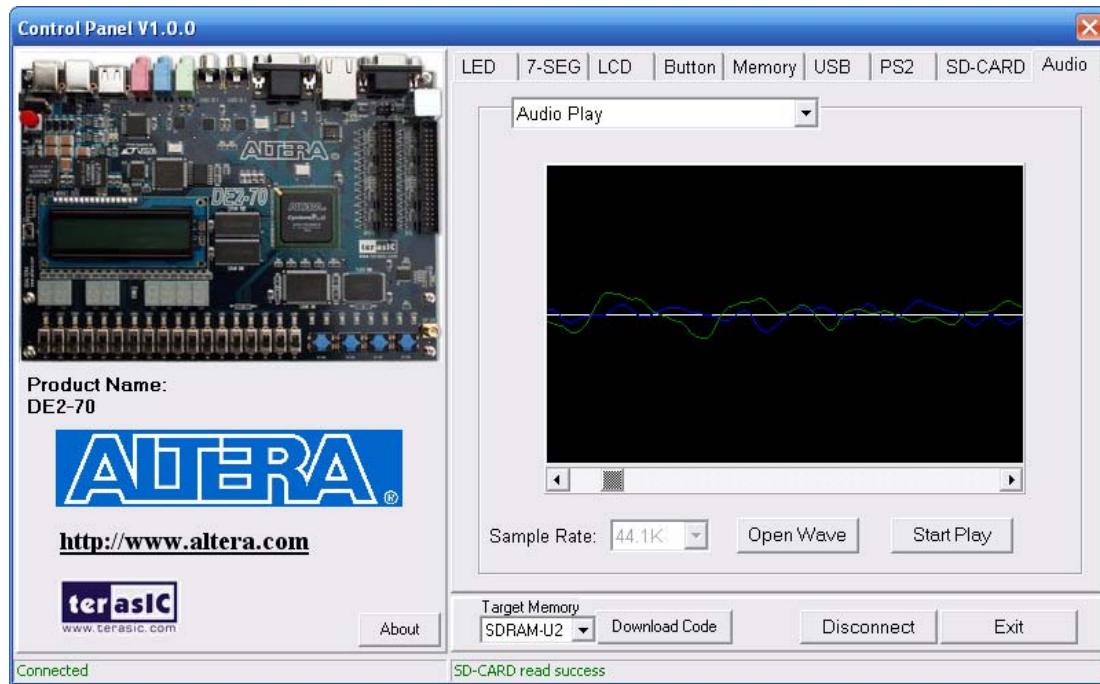


Figure 3.11. Playing audio from a selected wave file

To record sound using a microphone, please follow the steps below:

1. Plug a microphone to the MIC port on the board.
2. Select the “Record MIC” item in the com-box and select desired sampling rate, as shown in Figure 3.12.
3. Click “Start Record” to start the record process. The program will configure the audio chip for MIC recording, retrieve audio signal from the MIC port, and then save the audio signal into SDRAM-U1.
4. To stop recording, click “Stop Record”. Finally, audio signal saved in SDRAM-U1 will be uploaded to the host computer and displayed on the waveform window. Click “Save Wave” to save the waveform into a WAV file.

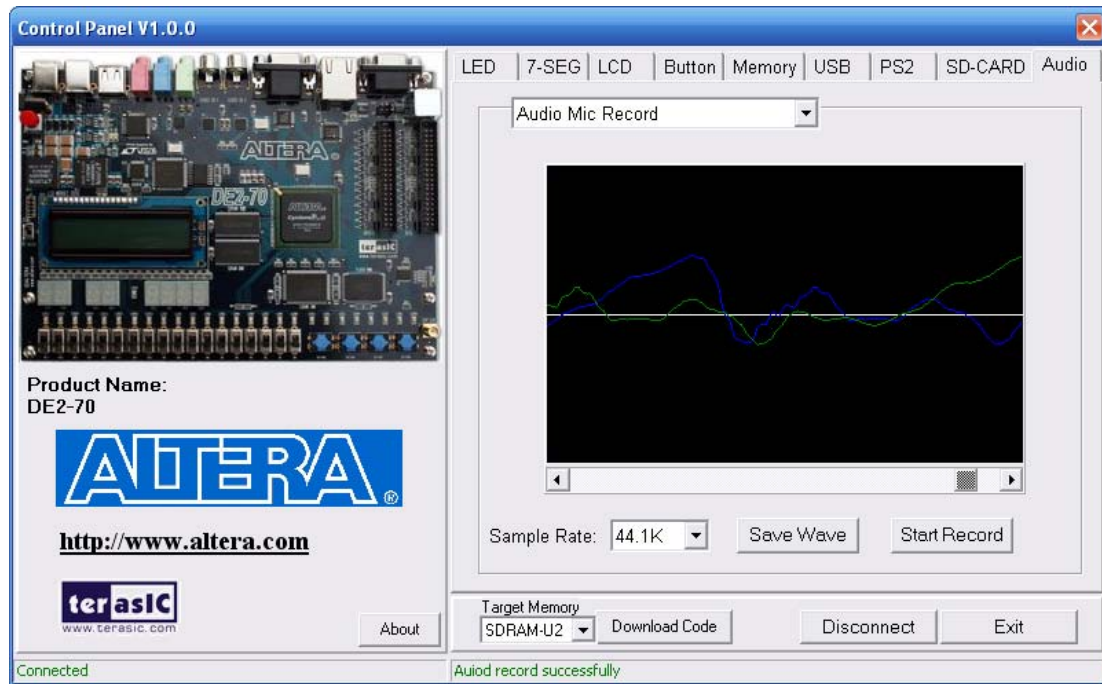


Figure 3.12. Audio Recording and Saving as a WAV file.

To record audio sound from LINE-IN port, please connect an audio source to the LINE-IN port on the board. The operation is as same as recording audio from MIC.

3.9 Overall Structure of the DE2-70 Control Panel

The DE2-70 Control Panel is based on a NIOS II system running in the Cyclone II FPGA with the SDRAM-U2 or SSRAM. The software part is implemented in C code; the hardware part is implemented in Verilog code with SOPC builder, which makes it possible for a knowledgeable user to change the functionality of the Control Panel. The code is located inside the *DE2_70_demonstrations* directory on the **DE2 System CD-ROM**.

To run the Control Panel, users must first configure it as explained in Section 3.1. Figure 3.13 depicts the structure of the Control Panel. Each input/output device is controlled by the NIOS II Processor instantiated in the FPGA chip. The communication with the PC is done via the USB Blaster link. The NIOS II interprets the commands sent from the PC and performs the corresponding actions.

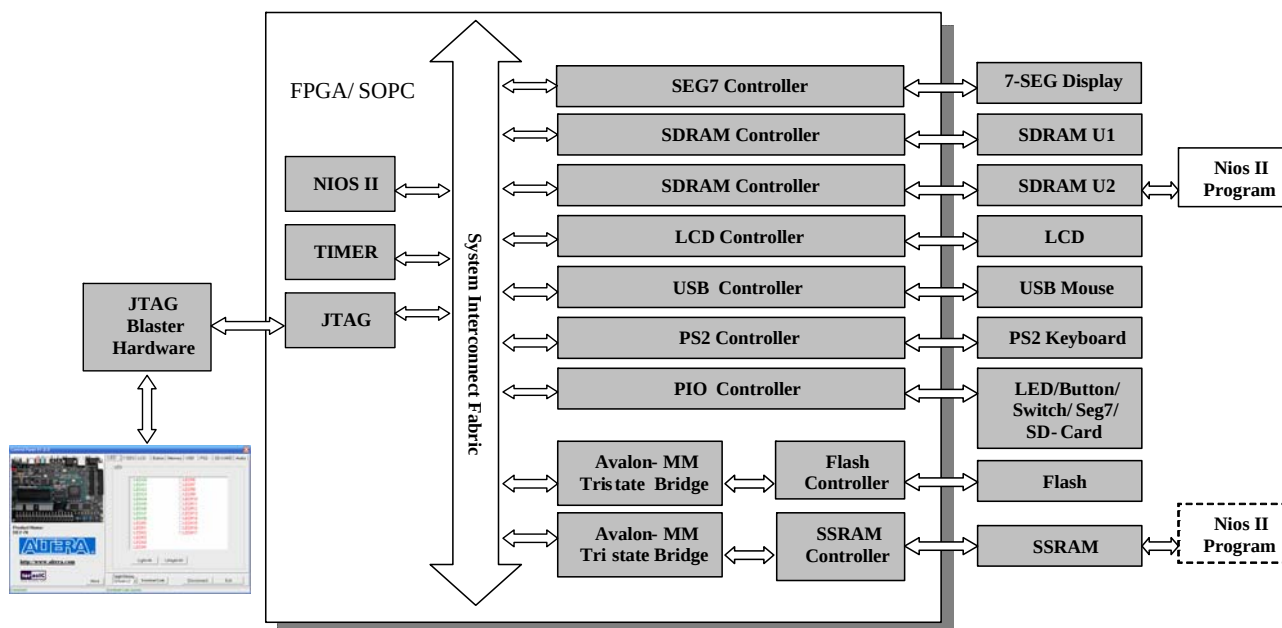


Figure 3.13. The block diagram of the DE2-70 control panel.

Chapter 4

DE2-70 Video Utility

The DE2-70 board comes with a video utility that allows users to access video components on the board from a host computer. The host computer communicates with the board through the USB-Blaster link. The facility can be used to verify the functionality of video components on the board, capture the video sent from the video-in ports, or display desired pattern on the VGA port.

This chapter first presents some basic functions of the Video Utility control panel, then describes its structure in block diagram form, and finally describes its capabilities.

4.1 Video Utility Setup

The Video Utility is located in the “*DE2_70_video utility/SW*” folder in the **DE2-70 System CD-ROM**. To install it, just copy the whole folder to your host computer. Launch the Video Utility by executing the “*DE2_70_VIDEO.exe*”.

Specific configuration files should be downloaded to your FPGA board before the Control Panel can request it to perform required tasks. The configuration files include one *.sof* file and one *.elf* file. To download the codes, simply click the “**Download Code**” button on the program. The program will call Quartus II and Nios II tools to download the control codes to the FPGA board through USB-Blaster[USB-0] connection. The *.sof* file is downloaded to FPGA. The *.elf* file is downloaded to SDRAM-U1.

To activate the Video Utility, perform the following steps:

1. Make sure Quartus II and Nios II are installed successfully on your PC.
2. Connect the supplied USB cable to the USB Blaster port, connect the 12V power supply, and turn the power switch ON
3. Set the **RUN/PROG** switch to the RUN position
4. Start the executable *DE2_70_VIDEO.exe* on the host computer. The Video Utility user interface shown in Figure 4.1 will appear.
5. Click the “**Download Code**” button. **The Control Panel will occupy the USB port until you close that port; you cannot use Quartus II to download a configuration file into the FPGA until you close the USB port.**
6. The Video Utility is now ready for use.

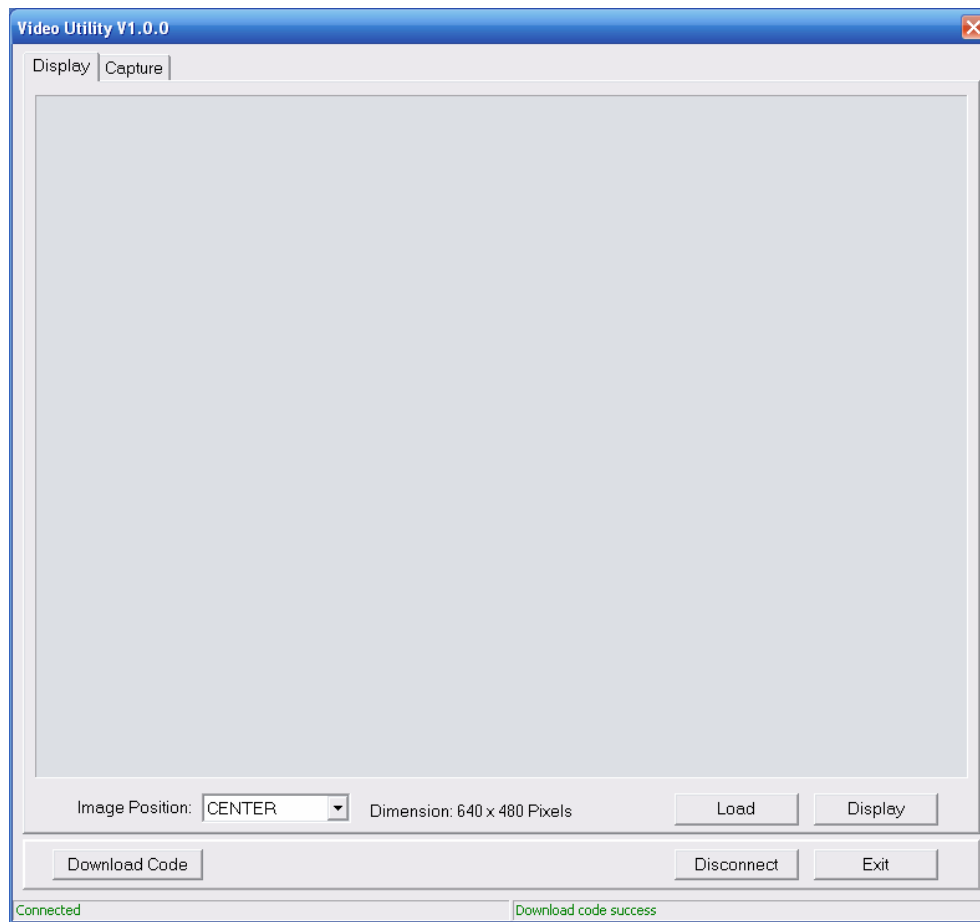


Figure 4.1. The DE2-70 Video Utility window.

4.2 VGA Display

Choosing the **Display** tab in the DE2-70 Video Utility leads to the window shown in Figure 4.2. The function is designed to download an image from the host computer to the FPGA board and output the image through the VGA interface with resolution 640x480.

Please follow the steps below to exercise the Video Utility:

1. Connect a VGA monitor to the VGA port of the board.
2. Click **Load** button and specify an image file for displaying. It can be a bitmap or jpeg file. The selected image file will be displayed on the display window of the Video Utility.
3. Select the desired **Image Positioning** method to fit the image to the VGA 640x480 display dimension.
4. Click **Display** button to start downloading the image to the DE2-70 board.
5. After finish downloading, you will see the desired image shown on the screen of the VGA monitor.



Figure 4.2. Displaying selected image file on VGA Monitor.

4.3 Video Capture

Choosing the **Capture** tab leads to the window in Figure 4.3. The function is designed to capture an image from the video sources, and sent the image from the FPGA board to the host computer. The input video source can be PAL or NTSC signals.

Please follow the steps below to capture an image from a video source:

1. Connect a video source, such as a VCD/DVD player or NTSC/PAL camera, to VIDEO IN 1 or VIDEO IN 2 port on the board.
2. Specify **Video Source** as VIDEO IN 1 or VIDEO IN 2.
3. Click **Capture** button to start capturing process. Then, you will see the captured image shown in the display window of the Video Utility. The image dimension of the captured image is also displayed.
4. Users can click **Save** button to save the captured image as a bitmap or jpeg file.

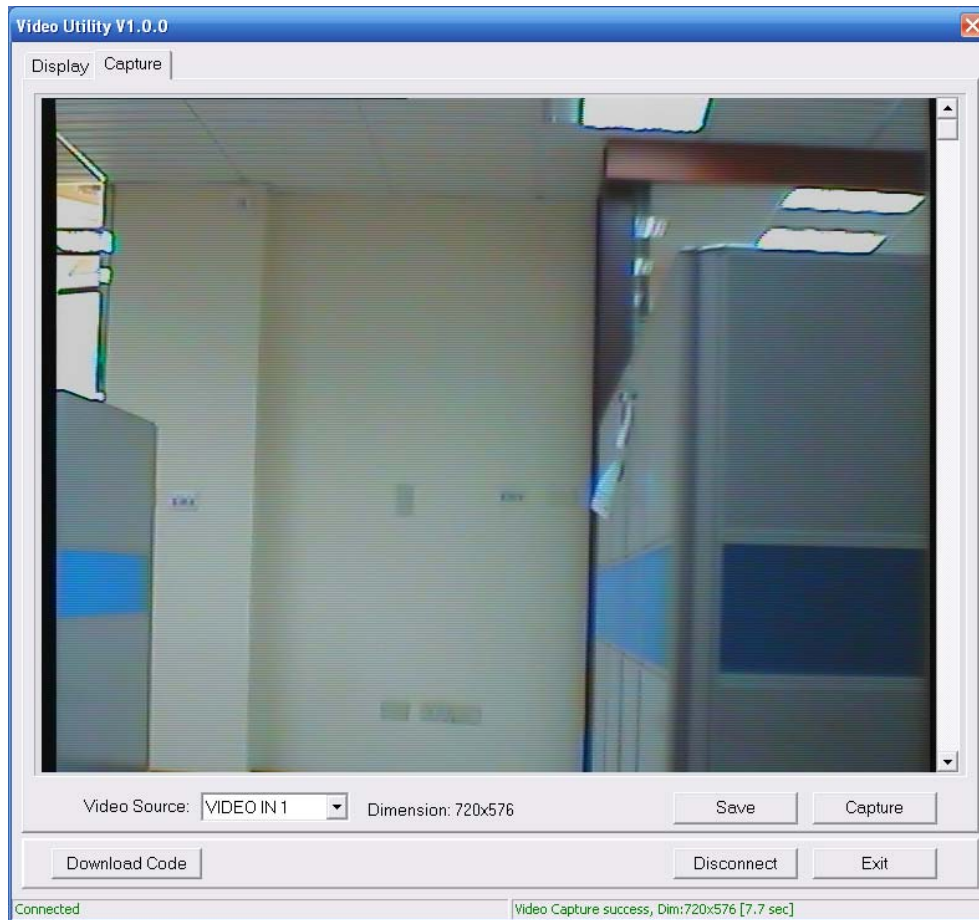


Figure 4.3. Video Capturing Tool.

4.4 Overall Structure of the DE2-70 Video Utility

The DE2-70 Video Utility is based on a NIOS II system running in the Cyclone II FPGA with the SDRAM-U2 or SSRAM. The software part is implemented in C code; the hardware part is implemented in Verilog code with SOPC builder, which makes it possible for a knowledgeable user to change the functionality of the Video Utility. The code is located inside the *DE2_70_demonstrations* directory on the **DE2-70 System CD-ROM**.

Figure 4.4 depicts the block diagram of the Video Utility. Each input/output device is controlled by the NIOS II Processor instantiated. The communication between the DE2-70 board and the host PC is via the USB Blaster link. The NIOS II processor interprets the commands sent from the PC and performs the appropriate actions.

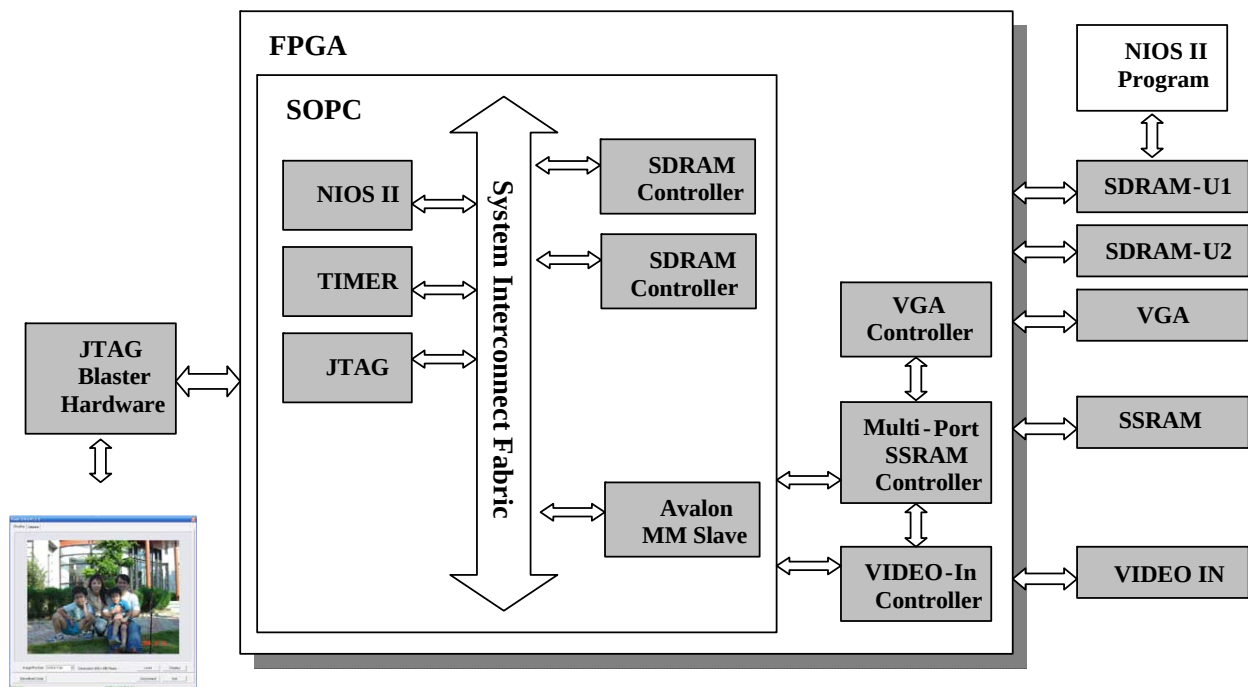


Figure 4.4. Video Capture Block Diagram.

The control flow for video displaying is described below:

1. Host computer downloads the raw image data to SDRAM-U2.
2. Host issues a “display” command to Nios II processor.
3. Nios II processor interprets the command received and moves the raw image data from the SDRAM to SSRAM through the Multi-Port SSRAM controller.
4. VGA Controller continuously reads the raw image data from the SSRAM and sends them to the VGA port.

The control flow for video capturing is described below:

1. Host computer issues a “capture” command to Nios II processor.
2. Nios II processor interprets the command and controls Video-In controller to capture the raw image data into the SSRAM. After capturing is done, Nios II processor copies the raw image data from the SSRAM to SDRAM-U2.
3. Host computer reads the raw image data from the SDRAM-U2
4. Host computer converts the raw image data to RGB color space and displays it.

Chapter 5

Using the DE2-70 Board

This chapter gives instructions for using the DE2-70 board and describes each of its I/O devices.

5.1 Configuring the Cyclone II FPGA

The procedure for downloading a circuit from a host computer to the DE2-70 board is described in the tutorial *Quartus II Introduction*. This tutorial is found in the *DE2_70_tutorials* folder on the **DE2-70 System CD-ROM**. The user is encouraged to read the tutorial first, and to treat the information below as a short reference.

The DE2-70 board contains a serial EEPROM chip that stores configuration data for the Cyclone II FPGA. This configuration data is automatically loaded from the EEPROM chip into the FPGA each time power is applied to the board. Using the Quartus II software, it is possible to reprogram the FPGA at any time, and it is also possible to change the non-volatile data that is stored in the serial EEPROM chip. Both types of programming methods are described below.

1. *JTAG* programming: In this method of programming, named after the IEEE standards *Joint Test Action Group*, the configuration bit stream is downloaded directly into the Cyclone II FPGA. The FPGA will retain this configuration as long as power is applied to the board; the configuration is lost when the power is turned off.
2. *AS* programming: In this method, called *Active Serial* programming, the configuration bit stream is downloaded into the Altera EPCS16 serial EEPROM chip. It provides non-volatile storage of the bit stream, so that the information is retained even when the power supply to the DE2-70 board is turned off. When the board's power is turned on, the configuration data in the EPCS16 device is automatically loaded into the Cyclone II FPGA.

The sections below describe the steps used to perform both JTAG and AS programming. For both methods the DE2-70 board is connected to a host computer via a USB cable. Using this connection, the board will be identified by the host computer as an Altera *USB Blaster* device. The process for installing on the host computer the necessary software device driver that communicates with the USB Blaster is described in the tutorial *Getting Started with Altera's DE2-70 Board*. This tutorial is available on the **DE2-70 System CD-ROM**.

Configuring the FPGA in JTAG Mode

Figure 5.1 illustrates the JTAG configuration setup. To download a configuration bit stream into the Cyclone II FPGA, perform the following steps:

- Ensure that power is applied to the DE2-70 board
- Connect the supplied USB cable to the USB Blaster port on the DE2-70 board (see Figure 2.1)
- Configure the JTAG programming circuit by setting the RUN/PROG switch (on the left side of the board) to the RUN position.
- The FPGA can now be programmed by using the Quartus II Programmer module to select a configuration bit stream file with the *.sof* filename extension

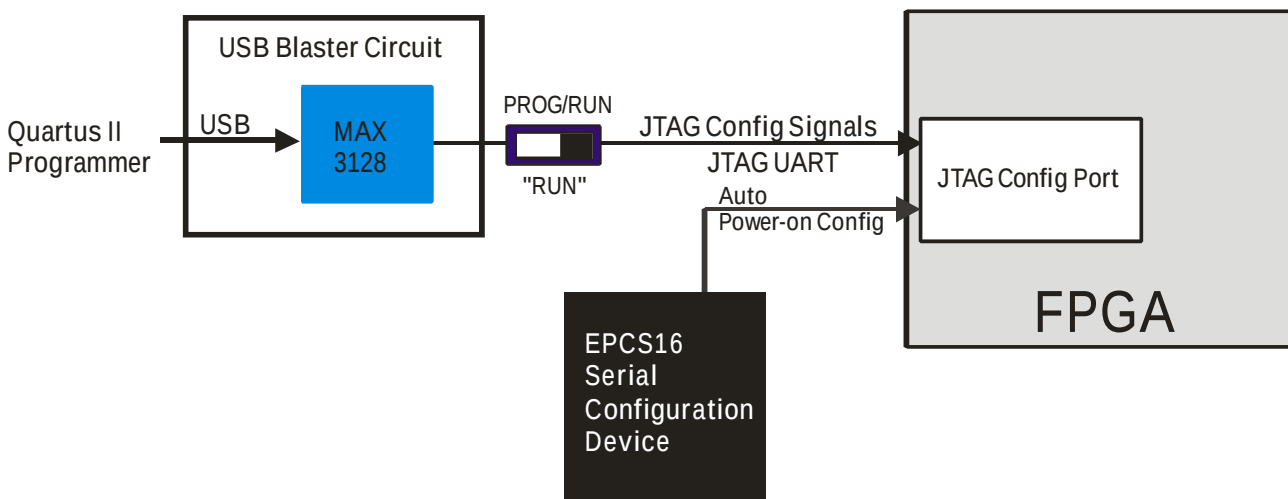


Figure 5.1. The JTAG configuration scheme.

Configuring the EPCS16 in AS Mode

Figure 5.2 illustrates the AS configuration set up. To download a configuration bit stream into the EPCS16 serial EEPROM device, perform the following steps:

- Ensure that power is applied to the DE2-70 board
- Connect the supplied USB cable to the USB Blaster port on the DE2-70 board (see Figure 2.1)
- Configure the JTAG programming circuit by setting the RUN/PROG switch (on the left side of the board) to the PROG position.
- The EPCS16 chip can now be programmed by using the Quartus II Programmer module to select a configuration bit stream file with the *.pof* filename extension
- Once the programming operation is finished, set the RUN/PROG switch back to the RUN

position and then reset the board by turning the power switch off and back on; this action causes the new configuration data in the EPCS16 device to be loaded into the FPGA chip.

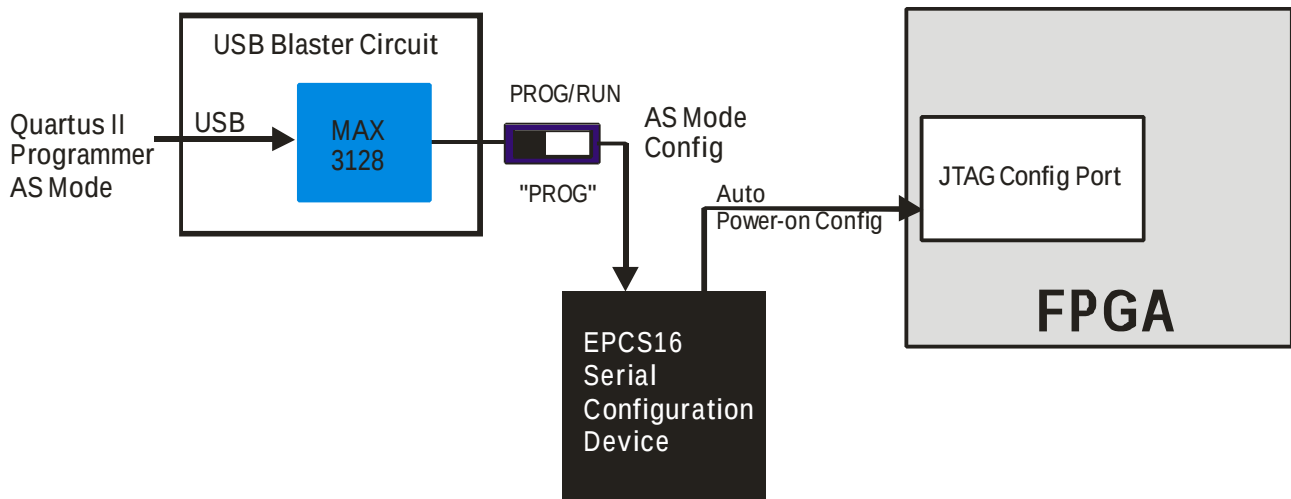


Figure 5.2. The AS configuration scheme.

In addition to its use for JTAG and AS programming, the USB Blaster port on the DE2-70 board can also be used to control some of the board's features remotely from a host computer. Details that describe this method of using the USB Blaster port are given in Chapter 3.

5.2 Using the LEDs and Switches

The DE2-70 board provides four pushbutton switches. Each of these switches is debounced using a Schmitt Trigger circuit, as indicated in Figure 5.3. The four outputs called *KEY0*, *KEY1*, *KEY2*, and *KEY3* of the Schmitt Trigger devices are connected directly to the Cyclone II FPGA. Each switch provides a high logic level (3.3 volts) when it is not pressed, and provides a low logic level (0 volts) when depressed. Since the pushbutton switches are debounced, they are appropriate for use as clock or reset inputs in a circuit.

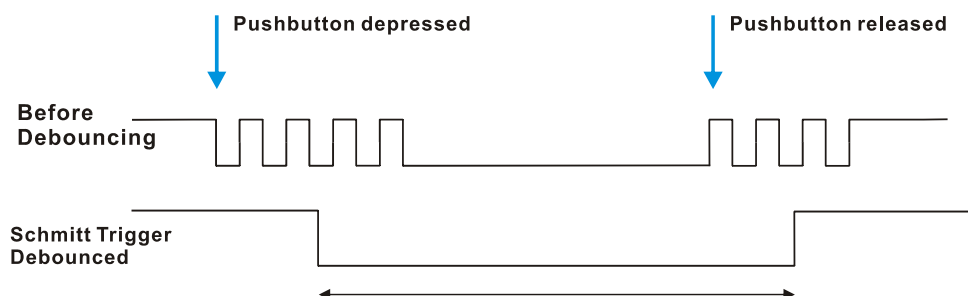


Figure 5.3. Switch debouncing.

There are also 18 toggle switches (sliders) on the DE2-70 board. These switches are not debounced, and are intended for use as level-sensitive data inputs to a circuit. Each switch is connected directly to a pin on the Cyclone II FPGA. When a switch is in the DOWN position (closest to the edge of the board) it provides a low logic level (0 volts) to the FPGA, and when the switch is in the UP position it provides a high logic level (3.3 volts).

There are 27 user-controllable LEDs on the DE2-70 board. Eighteen red LEDs are situated above the 18 toggle switches, and eight green LEDs are found above the pushbutton switches (the 9th green LED is in the middle of the 7-segment displays). Each LED is driven directly by a pin on the Cyclone II FPGA; driving its associated pin to a high logic level turns the LED on, and driving the pin low turns it off. A schematic diagram that shows the pushbutton and toggle switches is given in Figure 5.4. A schematic diagram that shows the LED circuitry appears in Figure 5.5.

A list of the pin names on the Cyclone II FPGA that are connected to the toggle switches is given in Table 5.1. Similarly, the pins used to connect to the pushbutton switches and LEDs are displayed in Tables 5.2 and 5.3, respectively.

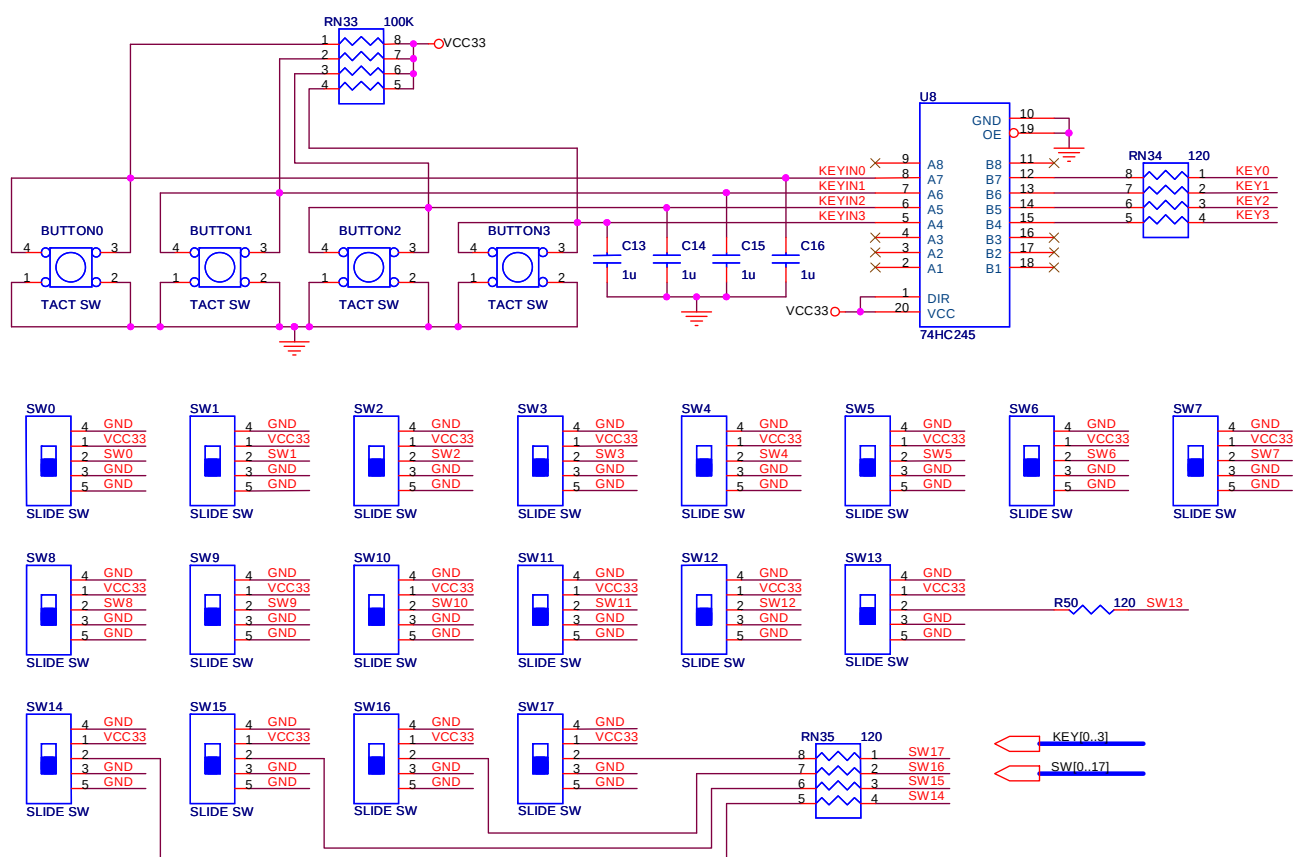


Figure 5.4. Schematic diagram of the pushbutton and toggle switches.

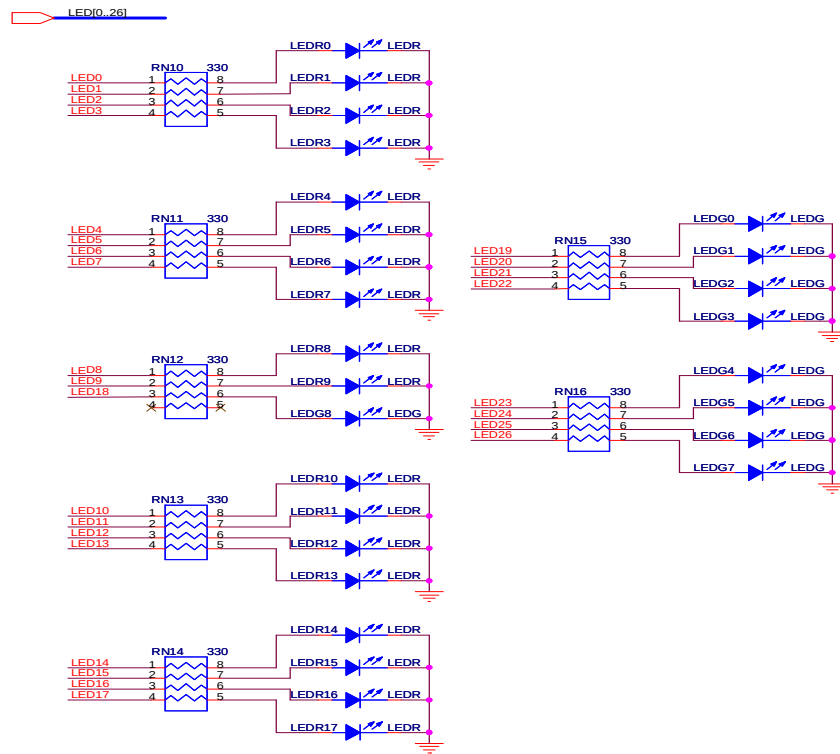


Figure 5.5. Schematic diagram of the LEDs.

Signal Name	FPGA Pin No.	Description
SW[0]	PIN_AA23	Toggle Switch[0]
SW[1]	PIN_AB26	Toggle Switch[1]
SW[2]	PIN_AB25	Toggle Switch[2]
SW[3]	PIN_AC27	Toggle Switch[3]
SW[4]	PIN_AC26	Toggle Switch[4]
SW[5]	PIN_AC24	Toggle Switch[5]
SW[6]	PIN_AC23	Toggle Switch[6]
SW[7]	PIN_AD25	Toggle Switch[7]
SW[8]	PIN_AD24	Toggle Switch[8]
SW[9]	PIN_AE27	Toggle Switch[9]
SW[10]	PIN_W5	Toggle Switch[10]
SW[11]	PIN_V10	Toggle Switch[11]
SW[12]	PIN_U9	Toggle Switch[12]
SW[13]	PIN_T9	Toggle Switch[13]
SW[14]	PIN_L5	Toggle Switch[14]
SW[15]	PIN_L4	Toggle Switch[15]

SW[16]	PIN_L7	Toggle Switch[16]
SW[17]	PIN_L8	Toggle Switch[17]

Table 5.1. Pin assignments for the toggle switches.

Signal Name	FPGA Pin No.	Description
KEY[0]	PIN_T29	Pushbutton[0]
KEY[1]	PIN_T28	Pushbutton[1]
KEY[2]	PIN_U30	Pushbutton[2]
KEY[3]	PIN_U29	Pushbutton[3]

Table 5.2. Pin assignments for the pushbutton switches.

Signal Name	FPGA Pin No.	Description
LEDR[0]	PIN_AJ6	LED Red[0]
LEDR[1]	PIN_AK5	LED Red[1]
LEDR[2]	PIN_AJ5	LED Red[2]
LEDR[3]	PIN_AJ4	LED Red[3]
LEDR[4]	PIN_AK3	LED Red[4]
LEDR[5]	PIN_AH4	LED Red[5]
LEDR[6]	PIN_AJ3	LED Red[6]
LEDR[7]	PIN_AJ2	LED Red[7]
LEDR[8]	PIN_AH3	LED Red[8]
LEDR[9]	PIN_AD14	LED Red[9]
LEDR[10]	PIN_AC13	LED Red[10]
LEDR[11]	PIN_AB13	LED Red[11]
LEDR[12]	PIN_AC12	LED Red[12]
LEDR[13]	PIN_AB12	LED Red[13]
LEDR[14]	PIN_AC11	LED Red[14]
LEDR[15]	PIN_AD9	LED Red[15]
LEDR[16]	PIN_AD8	LED Red[16]
LEDR[17]	PIN_AJ7	LED Red[17]
LEDG[0]	PIN_W27	LED Green[0]
LEDG[1]	PIN_W25	LED Green[1]
LEDG[2]	PIN_W23	LED Green[2]
LEDG[3]	PIN_Y27	LED Green[3]
LEDG[4]	PIN_Y24	LED Green[4]
LEDG[5]	PIN_Y23	LED Green[5]
LEDG[6]	PIN_AA27	LED Green[6]

LEDG[7]	PIN_AA24	LED Green[7]
LEDG[8]	PIN_AC14	LED Green[8]

Table 5.3. Pin assignments for the LEDs.

5.3 Using the 7-segment Displays

The DE2-70 Board has eight 7-segment displays. These displays are arranged into two pairs and a group of four, with the intent of displaying numbers of various sizes. As indicated in the schematic in Figure 5.6, the seven segments are connected to pins on the Cyclone II FPGA. Applying a low logic level to a segment causes it to light up, and applying a high logic level turns it off.

Each segment in a display is identified by an index from 0 to 6, with the positions given in Figure 5.7. In addition, the decimal point is identified as DP. Table 5.4 shows the assignments of FPGA pins to the 7-segment displays.

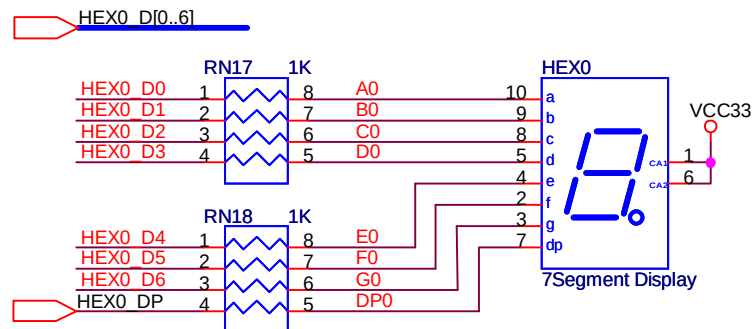


Figure 5.6. Schematic diagram of the 7-segment displays.

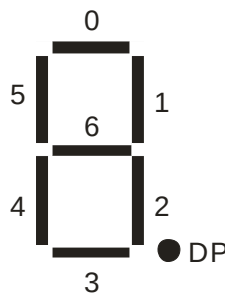


Figure 5.7. Position and index of each segment in a 7-segment display.

Signal Name	FPGA Pin No.	Description
HEX0_D[0]	PIN_AE8	Seven Segment Digit 0[0]
HEX0_D[1]	PIN_AF9	Seven Segment Digit 0[1]
HEX0_D[2]	PIN_AH9	Seven Segment Digit 0[2]

HEX0_D[3]	PIN_AD10	Seven Segment Digit 0[3]
HEX0_D[4]	PIN_AF10	Seven Segment Digit 0[4]
HEX0_D[5]	PIN_AD11	Seven Segment Digit 0[5]
HEX0_D[6]	PIN_AD12	Seven Segment Digit 0[6]
HEX0_DP	PIN_AF12	Seven Segment Decimal Point 0
HEX1_D[0]	PIN_AG13	Seven Segment Digit 1[0]
HEX1_D[1]	PIN_AE16	Seven Segment Digit 1[1]
HEX1_D[2]	PIN_AF16	Seven Segment Digit 1[2]
HEX1_D[3]	PIN_AG16	Seven Segment Digit 1[3]
HEX1_D[4]	PIN_AE17	Seven Segment Digit 1[4]
HEX1_D[5]	PIN_AF17	Seven Segment Digit 1[5]
HEX1_D[6]	PIN_AD17	Seven Segment Digit 1[6]
HEX1_DP	PIN_AC17	Seven Segment Decimal Point 1
HEX2_D[0]	PIN_AE7	Seven Segment Digit 2[0]
HEX2_D[1]	PIN_AF7	Seven Segment Digit 2[1]
HEX2_D[2]	PIN_AH5	Seven Segment Digit 2[2]
HEX2_D[3]	PIN_AG4	Seven Segment Digit 2[3]
HEX2_D[4]	PIN_AB18	Seven Segment Digit 2[4]
HEX2_D[5]	PIN_AB19	Seven Segment Digit 2[5]
HEX2_D[6]	PIN_AE19	Seven Segment Digit 2[6]
HEX2_DP	PIN_AC19	Seven Segment Decimal Point 2
HEX3_D[0]	PIN_P6	Seven Segment Digit 3[0]
HEX3_D[1]	PIN_P4	Seven Segment Digit 3[1]
HEX3_D[2]	PIN_N10	Seven Segment Digit 3[2]
HEX3_D[3]	PIN_N7	Seven Segment Digit 3[3]
HEX3_D[4]	PIN_M8	Seven Segment Digit 3[4]
HEX3_D[5]	PIN_M7	Seven Segment Digit 3[5]
HEX3_D[6]	PIN_M6	Seven Segment Digit 3[6]
HEX3_DP	PIN_M4	Seven Segment Decimal Point 3
HEX4_D[0]	PIN_P1	Seven Segment Digit 4[0]
HEX4_D[1]	PIN_P2	Seven Segment Digit 4[1]
HEX4_D[2]	PIN_P3	Seven Segment Digit 4[2]
HEX4_D[3]	PIN_N2	Seven Segment Digit 4[3]
HEX4_D[4]	PIN_N3	Seven Segment Digit 4[4]
HEX4_D[5]	PIN_M1	Seven Segment Digit 4[5]
HEX4_D[6]	PIN_M2	Seven Segment Digit 4[6]
HEX4_DP	PIN_L6	Seven Segment Decimal Point 4

HEX5_D[0]	PIN_M3	Seven Segment Digit 5[0]
HEX5_D[1]	PIN_L1	Seven Segment Digit 5[1]
HEX5_D[2]	PIN_L2	Seven Segment Digit 5[2]
HEX5_D[3]	PIN_L3	Seven Segment Digit 5[3]
HEX5_D[4]	PIN_K1	Seven Segment Digit 5[4]
HEX5_D[5]	PIN_K4	Seven Segment Digit 5[5]
HEX5_D[6]	PIN_K5	Seven Segment Digit 5[6]
HEX5_DP	PIN_K6	Seven Segment Decimal Point 5
HEX6_D[0]	PIN_H6	Seven Segment Digit 6[0]
HEX6_D[1]	PIN_H4	Seven Segment Digit 6[1]
HEX6_D[2]	PIN_H7	Seven Segment Digit 6[2]
HEX6_D[3]	PIN_H8	Seven Segment Digit 6[3]
HEX6_D[4]	PIN_G4	Seven Segment Digit 6[4]
HEX6_D[5]	PIN_F4	Seven Segment Digit 6[5]
HEX6_D[6]	PIN_E4	Seven Segment Digit 6[6]
HEX6_DP	PIN_K2	Seven Segment Decimal Point 6
HEX7_D[0]	PIN_K3	Seven Segment Digit 7[0]
HEX7_D[1]	PIN_J1	Seven Segment Digit 7[1]
HEX7_D[2]	PIN_J2	Seven Segment Digit 7[2]
HEX7_D[3]	PIN_H1	Seven Segment Digit 7[3]
HEX7_D[4]	PIN_H2	Seven Segment Digit 7[4]
HEX7_D[5]	PIN_H3	Seven Segment Digit 7[5]
HEX7_D[6]	PIN_G1	Seven Segment Digit 7[6]
HEX7_DP	PIN_G2	Seven Segment Decimal Point 7

Table 5.4. Pin assignments for the 7-segment displays.

5.4 Clock Circuitry

The DE2-70 board includes two oscillators that produce 28.86 MHz and 50 MHz clock signals. Both two clock signals are connected to the FPGA that are used for clocking the user logic. Also, the 28.86 MHz oscillator is used to drive the two TV decoders. The board also includes an SMA connector which can be used to connect an external clock source to the board. In addition, all these clock inputs are connected to the phase lock loops (PLL) clock input pin of the FPGA allowed users can use these clocks as a source clock for the PLL circuit.

The clock distribution on the DE2-70 board is shown in Figure 5.8. The associated pin assignments for clock inputs to FPGA I/O pins are listed in Table 5.5.

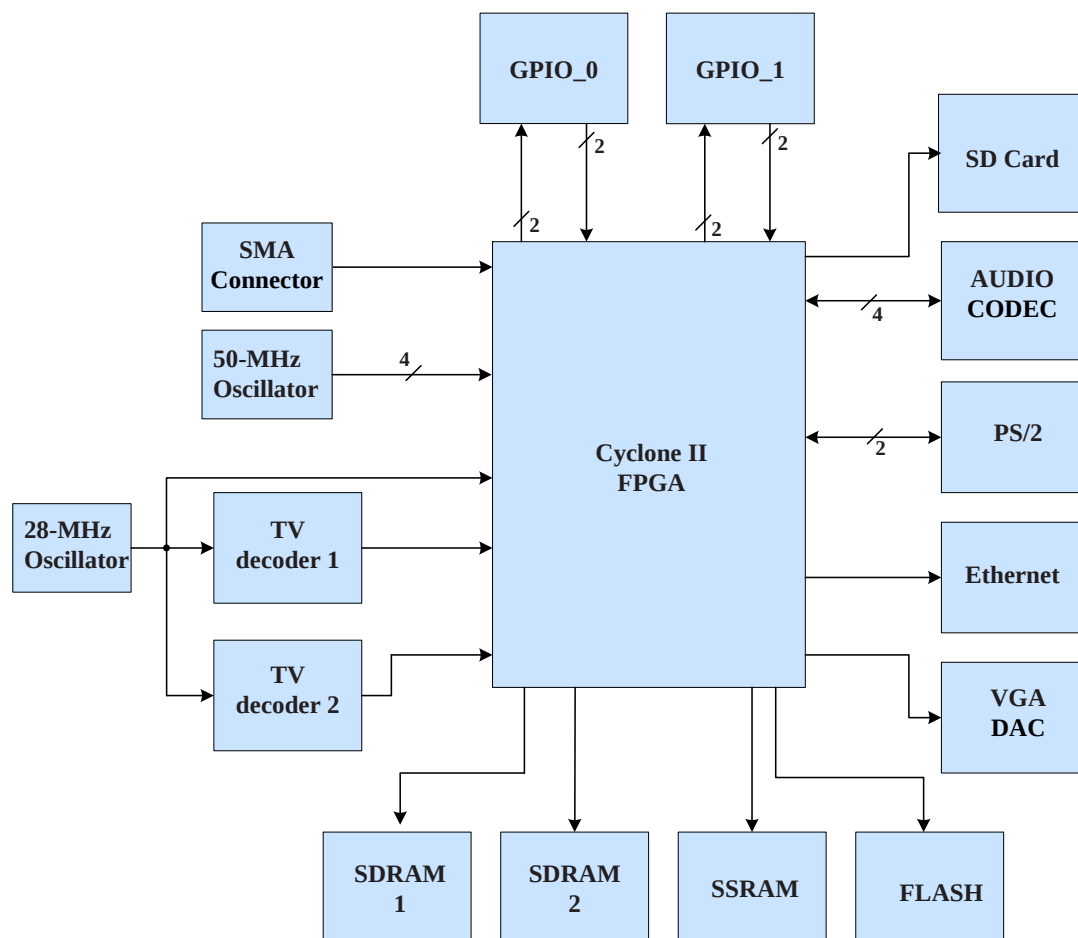


Figure 5.8. Block diagram of the clock distribution.

Signal Name	FPGA Pin No.	Description
CLK_28	PIN_E16	28 MHz clock input
CLK_50	PIN_AD15	50 MHz clock input
CLK_50_2	PIN_D16	50 MHz clock input
CLK_50_3	PIN_R28	50 MHz clock input
CLK_50_4	PIN_R3	50 MHz clock input
EXT_CLOCK	PIN_R29	External (SMA) clock input

Table 5.5. Pin assignments for the clock inputs.

5.5 Using the LCD Module

The LCD module has built-in fonts and can be used to display text by sending appropriate commands to the display controller, which is called HD44780. Detailed information for using the display is available in its datasheet, which can be found on the manufacturer's web site, and from the *Datasheet/LCD* folder on the **DE2-70 System CD-ROM**. A schematic diagram of the LCD module showing connections to the Cyclone II FPGA is given in Figure 5.9. The associated pin assignments appear in Table 5.6.

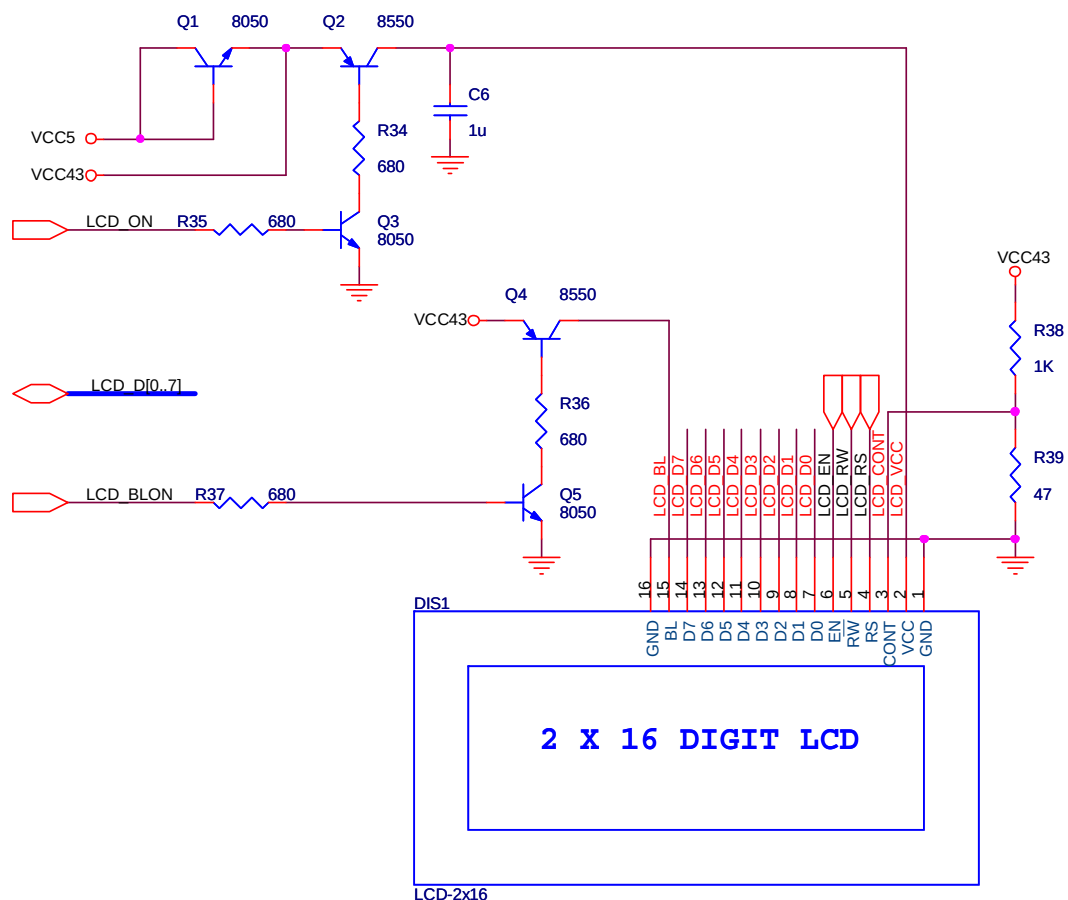


Figure 5.9. Schematic diagram of the LCD module.

Signal Name	FPGA Pin No.	Description
LCD_DATA[0]	PIN_E1	LCD Data[0]
LCD_DATA[1]	PIN_E3	LCD Data[1]
LCD_DATA[2]	PIN_D2	LCD Data[2]
LCD_DATA[3]	PIN_D3	LCD Data[3]
LCD_DATA[4]	PIN_C1	LCD Data[4]

LCD_DATA[5]	PIN_C2	LCD Data[5]
LCD_DATA[6]	PIN_C3	LCD Data[6]
LCD_DATA[7]	PIN_B2	LCD Data[7]
LCD_RW	PIN_F3	LCD Read/Write Select, 0 = Write, 1 = Read
LCD_EN	PIN_E2	LCD Enable
LCD_RS	PIN_F2	LCD Command/Data Select, 0 = Command, 1 = Data
LCD_ON	PIN_F1	LCD Power ON/OFF
LCD_BLON	PIN_G3	LCD Back Light ON/OFF

Table 5.6. Pin assignments for the LCD module.

Note that the current LCD modules used on DE2/DE2-70 boards do not have backlight. Therefore the LCD_BLON signal should not be used in users' design projects.

5.6 Using the Expansion Header

The DE2-70 Board provides two 40-pin expansion headers. Each header connects directly to 36 pins of the Cyclone II FPGA, and also provides DC +5V (VCC5), DC +3.3V (VCC33), and two GND pins. Among these 36 I/O pins, 4 pins are connected to the PLL clock input and output pins of the FPGA allowing the expansion daughter cards to access the PLL blocks in the FPGA.

The voltage level of the I/O pins on the expansion headers can be adjusted to 3.3V, 2.5V, or 1.8V using JP1. Because the expansion I/Os are connected to the BANK 5 of the FPGA and the VCCIO voltage (VCCIO5) of this bank is controlled by the header JP1, users can use a jumper to select the input voltage of VCCIO5 to 3.3V, 2.5V, and 1.8V to control the voltage level of the I/O pins. Table 5.7 lists the jumper settings of the JP1. The pin-outs of the JP1 appear in the Figure 5.10.

Finally, Figure 5.11 shows the related schematics. Each pin on the expansion headers is connected to two diodes and a resistor that provide protection from high and low voltages. The figure shows the protection circuitry for only two of the pins on each header, but this circuitry is included for all 72 data pins. Table 5.8 gives the pin assignments.

JP1 Jumper Settings	Supplied Voltage to VCCIO5	IO Voltage of Expansion Headers (J4/J5)
Short Pins 1 and 2	1.8V	1.8V
Short Pins 3 and 4	2.5V	2.5V
Short Pins 5 and 6	3.3V	3.3V

Table 5.7. Voltage level setting of the expansion headers using JP1.

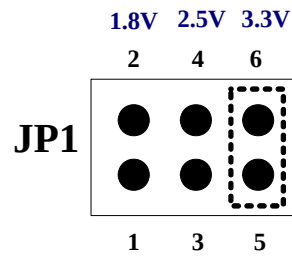


Figure 5.10. JP1 pin settings.

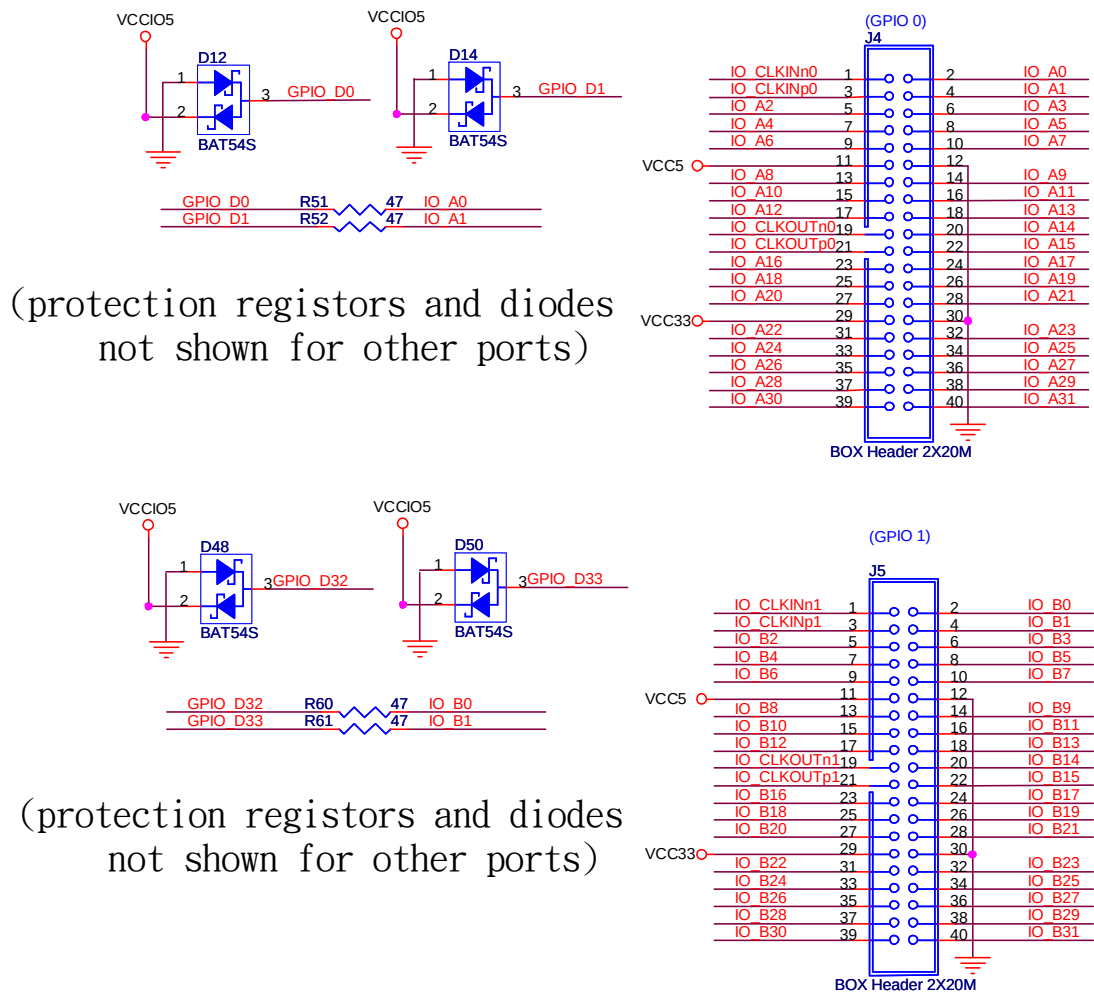


Figure 5.11. Schematic diagram of the expansion headers.

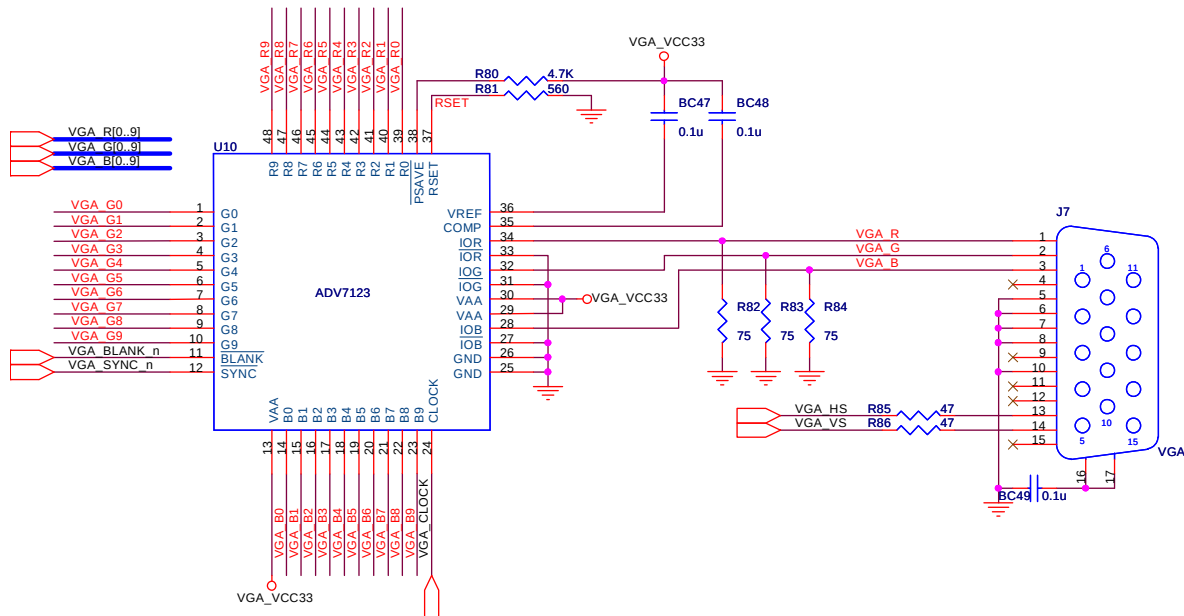
Signal Name	FPGA Pin No.	Description
IO_A [0]	PIN_C30	GPIO Connection 0 IO[0]
IO_A [1]	PIN_C29	GPIO Connection 0 IO[1]
IO_A [2]	PIN_E28	GPIO Connection 0 IO[2]

IO_A [3]	PIN_D29	GPIO Connection 0 IO[3]
IO_A [4]	PIN_E27	GPIO Connection 0 IO[4]
IO_A [5]	PIN_D28	GPIO Connection 0 IO[5]
IO_A [6]	PIN_E29	GPIO Connection 0 IO[6]
IO_A [7]	PIN_G25	GPIO Connection 0 IO[7]
IO_A [8]	PIN_E30	GPIO Connection 0 IO[8]
IO_A [9]	PIN_G26	GPIO Connection 0 IO[9]
IO_A [10]	PIN_F29	GPIO Connection 0 IO[10]
IO_A [11]	PIN_G29	GPIO Connection 0 IO[11]
IO_A [12]	PIN_F30	GPIO Connection 0 IO[12]
IO_A [13]	PIN_G30	GPIO Connection 0 IO[13]
IO_A [14]	PIN_H29	GPIO Connection 0 IO[14]
IO_A [15]	PIN_H30	GPIO Connection 0 IO[15]
IO_A [16]	PIN_J29	GPIO Connection 0 IO[16]
IO_A [17]	PIN_H25	GPIO Connection 0 IO[17]
IO_A [18]	PIN_J30	GPIO Connection 0 IO[18]
IO_A [19]	PIN_H24	GPIO Connection 0 IO[19]
IO_A [20]	PIN_J25	GPIO Connection 0 IO[20]
IO_A [21]	PIN_K24	GPIO Connection 0 IO[21]
IO_A [22]	PIN_J24	GPIO Connection 0 IO[22]
IO_A [23]	PIN_K25	GPIO Connection 0 IO[23]
IO_A [24]	PIN_L22	GPIO Connection 0 IO[24]
IO_A [25]	PIN_M21	GPIO Connection 0 IO[25]
IO_A [26]	PIN_L21	GPIO Connection 0 IO[26]
IO_A [27]	PIN_M22	GPIO Connection 0 IO[27]
IO_A [28]	PIN_N22	GPIO Connection 0 IO[28]
IO_A [29]	PIN_N25	GPIO Connection 0 IO[29]
IO_A [30]	PIN_N21	GPIO Connection 0 IO[30]
IO_A [31]	PIN_N24	GPIO Connection 0 IO[31]
IO_CLKINN0	PIN_T25	GPIO Connection 0 PLL In
IO_CLKINP0	PIN_T24	GPIO Connection 0 PLL In
IO_CLKOUTN0	PIN_H23	GPIO Connection 0 PLL Out
IO_CLKOUTP0	PIN_G24	GPIO Connection 0 PLL Out
IO_B [0]	PIN_G27	GPIO Connection 1 IO[0]
IO_B [1]	PIN_G28	GPIO Connection 1 IO[1]
IO_B [2]	PIN_H27	GPIO Connection 1 IO[2]
IO_B [3]	PIN_L24	GPIO Connection 1 IO[3]

IO_B [4]	PIN_H28	GPIO Connection 1 IO[4]
IO_B [5]	PIN_L25	GPIO Connection 1 IO[5]
IO_B [6]	PIN_K27	GPIO Connection 1 IO[6]
IO_B [7]	PIN_L28	GPIO Connection 1 IO[7]
IO_B [8]	PIN_K28	GPIO Connection 1 IO[8]
IO_B [9]	PIN_L27	GPIO Connection 1 IO[9]
IO_B [10]	PIN_K29	GPIO Connection 1 IO[10]
IO_B [11]	PIN_M25	GPIO Connection 1 IO[11]
IO_B [12]	PIN_K30	GPIO Connection 1 IO[12]
IO_B [13]	PIN_M24	GPIO Connection 1 IO[13]
IO_B [14]	PIN_L29	GPIO Connection 1 IO[14]
IO_B [15]	PIN_L30	GPIO Connection 1 IO[15]
IO_B [16]	PIN_P26	GPIO Connection 1 IO[16]
IO_B [17]	PIN_P28	GPIO Connection 1 IO[17]
IO_B [18]	PIN_P25	GPIO Connection 1 IO[18]
IO_B [19]	PIN_P27	GPIO Connection 1 IO[19]
IO_B [20]	PIN_M29	GPIO Connection 1 IO[20]
IO_B [21]	PIN_R26	GPIO Connection 1 IO[21]
IO_B [22]	PIN_M30	GPIO Connection 1 IO[22]
IO_B [23]	PIN_R27	GPIO Connection 1 IO[23]
IO_B [24]	PIN_P24	GPIO Connection 1 IO[24]
IO_B [25]	PIN_N28	GPIO Connection 1 IO[25]
IO_B [26]	PIN_P23	GPIO Connection 1 IO[26]
IO_B [27]	PIN_N29	GPIO Connection 1 IO[27]
IO_B [28]	PIN_R23	GPIO Connection 1 IO[28]
IO_B [29]	PIN_P29	GPIO Connection 1 IO[29]
IO_B [30]	PIN_R22	GPIO Connection 1 IO[30]
IO_B [31]	PIN_P30	GPIO Connection 1 IO[31]
GPIO_CLKINN1	PIN_AH14	GPIO Connection 1 PLL In
GPIO_CLKINP1	PIN_AG15	GPIO Connection 1 PLL In
GPIO_CLKOUTN1	PIN_AF27	GPIO Connection 1 PLL Out
GPIO_CLKOUTP1	PIN_AF28	GPIO Connection 1 PLL Out

Table 5.8. Pin assignments for the expansion headers.

The DE2-70 board includes a 16-pin D-SUB connector for VGA output. The VGA synchronization signals are provided directly from the Cyclone II FPGA, and the Analog Devices ADV7123 triple 10-bit high-speed video DAC is used to produce the analog data signals (red, green, and blue). The associated schematic is given in Figure 5.12 and can support resolutions of up to 1600 x 1200 pixels, at 100 MHz.



The timing specification for VGA synchronization and RGB (red, green, blue) data can be found on various educational web sites (for example, search for “VGA signal timing”). Figure 5.13 illustrates the basic timing requirements for each row (horizontal) that is displayed on a VGA monitor. An active-low pulse of specific duration (time a in the figure) is applied to the horizontal synchronization ($hsync$) input of the monitor, which signifies the end of one row of data and the start of the next. The data (RGB) inputs on the monitor must be off (driven to 0 V) for a time period called the *back porch* (b) after the $hsync$ pulse occurs, which is followed by the display interval (c). During the data display interval the RGB data drives each pixel in turn across the row being displayed. Finally, there is a time period called the *front porch* (d) where the RGB signals must again be off before the next $hsync$ pulse can occur. The timing of the vertical synchronization ($vsync$) is the same as shown in Figure 5.13, except that a $vsync$ pulse signifies the end of one frame and the start of the next, and the data refers to the set of rows in the frame (horizontal timing). Table 5.9 and 5.10 show, for different resolutions, the durations of time periods a , b , c , and d for both horizontal and vertical timing.

45

found on the manufacturer's web site, or in the *Datasheet/VGA DAC* folder on the **DE2-70 System CD-ROM**. The pin assignments between the Cyclone II FPGA and the ADV7123 are listed in Table 5.11. An example of code that drives a VGA display is described in Sections 6.2, 6.3 and 6.4.

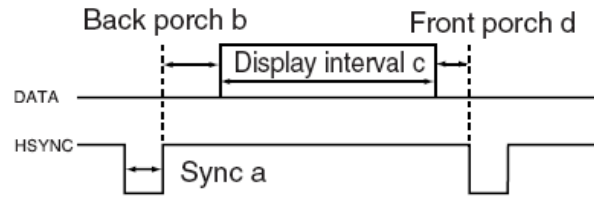


Figure 5.13. VGA horizontal timing specification.

VGA mode		Horizontal Timing Spec				
Configuration	Resolution(HxV)	a(us)	b(us)	c(us)	d(us)	Pixel clock(Mhz)
VGA(60Hz)	640x480	3.8	1.9	25.4	0.6	25 (640/c)
VGA(85Hz)	640x480	1.6	2.2	17.8	1.6	36 (640/c)
SVGA(60Hz)	800x600	3.2	2.2	20	1	40 (800/c)
SVGA(75Hz)	800x600	1.6	3.2	16.2	0.3	49 (800/c)
SVGA(85Hz)	800x600	1.1	2.7	14.2	0.6	56 (800/c)
XGA(60Hz)	1024x768	2.1	2.5	15.8	0.4	65 (1024/c)
XGA(70Hz)	1024x768	1.8	1.9	13.7	0.3	75 (1024/c)
XGA(85Hz)	1024x768	1.0	2.2	10.8	0.5	95 (1024/c)
1280x1024(60Hz)	1280x1024	1.0	2.3	11.9	0.4	108 (1280/c)

Table 5.9. VGA horizontal timing specification.

VGA mode		Vertical Timing Spec			
Configuration	Resolution (HxV)	a(lines)	b(lines)	c(lines)	d(lines)
VGA(60Hz)	640x480	2	33	480	10
VGA(85Hz)	640x480	3	25	480	1
SVGA(60Hz)	800x600	4	23	600	1
SVGA(75Hz)	800x600	3	21	600	1
SVGA(85Hz)	800x600	3	27	600	1
XGA(60Hz)	1024x768	6	29	768	3
XGA(70Hz)	1024x768	6	29	768	3
XGA(85Hz)	1024x768	3	36	768	1
1280x1024(60Hz)	1280x1024	3	38	1024	1

Table 5.10. VGA vertical timing specification.

Signal Name	FPGA Pin No.	Description
VGA_R[0]	PIN_D23	VGA Red[0]
VGA_R[1]	PIN_E23	VGA Red[1]
VGA_R[2]	PIN_E22	VGA Red[2]
VGA_R[3]	PIN_D22	VGA Red[3]
VGA_R[4]	PIN_H21	VGA Red[4]
VGA_R[5]	PIN_G21	VGA Red[5]
VGA_R[6]	PIN_H20	VGA Red[6]
VGA_R[7]	PIN_F20	VGA Red[7]
VGA_R[8]	PIN_E20	VGA Red[8]
VGA_R[9]	PIN_G20	VGA Red[9]
VGA_G[0]	PIN_A10	VGA Green[0]
VGA_G[1]	PIN_B11	VGA Green[1]
VGA_G[2]	PIN_A11	VGA Green[2]
VGA_G[3]	PIN_C12	VGA Green[3]
VGA_G[4]	PIN_B12	VGA Green[4]
VGA_G[5]	PIN_A12	VGA Green[5]
VGA_G[6]	PIN_C13	VGA Green[6]
VGA_G[7]	PIN_B13	VGA Green[7]
VGA_G[8]	PIN_B14	VGA Green[8]
VGA_G[9]	PIN_A14	VGA Green[9]
VGA_B[0]	PIN_B16	VGA Blue[0]
VGA_B[1]	PIN_C16	VGA Blue[1]
VGA_B[2]	PIN_A17	VGA Blue[2]
VGA_B[3]	PIN_B17	VGA Blue[3]
VGA_B[4]	PIN_C18	VGA Blue[4]
VGA_B[5]	PIN_B18	VGA Blue[5]
VGA_B[6]	PIN_B19	VGA Blue[6]
VGA_B[7]	PIN_A19	VGA Blue[7]
VGA_B[8]	PIN_C19	VGA Blue[8]
VGA_B[9]	PIN_D19	VGA Blue[9]
VGA_CLK	PIN_D24	VGA Clock
VGA_BLANK_N	PIN_C15	VGA BLANK
VGA_HS	PIN_J19	VGA H_SYNC
VGA_VS	PIN_H19	VGA V_SYNC
VGA_SYNC	PIN_B15	VGA SYNC

Table 5.11. ADV7123 pin assignments.

5.8 Using the 24-bit Audio CODEC

The DE2-70 board provides high-quality 24-bit audio via the Wolfson WM8731 audio CODEC (enCOder/DECOder). This chip supports microphone-in, line-in, and line-out ports, with a sample rate adjustable from 8 kHz to 96 kHz. The WM8731 is controlled by a serial I2C bus interface, which is connected to pins on the Cyclone II FPGA. A schematic diagram of the audio circuitry is shown in Figure 5.14, and the FPGA pin assignments are listed in Table 5.12. Detailed information for using the WM8731 codec is available in its datasheet, which can be found on the manufacturer's web site, or in the *Datasheet/Audio CODEC* folder on the **DE2-70 System CD-ROM**.

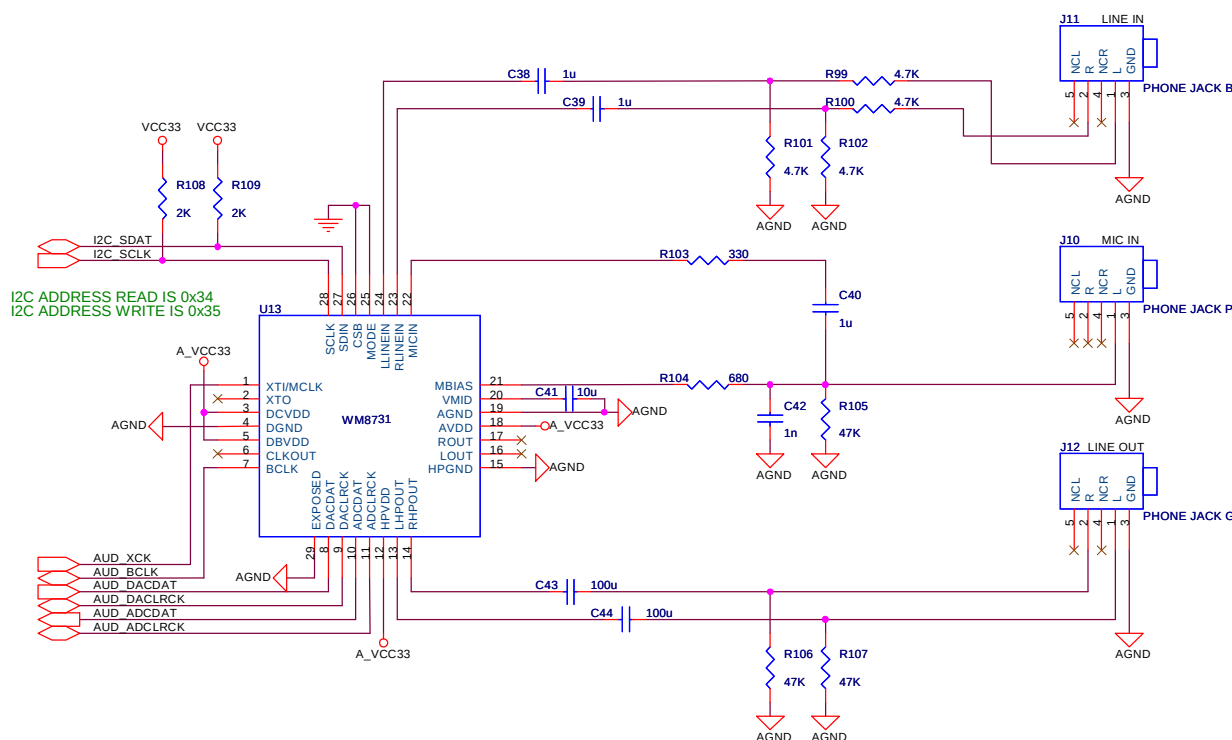


Figure 5.14. Audio CODEC schematic.

Signal Name	FPGA Pin No.	Description
AUD_ADCLRCK	PIN_F19	Audio CODEC ADC LR Clock
AUD_ADCDAT	PIN_E19	Audio CODEC ADC Data
AUD_DACLCK	PIN_G18	Audio CODEC DAC LR Clock
AUD_DACDAT	PIN_F18	Audio CODEC DAC Data
AUD_XCK	PIN_D17	Audio CODEC Chip Clock
AUD_BCLK	PIN_E17	Audio CODEC Bit-Stream Clock
I2C_SCLK	PIN_J18	I2C Data
I2C_SDAT	PIN_H18	I2C Clock

Table 5.12. Audio CODEC pin assignments.

5.9 RS-232 Serial Port

The DE2-70 board uses the ADM3202 transceiver chip and a 9-pin D-SUB connector for RS-232 communications. For detailed information on how to use the transceiver refer to the datasheet, which is available on the manufacturer's web site, or in the *Datasheet/RS232* folder on the **DE2-70 System CD-ROM**. Figure 5.15 shows the related schematics, and Table 5.13 lists the Cyclone II FPGA pin assignments.

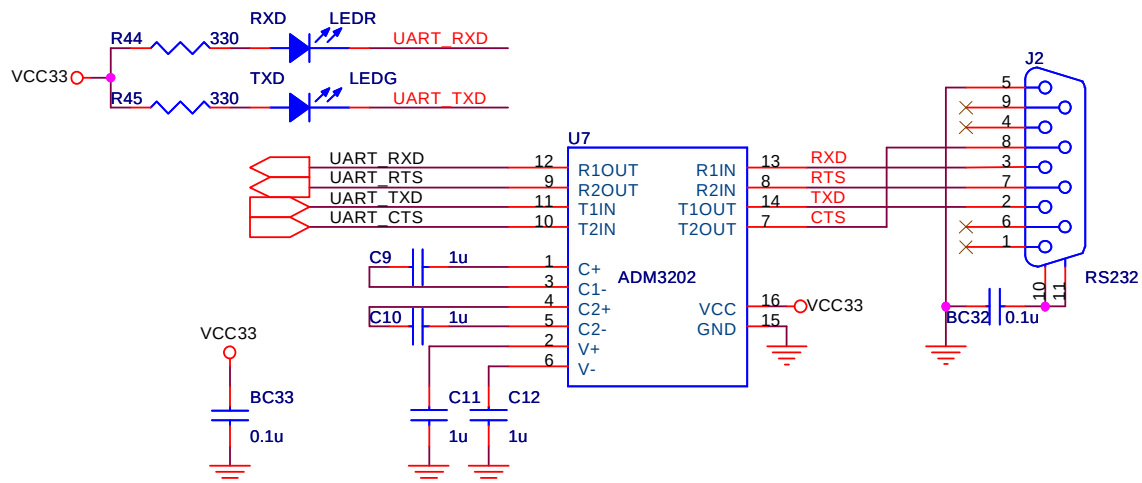


Figure 5.15. MAX232 (RS-232) chip schematic.

Signal Name	FPGA Pin No.	Description
UART_RXD	PIN_D21	UART Receiver
UART_TXD	PIN_E21	UART Transmitter
UART_CTS	PIN_G22	UART Clear to Send
UART_RTS	PIN_F23	UART Request to Send

Table 5.13. RS-232 pin assignments.

5.10 PS/2 Serial Port

The DE2-70 board includes a standard PS/2 interface and a connector for a PS/2 keyboard or mouse. In addition, users can use the PS/2 keyboard and mouse on the DE2-70 board simultaneously by an plug an extension PS/2 Y-Cable. Figure 5.16 shows the schematic of the PS/2 circuit. Instructions for using a PS/2 mouse or keyboard can be found by performing an appropriate search on various educational web sites. The pin assignments for the associated interface are shown in Table 5.14.

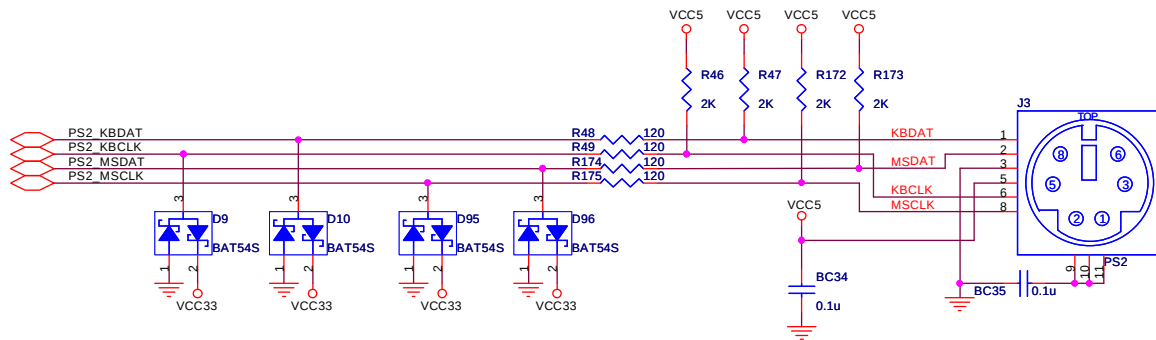


Figure 5.16. PS/2 schematic.

Signal Name	FPGA Pin No.	Description
PS2_KBCLK	PIN_F24	PS/2 Clock
PS2_KBDAT	PIN_E24	PS/2 Data
PS2_MSCLK	PIN_D26	PS/2 Clock (reserved for second PS/2 device)
PS2_MSDAT	PIN_D25	PS/2 Data(reserved for second PS/2 device)

Table 5.14. PS/2 pin assignments.

5.11 Fast Ethernet Network Controller

The DE2-70 board provides Ethernet support via the Davicom DM9000A Fast Ethernet controller chip. The DM9000A includes a general processor interface, 16 Kbytes SRAM, a media access control (MAC) unit, and a 10/100M PHY transceiver. Figure 5.17 shows the schematic for the Fast Ethernet interface, and the associated pin assignments are listed in Table 5.15. For detailed information on how to use the DM9000A refer to its datasheet and application note, which are available on the manufacturer's web site, or in the *Datasheet/Ethernet* folder on the **DE2-70 System CD-ROM**.

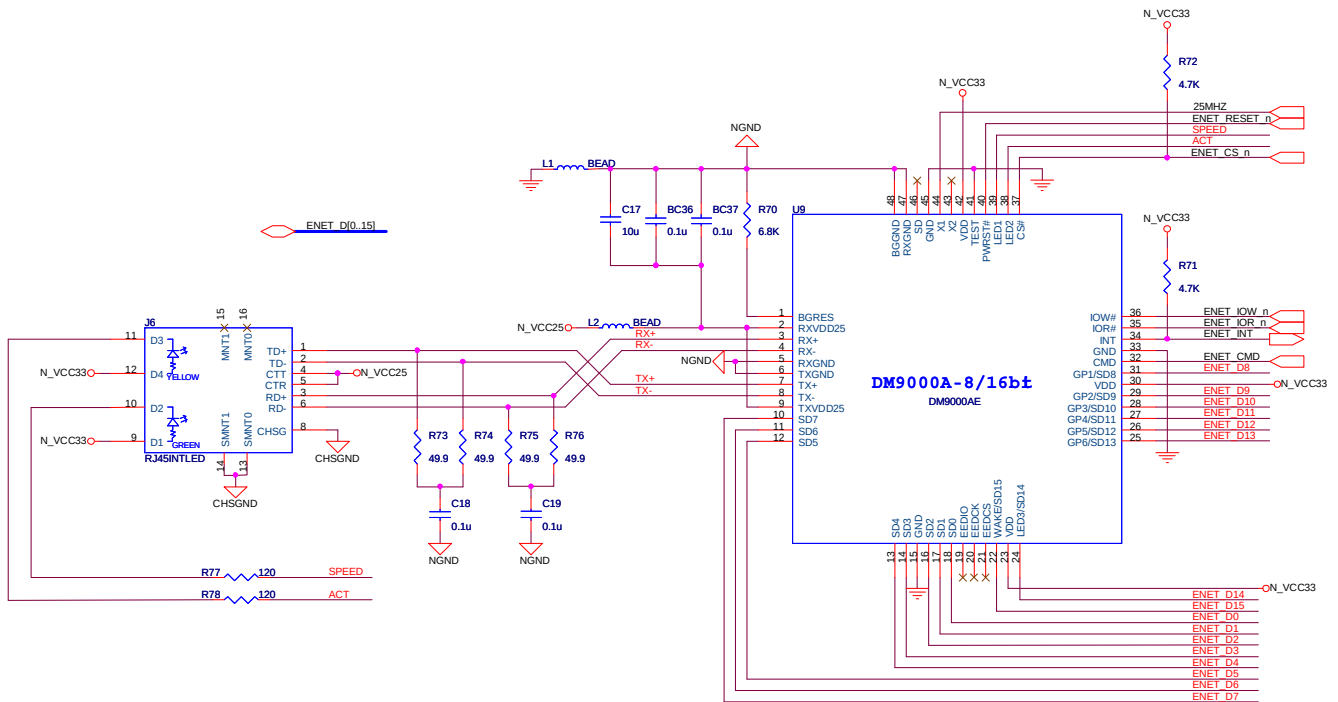


Figure 5.17. Fast Ethernet schematic.

Signal Name	FPGA Pin No.	Description
ENET_DATA[0]	PIN_A23	DM9000A DATA[0]
ENET_DATA[1]	PIN_C22	DM9000A DATA[1]
ENET_DATA[2]	PIN_B22	DM9000A DATA[2]
ENET_DATA[3]	PIN_A22	DM9000A DATA[3]
ENET_DATA[4]	PIN_B21	DM9000A DATA[4]
ENET_DATA[5]	PIN_A21	DM9000A DATA[5]
ENET_DATA[6]	PIN_B20	DM9000A DATA[6]
ENET_DATA[7]	PIN_A20	DM9000A DATA[7]
ENET_DATA[8]	PIN_B26	DM9000A DATA[8]
ENET_DATA[9]	PIN_A26	DM9000A DATA[9]
ENET_DATA[10]	PIN_B25	DM9000A DATA[10]
ENET_DATA[11]	PIN_A25	DM9000A DATA[11]
ENET_DATA[12]	PIN_C24	DM9000A DATA[12]
ENET_DATA[13]	PIN_B24	DM9000A DATA[13]
ENET_DATA[14]	PIN_A24	DM9000A DATA[14]
ENET_DATA[15]	PIN_B23	DM9000A DATA[15]
ENET_CLK	PIN_D27	DM9000A Clock 25 MHz
ENET_CMD	PIN_B27	DM9000A Command/Data Select, 0 = Command, 1 = Data

ENET_CS_N	PIN_C28	DM9000A Chip Select
ENET_INT	PIN_C27	DM9000A Interrupt
ENET_IOR_N	PIN_A28	DM9000A Read
ENET_IOW_N	PIN_B28	DM9000A Write
ENET_RESET_N	PIN_B29	DM9000A Reset

Table 5.15. Fast Ethernet pin assignments.

5.12 TV Decoder

The DE2-70 board is equipped with two Analog Devices ADV7180 TV decoder chips. The ADV7180 is an integrated video decoder that automatically detects and converts a standard analog baseband television signal (NTSC, PAL, and SECAM) into 4:2:2 component video data compatible with the 8-bit ITU-R BT.656 interface standard. The ADV7180 is compatible with a broad range of video devices, including DVD players, tape-based sources, broadcast sources, and security/surveillance cameras.

The registers in both of the TV decoders can be programmed by a serial I2C bus, which is connected to the Cyclone II FPGA as indicated in Figure 5.18. Note that the I2C address of the TV decoder 1(U11) and TV decoder 2(U12) are 0x40 and 0x42 respectively. The pin assignments are listed in Table 5.16. Detailed information on the ADV7180 is available on the manufacturer's web site, or in the *Datasheet/TV Decoder* folder on the **DE2-70 System CD-ROM**.

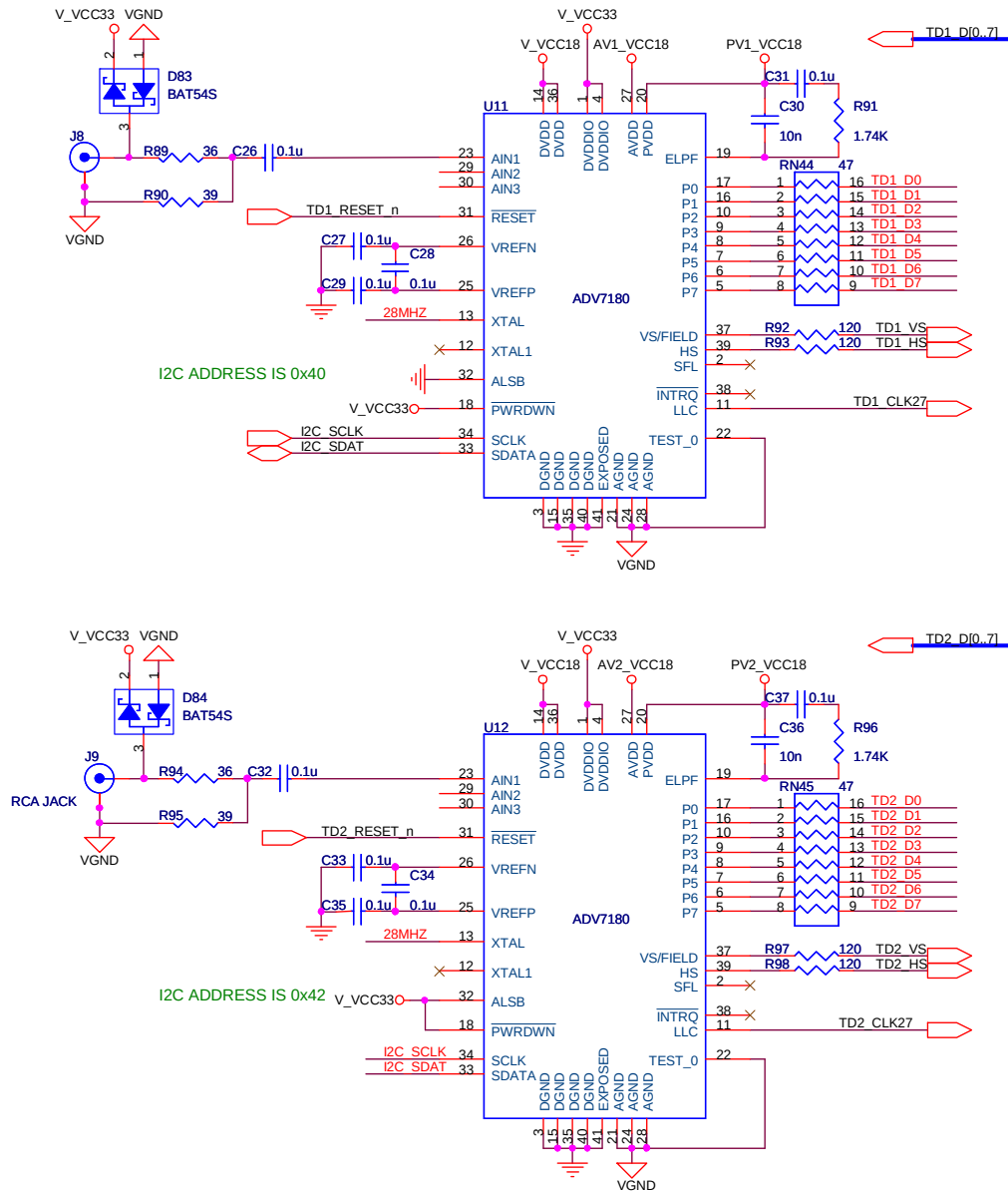


Figure 5.18. TV Decoder schematic.

Signal Name	FPGA Pin No.	Description
TD1_D[0]	PIN_A6	TV Decoder 1 Data[0]
TD1_D[1]	PIN_B6	TV Decoder 1 Data[1]
TD1_D[2]	PIN_A5	TV Decoder 1 Data[2]
TD1_D[3]	PIN_B5	TV Decoder 1 Data[3]
TD1_D[4]	PIN_B4	TV Decoder 1 Data[4]
TD1_D[5]	PIN_C4	TV Decoder 1 Data[5]
TD1_D[6]	PIN_A3	TV Decoder 1 Data[6]
TD1_D[7]	PIN_B3	TV Decoder 1 Data[7]
TD1_HS	PIN_E13	TV Decoder 1 H_SYNC
TD1_VS	PIN_E14	TV Decoder 1 V_SYNC

TD1_CLK27	PIN_G15	TV Decoder 1 Clock Input.
TD1_RESET_N	PIN_D14	TV Decoder 1 Reset
TD2_D[0]	PIN_C10	TV Decoder 2 Data[0]
TD2_D[1]	PIN_A9	TV Decoder 2 Data[1]
TD2_D[2]	PIN_B9	TV Decoder 2 Data[2]
TD2_D[3]	PIN_C9	TV Decoder 2 Data[3]
TD2_D[4]	PIN_A8	TV Decoder 2 Data[4]
TD2_D[5]	PIN_B8	TV Decoder 2 Data[5]
TD2_D[6]	PIN_A7	TV Decoder 2 Data[6]
TD2_D[7]	PIN_B7	TV Decoder 2 Data[7]
TD2_HS	PIN_E15	TV Decoder 2 H_SYNC
TD2_VS	PIN_D15	TV Decoder 2 V_SYNC
TD2_CLK27	PIN_H15	TV Decoder 2 Clock Input.
TD2_RESET_N	PIN_B10	TV Decoder 2 Reset
I2C_SCLK	PIN_J18	I2C Data
I2C_SDAT	PIN_H18	I2C Clock

Table 5.16. TV Decoder pin assignments.

5.13 Implementing a TV Encoder

Although the DE2-70 board does not include a TV encoder chip, the ADV7123 (10-bit high-speed triple ADCs) can be used to implement a professional-quality TV encoder with the digital processing part implemented in the Cyclone II FPGA. Figure 5.19 shows a block diagram of a TV encoder implemented in this manner.

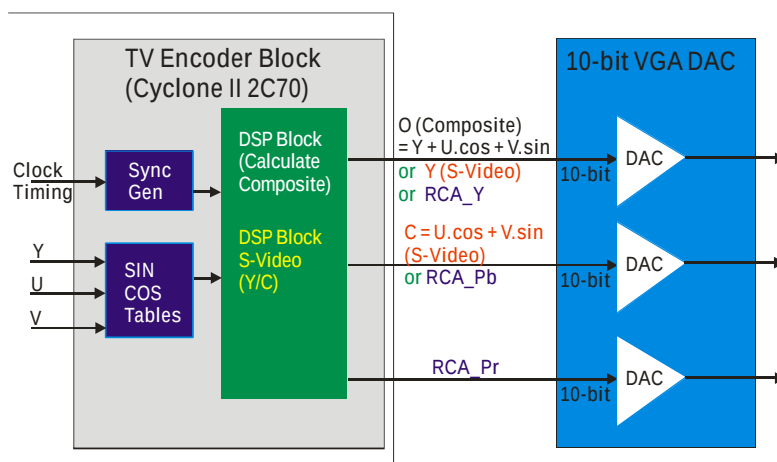


Figure 5.19. A TV Encoder that uses the Cyclone II FPGA and the ADV7123.

The DE2-70 board provides both USB host and device interfaces using the Philips ISP1362 single-chip USB controller. The host and device controllers are compliant with the Universal Serial Bus Specification Rev. 2.0, supporting data transfer at full-speed (12 Mbit/s) and low-speed (1.5 Mbit/s). Figure 5.20 shows the schematic diagram of the USB circuitry; the pin assignments for the associated interface are listed in Table 5.17.

Signal Name	FPGA Pin No.	Description
OTG_A[0]	PIN_E9	ISP1362 Address[0]
OTG_A[1]	PIN_D8	ISP1362 Address[1]
OTG_D[0]	PIN_H10	ISP1362 Data[0]
OTG_D[1]	PIN_G9	ISP1362 Data[1]
OTG_D[2]	PIN_G11	ISP1362 Data[2]
OTG_D[3]	PIN_F11	ISP1362 Data[3]

OTG_D[4]	PIN_J12	ISP1362 Data[4]
OTG_D[5]	PIN_H12	ISP1362 Data[5]
OTG_D[6]	PIN_H13	ISP1362 Data[6]
OTG_D[7]	PIN_G13	ISP1362 Data[7]
OTG_D[8]	PIN_D4	ISP1362 Data[8]
OTG_D[9]	PIN_D5	ISP1362 Data[9]
OTG_D[10]	PIN_D6	ISP1362 Data[10]
OTG_D[11]	PIN_E7	ISP1362 Data[11]
OTG_D[12]	PIN_D7	ISP1362 Data[12]
OTG_D[13]	PIN_E8	ISP1362 Data[13]
OTG_D[14]	PIN_D9	ISP1362 Data[14]
OTG_D[15]	PIN_G10	ISP1362 Data[15]
OTG_CS_N	PIN_E10	ISP1362 Chip Select
OTG_OE_N	PIN_D10	ISP1362 Read
OTG_WE_N	PIN_E11	ISP1362 Write
OTG_RESET_N	PIN_H14	ISP1362 Reset
OTG_INT0	PIN_F13	ISP1362 Interrupt 0
OTG_INT1	PIN_J13	ISP1362 Interrupt 1
OTG_DACK0_N	PIN_D12	ISP1362 DMA Acknowledge 0
OTG_DACK1_N	PIN_E12	ISP1362 DMA Acknowledge 1
OTG_DREQ0	PIN_G12	ISP1362 DMA Request 0
OTG_DREQ1	PIN_F12	ISP1362 DMA Request 1
OTG_FSPEED	PIN_F7	USB Full Speed, 0 = Enable, Z = Disable
OTG_LSPEED	PIN_F8	USB Low Speed, 0 = Enable, Z = Disable

Table 5.17. USB (ISP1362) pin assignments.

5.15 Using IrDA

The DE2-70 board provides a simple wireless communication media using the Agilent HSDL-3201 low power infrared transceiver. The datasheet for this device is provided in the *Datasheet\IrDA* folder on the **DE2-70 System CD-ROM**. Note that the highest transmission rate supported is 115.2 Kbit/s and both the TX and RX sides have to use the same transmission rate. Figure 5.21 shows the schematic of the IrDA communication link. Please refer to the following website for detailed information on how to send and receive data using the IrDA link:

http://techtrain.microchip.com/webseminars/documents/IrDA_BW.pdf

The pin assignments of the associated interface are listed in Table 5.18.

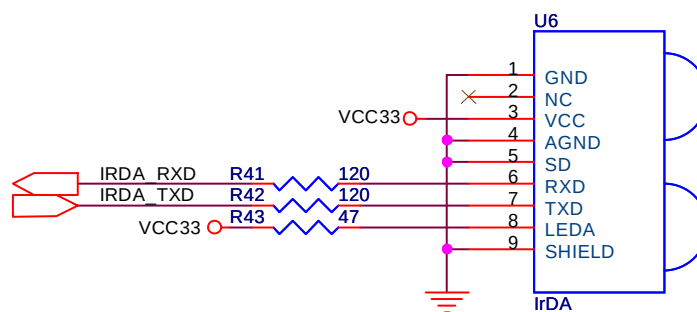


Figure 5.21. IrDA schematic.

Signal Name	FPGA Pin No.	Description
IRDA_TXD	PIN_W21	IRDA Transmitter
IRDA_RXD	PIN_W22	IRDA Receiver

Table 5.18. IrDA pin assignments.

5.16 Using SDRAM/SRAM/Flash

The DE2-70 board provides a 2-Mbyte SSRAM, 8-Mbyte Flash memory, and two 32-Mbyte SDRAM chips. Figures 5.22, 5.23, and 5.24 show the schematics of the memory chips. The pin assignments for each device are listed in Tables 5.19, 5.20, and 5.21. The datasheets for the memory chips are provided in the *Datasheet/Memory* folder on the **DE2-70 System CD-ROM**.

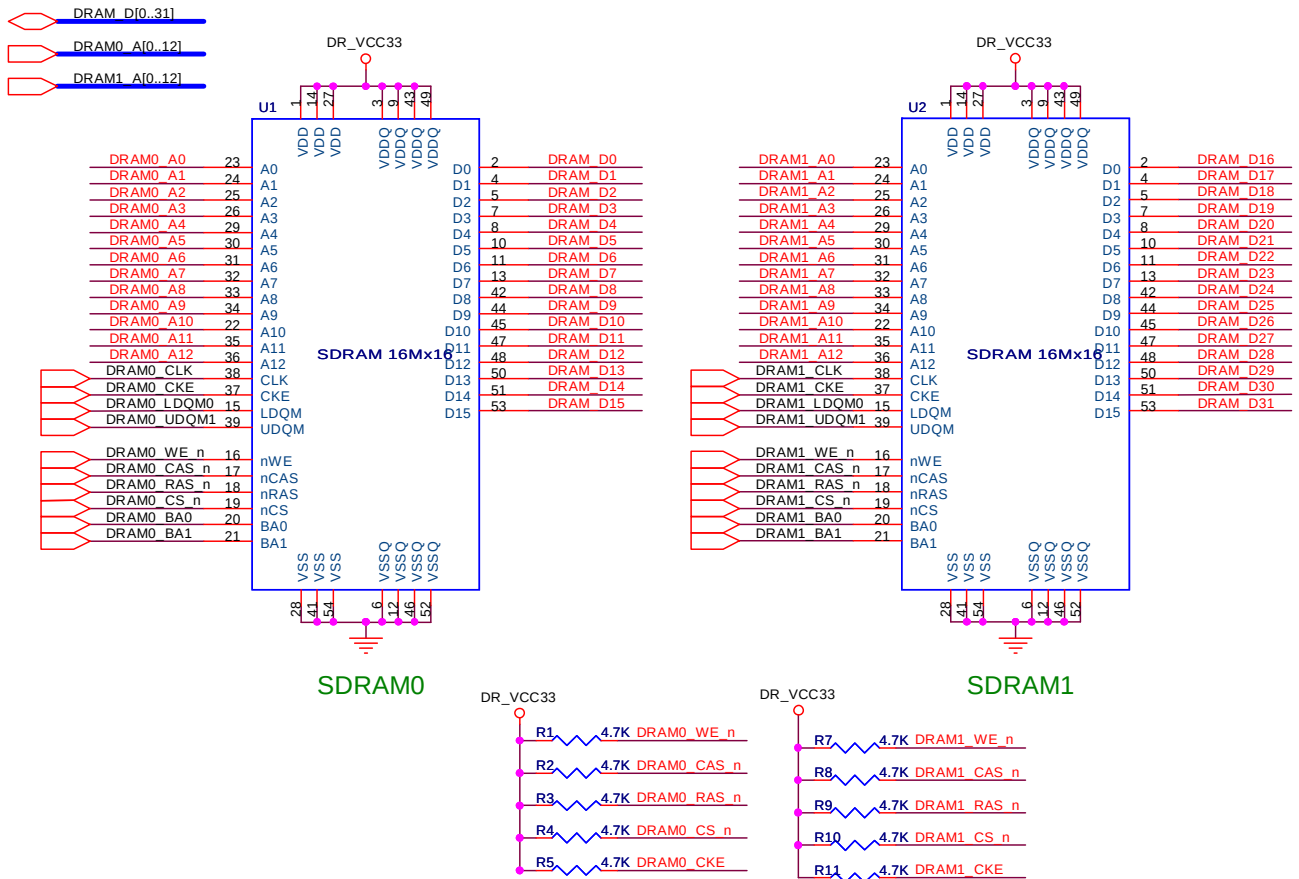


Figure 5.22. SDRAM schematic.

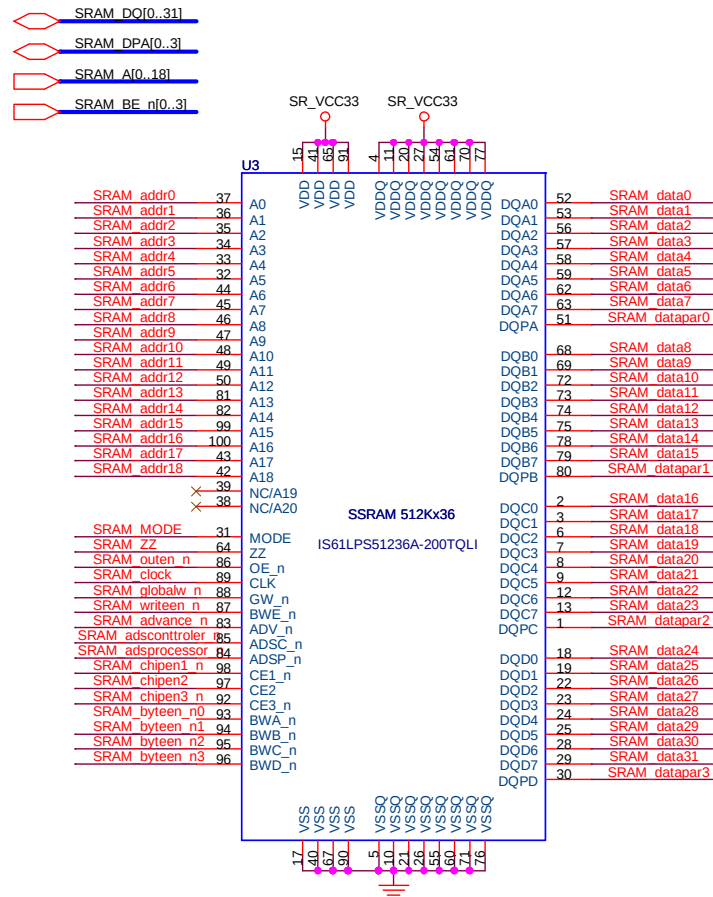


Figure 5.23. SSRAM schematic.

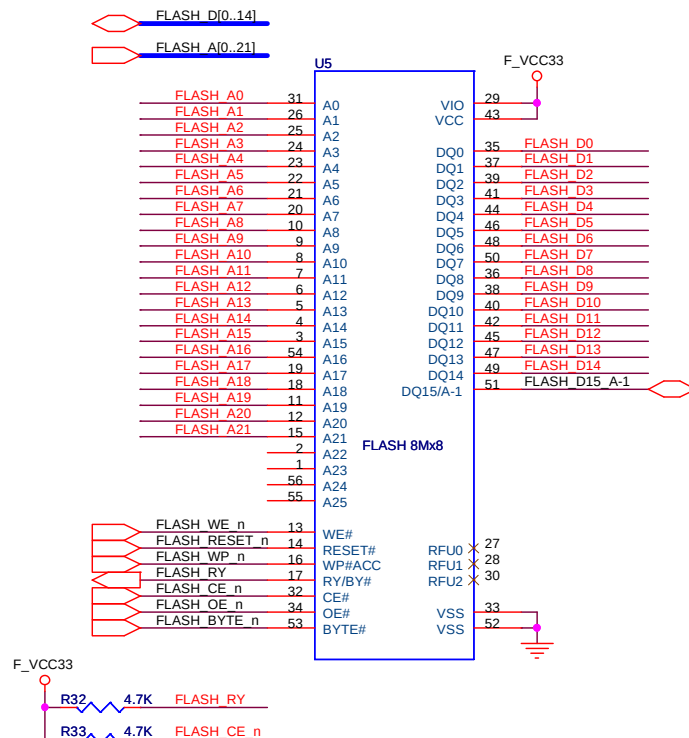


Figure 5.24. Flash schematic.

Signal Name	FPGA Pin No.	Description
DRAM0_A[0]	PIN_AA4	SDRAM 1 Address[0]
DRAM0_A[1]	PIN_AA5	SDRAM 1 Address[1]
DRAM0_A[2]	PIN_AA6	SDRAM 1 Address[2]
DRAM0_A[3]	PIN_AB5	SDRAM 1 Address[3]
DRAM0_A[4]	PIN_AB7	SDRAM 1 Address[4]
DRAM0_A[5]	PIN_AC4	SDRAM 1 Address[5]
DRAM0_A[6]	PIN_AC5	SDRAM 1 Address[6]
DRAM0_A[7]	PIN_AC6	SDRAM 1 Address[7]
DRAM0_A[8]	PIN_AD4	SDRAM 1 Address[8]
DRAM0_A[9]	PIN_AC7	SDRAM 1 Address[9]
DRAM0_A[10]	PIN_Y8	SDRAM 1 Address[10]
DRAM0_A[11]	PIN_AE4	SDRAM 1 Address[11]
DRAM0_A[12]	PIN_AF4	SDRAM 1 Address[12]
DRAM_D[0]	PIN_AC1	SDRAM 1 Data[0]
DRAM_D[1]	PIN_AC2	SDRAM 1 Data[1]
DRAM_D[2]	PIN_AC3	SDRAM 1 Data[2]
DRAM_D[3]	PIN_AD1	SDRAM 1 Data[3]
DRAM_D[4]	PIN_AD2	SDRAM 1 Data[4]
DRAM_D[5]	PIN_AD3	SDRAM 1 Data[5]
DRAM_D[6]	PIN_AE1	SDRAM 1 Data[6]
DRAM_D[7]	PIN_AE2	SDRAM 1 Data[7]
DRAM_D[8]	PIN_AE3	SDRAM 1 Data[8]
DRAM_D[9]	PIN_AF1	SDRAM 1 Data[9]
DRAM_D[10]	PIN_AF2	SDRAM 1 Data[10]
DRAM_D[11]	PIN_AF3	SDRAM 1 Data[11]
DRAM_D[12]	PIN_AG2	SDRAM 1 Data[12]
DRAM_D[13]	PIN_AG3	SDRAM 1 Data[13]
DRAM_D[14]	PIN_AH1	SDRAM 1 Data[14]
DRAM_D[15]	PIN_AH2	SDRAM 1 Data[15]
DRAM0_BA_0	PIN_AA9	SDRAM 1 Bank Address[0]
DRAM0_BA_1	PIN_AA10	SDRAM 1 Bank Address[1]
DRAM0_LDQM0	PIN_V9	SDRAM 1 Low-byte Data Mask
DRAM0_UDQM1	PIN_AB6	SDRAM 1 High-byte Data Mask
DRAM0_RAS_N	PIN_Y9	SDRAM 1 Row Address Strobe
DRAM0_CAS_N	PIN_W10	SDRAM 1 Column Address Strobe

DRAM0_CKE	PIN_AA8	SDRAM 1 Clock Enable
DRAM0_CLK	PIN_AD6	SDRAM 1 Clock
DRAM0_WE_N	PIN_W9	SDRAM 1 Write Enable
DRAM0_CS_N	PIN_Y10	SDRAM 1 Chip Select
DRAM1_A[0]	PIN_T5	SDRAM 2 Address[0]
DRAM1_A[1]	PIN_T6	SDRAM 2 Address[1]
DRAM1_A[2]	PIN_U4	SDRAM 2 Address[2]
DRAM1_A[3]	PIN_U6	SDRAM 2 Address[3]
DRAM1_A[4]	PIN_U7	SDRAM 2 Address[4]
DRAM1_A[5]	PIN_V7	SDRAM 2 Address[5]
DRAM1_A[6]	PIN_V8	SDRAM 2 Address[6]
DRAM1_A[7]	PIN_W4	SDRAM 2 Address[7]
DRAM1_A[8]	PIN_W7	SDRAM 2 Address[8]
DRAM1_A[9]	PIN_W8	SDRAM 2 Address[9]
DRAM1_A[10]	PIN_T4	SDRAM 2 Address[10]
DRAM1_A[11]	PIN_Y4	SDRAM 2 Address[11]
DRAM1_A[12]	PIN_Y7	SDRAM 2 Address[12]
DRAM_D[16]	PIN_U1	SDRAM 2 Data[0]
DRAM_D[17]	PIN_U2	SDRAM 2 Data[1]
DRAM_D[18]	PIN_U3	SDRAM 2 Data[2]
DRAM_D[19]	PIN_V2	SDRAM 2 Data[3]
DRAM_D[20]	PIN_V3	SDRAM 2 Data[4]
DRAM_D[21]	PIN_W1	SDRAM 2 Data[5]
DRAM_D[22]	PIN_W2	SDRAM 2 Data[6]
DRAM_D[23]	PIN_W3	SDRAM 2 Data[7]
DRAM_D[24]	PIN_Y1	SDRAM 2 Data[8]
DRAM_D[25]	PIN_Y2	SDRAM 2 Data[9]
DRAM_D[26]	PIN_Y3	SDRAM 2 Data[10]
DRAM_D[27]	PIN_AA1	SDRAM 2 Data[11]
DRAM_D[28]	PIN_AA2	SDRAM 2 Data[12]
DRAM_D[29]	PIN_AA3	SDRAM 2 Data[13]
DRAM_D[30]	PIN_AB1	SDRAM 2 Data[14]
DRAM_D[31]	PIN_AB2	SDRAM 2 Data[15]
DRAM1_BA_0	PIN_T7	SDRAM 2 Bank Address[0]
DRAM1_BA_1	PIN_T8	SDRAM 2 Bank Address[1]
DRAM1_LDQM0	PIN_M10	SDRAM 2 Low-byte Data Mask
DRAM1_UDQM1	PIN_U8	SDRAM 2 High-byte Data Mask

DRAM1_RAS_N	PIN_N9	SDRAM 2 Row Address Strobe
DRAM1_CAS_N	PIN_N8	SDRAM 2 Column Address Strobe
DRAM1_CKE	PIN_L10	SDRAM 2 Clock Enable
DRAM1_CLK	PIN_G5	SDRAM 2 Clock
DRAM1_WE_N	PIN_M9	SDRAM 2 Write Enable
DRAM1_CS_N	PIN_P9	SDRAM 2 Chip Select

Table 5.19. SDRAM pin assignments.

Signal Name	FPGA Pin No.	Description
SRAM_A[0]	PIN_AG8	SRAM Address[0]
SRAM_A[1]	PIN_AF8	SRAM Address[1]
SRAM_A[2]	PIN_AH7	SRAM Address[2]
SRAM_A[3]	PIN_AG7	SRAM Address[3]
SRAM_A[4]	PIN_AG6	SRAM Address[4]
SRAM_A[5]	PIN_AG5	SRAM Address[5]
SRAM_A[6]	PIN_AE12	SRAM Address[6]
SRAM_A[7]	PIN_AG12	SRAM Address[7]
SRAM_A[8]	PIN_AD13	SRAM Address[8]
SRAM_A[9]	PIN_AE13	SRAM Address[9]
SRAM_A[10]	PIN_AF14	SRAM Address[10]
SRAM_A[11]	PIN_AG14	SRAM Address[11]
SRAM_A[12]	PIN_AE15	SRAM Address[12]
SRAM_A[13]	PIN_AF15	SRAM Address[13]
SRAM_A[14]	PIN_AC16	SRAM Address[14]
SRAM_A[15]	PIN_AF20	SRAM Address[15]
SRAM_A[16]	PIN_AG20	SRAM Address[16]
SRAM_A[17]	PIN_AE11	SRAM Address[17]
SRAM_A[18]	PIN_AF11	SRAM Address[18]
SRAM_DQ[0]	PIN_AH10	SRAM Data[0]
SRAM_DQ[1]	PIN_AJ10	SRAM Data[1]
SRAM_DQ[2]	PIN_AK10	SRAM Data[2]
SRAM_DQ[3]	PIN_AJ11	SRAM Data[3]
SRAM_DQ[4]	PIN_AK11	SRAM Data[4]
SRAM_DQ[5]	PIN_AH12	SRAM Data[5]
SRAM_DQ[6]	PIN_AJ12	SRAM Data[6]
SRAM_DQ[7]	PIN_AH16	SRAM Data[7]

SRAM_DQ[8]	PIN_AK17	SRAM Data[8]
SRAM_DQ[9]	PIN_AJ17	SRAM Data[9]
SRAM_DQ[10]	PIN_AH17	SRAM Data[10]
SRAM_DQ[11]	PIN_AJ18	SRAM Data[11]
SRAM_DQ[12]	PIN_AH18	SRAM Data[12]
SRAM_DQ[13]	PIN_AK19	SRAM Data[13]
SRAM_DQ[14]	PIN_AJ19	SRAM Data[14]
SRAM_DQ[15]	PIN_AK23	SRAM Data[15]
SRAM_DQ[16]	PIN_AJ20	SRAM Data[16]
SRAM_DQ[17]	PIN_AK21	SRAM Data[17]
SRAM_DQ[18]	PIN_AJ21	SRAM Data[18]
SRAM_DQ[19]	PIN_AK22	SRAM Data[19]
SRAM_DQ[20]	PIN_AJ22	SRAM Data[20]
SRAM_DQ[21]	PIN_AH15	SRAM Data[21]
SRAM_DQ[22]	PIN_AJ15	SRAM Data[22]
SRAM_DQ[23]	PIN_AJ16	SRAM Data[23]
SRAM_DQ[24]	PIN_AK14	SRAM Data[24]
SRAM_DQ[25]	PIN_AJ14	SRAM Data[25]
SRAM_DQ[26]	PIN_AJ13	SRAM Data[26]
SRAM_DQ[27]	PIN_AH13	SRAM Data[27]
SRAM_DQ[28]	PIN_AK12	SRAM Data[28]
SRAM_DQ[29]	PIN_AK7	SRAM Data[29]
SRAM_DQ[30]	PIN_AJ8	SRAM Data[30]
SRAM_DQ[31]	PIN_AK8	SRAM Data[31]
SRAM_ADSC_N	PIN_AG17	SRAM Controller Address Status
SRAM_ADSP_N	PIN_AC18	SRAM Processor Address Status
SRAM_ADV_N	PIN_AD16	SRAM Burst Address Advance
SRAM_BE_N0	PIN_AC21	SRAM Byte Write Enable[0]
SRAM_BE_N1	PIN_AC20	SRAM Byte Write Enable[1]
SRAM_BE_N2	PIN_AD20	SRAM Byte Write Enable[2]
SRAM_BE_N3	PIN_AH20	SRAM Byte Write Enable[3]
SRAM_CE1_N	PIN_AH19	SRAM Chip Enable 1
SRAM_CE2	PIN_AG19	SRAM Chip Enable 2
SRAM_CE3_N	PIN_AD22	SRAM Chip Enable 3
SRAM_CLK	PIN_AD7	SRAM Clock
SRAM_DPA0	PIN_AK9	SRAM Parity Data[0]
SRAM_DPA1	PIN_AJ23	SRAM Parity Data[1]

SRAM_DPA2	PIN_AK20	SRAM Parity Data[2]
SRAM_DPA3	PIN_AJ9	SRAM Parity Data[3]
SRAM_GW_N	PIN_AG18	SRAM Global Write Enable
SRAM_OE_N	PIN_AD18	SRAM Output Enable
SRAM_WE_N	PIN_AF18	SRAM Write Enable

Table 5.20. SSRAM pin assignments.

Signal Name	FPGA Pin No.	Description
FLASH_A[0]	PIN_AF24	FLASH Address[0]
FLASH_A[1]	PIN_AG24	FLASH Address[1]
FLASH_A[2]	PIN_AE23	FLASH Address[2]
FLASH_A[3]	PIN_AG23	FLASH Address[3]
FLASH_A[4]	PIN_AF23	FLASH Address[4]
FLASH_A[5]	PIN_AG22	FLASH Address[5]
FLASH_A[6]	PIN_AH22	FLASH Address[6]
FLASH_A[7]	PIN_AF22	FLASH Address[7]
FLASH_A[8]	PIN_AH27	FLASH Address[8]
FLASH_A[9]	PIN_AJ27	FLASH Address[9]
FLASH_A[10]	PIN_AH26	FLASH Address[10]
FLASH_A[11]	PIN_AJ26	FLASH Address[11]
FLASH_A[12]	PIN_AK26	FLASH Address[12]
FLASH_A[13]	PIN_AJ25	FLASH Address[13]
FLASH_A[14]	PIN_AK25	FLASH Address[14]
FLASH_A[15]	PIN_AH24	FLASH Address[15]
FLASH_A[16]	PIN_AG25	FLASH Address[16]
FLASH_A[17]	PIN_AF21	FLASH Address[17]
FLASH_A[18]	PIN_AD21	FLASH Address[18]
FLASH_A[19]	PIN_AK28	FLASH Address[19]
FLASH_A[20]	PIN_AJ28	FLASH Address[20]
FLASH_A[21]	PIN_AE20	FLASH Address[21]
FLASH_DQ[0]	PIN_AF29	FLASH Data[0]
FLASH_DQ[1]	PIN_AE28	FLASH Data[1]
FLASH_DQ[2]	PIN_AE30	FLASH Data[2]
FLASH_DQ[3]	PIN_AD30	FLASH Data[3]
FLASH_DQ[4]	PIN_AC29	FLASH Data[4]

FLASH_DQ[5]	PIN_AB29	FLASH Data[5]
FLASH_DQ[6]	PIN_AA29	FLASH Data[6]
FLASH_DQ[7]	PIN_Y28	FLASH Data[7]
FLASH_DQ[8]	PIN_AF30	FLASH Data[8]
FLASH_DQ[9]	PIN_AE29	FLASH Data[9]
FLASH_DQ[10]	PIN_AD29	FLASH Data[10]
FLASH_DQ[11]	PIN_AC28	FLASH Data[11]
FLASH_DQ[12]	PIN_AC30	FLASH Data[12]
FLASH_DQ[13]	PIN_AB30	FLASH Data[13]
FLASH_DQ[14]	PIN_AA30	FLASH Data[14]
FLASH_DQ15_AM1	PIN_AE24	FLASH Data[15]
FLASH_BYTE_N	PIN_Y29	FLASH Byte/Word Mode Configuration
FLASH_CE_N	PIN_AG28	FLASH Chip Enable
FLASH_OE_N	PIN_AG29	FLASH Output Enable
FLASH_RESET_N	PIN_AH28	FLASH Reset
FLASH_RY	PIN_AH30	LASH Ready/Busy output
FLASH_WE_N	PIN_AJ29	FLASH Write Enable
FLASH_WP_N	PIN_AH29	FLASH Write Protect /Programming Acceleration

Table 5.21. Flash pin assignments.

Chapter 6

Examples of Advanced Demonstrations

This chapter provides a number of examples of advanced circuits implemented on the DE2-70 board. These circuits provide demonstrations of the major features on the board, such as its audio and video capabilities, and USB and Ethernet connectivity. For each demonstration the Cyclone II FPGA (or EPCS16 serial EEPROM) configuration file is provided, as well as the full source code in Verilog HDL code. All of the associated files can be found in the *DE2_70_demonstrations* folder from the **DE2-70 System CD-ROM**. For each of demonstrations described in the following sections, we give the name of the project directory for its files, which are subdirectories of the *DE2-70_demonstrations* folder.

Installing the Demonstrations

To install the demonstrations on your computer, perform the following

1. Copy the directory *DE2_70_demonstrations* into a local directory of your choice. **It is important to ensure that the path to your local directory contains no spaces – otherwise, the Nios II software will not work.**

6.1 DE2-70 Factory Configuration

The DE2-70 board is shipped from the factory with a default configuration that demonstrates some of the basic features of the board. The setup required for this demonstration, and the locations of its files are shown below.

Demonstration Setup, File Locations, and Instructions

- Project directory: *DE2_70_Default*
- Bit stream used: *DE2_70_Default.sof* or *DE2_70_Default.pof*
- Power on the DE2-70 board, with the USB cable connected to the USB Blaster port. If necessary (that is, if the default factory configuration of the DE2-70 board is not currently stored in EPCS16 device), download the bit stream to the board by using either JTAG or AS programming
- You should now be able to observe that the 7-segment displays are displaying a sequence of characters, and the red and green LEDs are flashing. Also, **Welcome to the Altera DE2-70** is shown on the LCD display

- Optionally connect a VGA display to the VGA D-SUB connector. When connected, the VGA display should show a pattern of colors
- Optionally connect a powered speaker to the stereo audio-out jack
- Place toggle switch SW17 in the UP position to hear a 1 kHz humming sound from the audio-out port. Alternatively, if switch SW17 is DOWN, the microphone-in port can be connected to a microphone to hear voice sounds, or the line-in port can be used to play audio from an appropriate sound source

The Verilog source code for this demonstration is provided in the *DE2_70_Default* folder, which also includes the necessary files for the corresponding Quartus II project. The top-level Verilog file, called *DE2_70_Default.v*, can be used as a template for other projects, because it defines ports that correspond to all of the user-accessible pins on the Cyclone II FPGA.

6.2 TV Box Demonstration

This demonstration plays video and audio input from a DVD player using the VGA output, audio CODEC, and one TV decoder (U11) on the DE2-70 board. Figure 6.1 shows the block diagram of the design. There are two major blocks in the circuit, called *I2C_AV_Config* and *TV_to_VGA*. The *TV_to_VGA* block consists of the *ITU-R 656 Decoder*, *SDRAM Frame Buffer*, *YUV422 to YUV444*, *YCrCb to RGB*, and *VGA Controller*. The figure also shows the TV Decoder (ADV7180) and the VGA DAC (ADV7123) chips used.

As soon as the bit stream is downloaded into the FPGA, the register values of the TV Decoder chip are used to configure the TV decoder via the *I2C_AV_Config* block, which uses the I2C protocol to communicate with the TV Decoder chip. Following the power-on sequence, the TV Decoder chip will be unstable for a time period; the *Lock Detector* is responsible for detecting this instability.

The *ITU-R 656 Decoder* block extracts *YCrCb 4:2:2 (YUV 4:2:2)* video signals from the *ITU-R 656* data stream sent from the TV Decoder. It also generates a data valid control signal indicating the valid period of data output. Because the video signal from the TV Decoder is interlaced, we need to perform de-interlacing on the data source. We used the *SDRAM Frame Buffer* and a *field selection multiplexer(MUX)* which is controlled by the *VGA controller* to perform the de-interlacing operation. Internally, the *VGA Controller* generates data request and odd/even selected signals to the *SDRAM Frame Buffer* and *field selection multiplexer(MUX)*. The *YUV422 to YUV444* block converts the selected *YCrCb 4:2:2 (YUV 4:2:2)* video data to the *YCrCb 4:4:4 (YUV 4:4:4)* video data format.

Finally, the *YCrCb_to_RGB* block converts the *YCrCb* data into RGB output. The *VGA Controller* block generates standard VGA sync signals *VGA_HS* and *VGA_VS* to enable the display on a VGA

monitor.

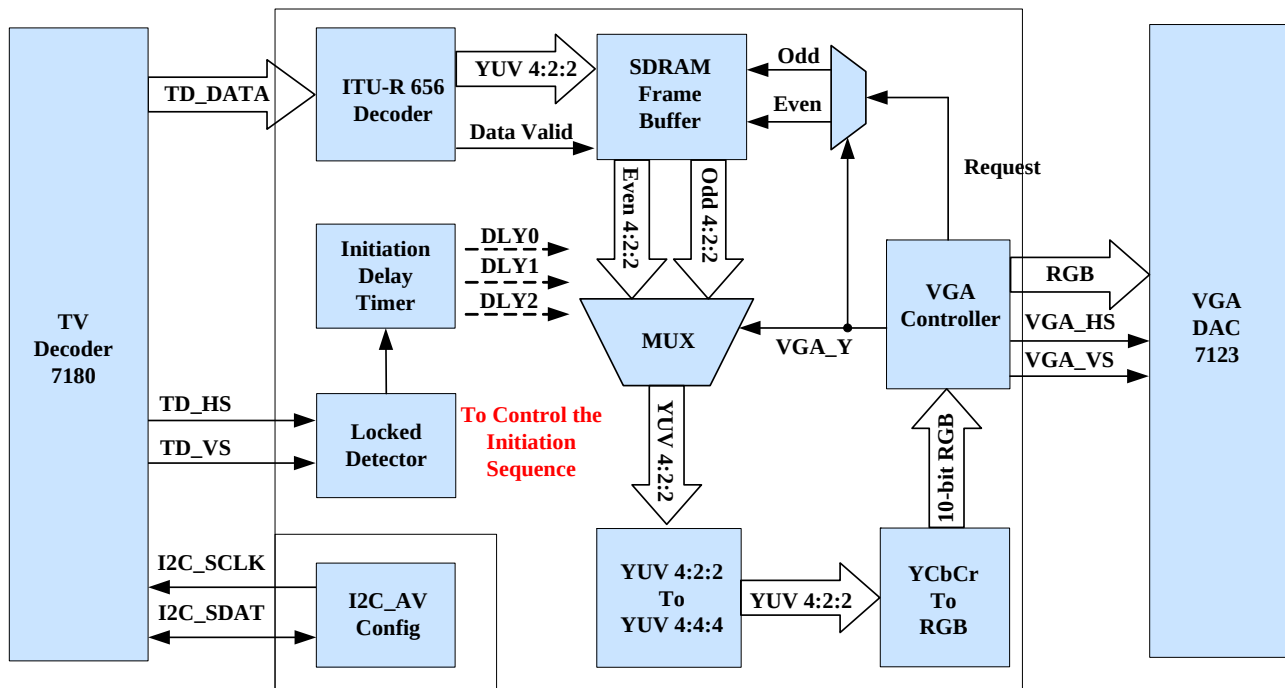


Figure 6.1. Block diagram of the TV box demonstration.

Demonstration Setup, File Locations, and Instructions

- Project directory: *DE2_70_TV*
- Bit stream used: *DE2_70_TV.sof* or *DE2_70_TV.pof*
- Connect a DVD player's composite video output (yellow plug) to the **Video-IN 1** RCA jack (J8) of the DE2-70 board. The DVD player has to be configured to provide
 - o NTSC output
 - o 60 Hz refresh rate
 - o 4:3 aspect ratio
 - o Non-progressive video
- Connect the VGA output of the DE2-70 board to a VGA monitor (both LCD and CRT type of monitors should work)
- Connect the audio output of the DVD player to the line-in port of the DE2-70 board and connect a speaker to the line-out port. If the audio output jacks from the DVD player are of RCA type, then an adaptor will be needed to convert to the mini-stereo plug supported on the DE2-70 board; this is the same type of plug supported on most computers
- Load the bit stream into FPGA. Press KEY0 on the DE2-70 board to reset the circuit

Figure 6.2 illustrates the setup for this demonstration.

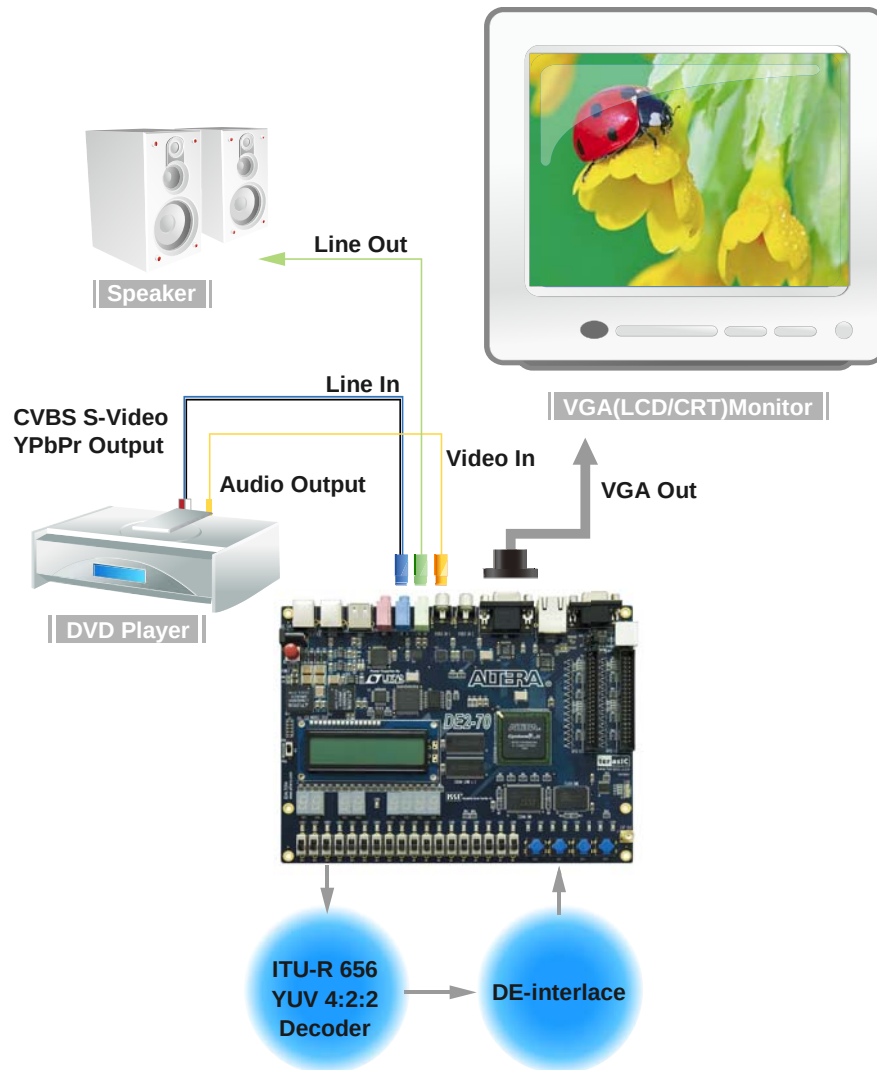


Figure 6.2. The setup for the TV box demonstration.

6.3 TV Box Picture in Picture (PIP) Demonstration

The DE2-70 board has two TV decoders and RCA jacks that allow users to process two video sources simultaneously using the 2C70 FPGA. This demonstration will multiplex two different video source signals from the TV decoders and display both video signals on the LCD/CRT monitor using picture in picture mode (PIP mode : One picture is displayed on the full screen and the other picture is displayed in a small sub window). Also, users can select which video is displayed in main/sub window via a toggle switch.

Figure 6.3 shows the basic block diagram of this demonstration. There are three major blocks in the circuit, called *Composite_to_VGA*, *PIP_Position_Controller*, and *VGA_Multiplexer*. The *Composite_to_VGA* block consists all of the function blocks in the TV box demonstration project

described in the section 6.2. The *Composite_to_VGA* block takes the video signals from the TV decoders as input and generate VGA-interfaced signals as output. The circuit in the FPGA is equipped with two *Composite_to_VGA* blocks converting the video signals from the TV decoder 1 and TV decoder 2 respectively. To display two video signals in PIP mode on the LCD/CRT monitor, the output VGA data rate of the *Composite_to_VGA* block for the sub window must be two times as fast as the rate of the *Composite_to_VGA* block for the main window. In addition, the output timing of the VGA interface signal from the *Composite_to_VGA* block is controlled by the *pip_position_controller* block that determines the starting position of the sub window. Finally, both of the two VGA interfaced signals will be multiplexed and sent to the LCD/CRT monitor via the *VGA_muxplexer* block.

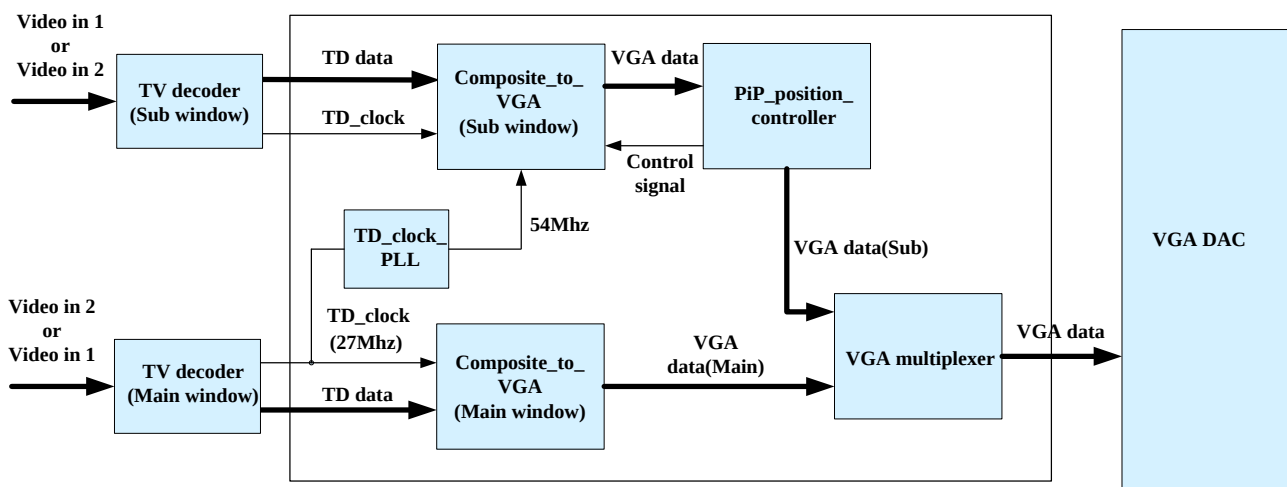


Figure 6.3. Block diagram of the TV PIP demonstration.

Demonstration Setup, File Locations, and Instructions

- Project directory: *DE2_70_TV_PIP*
- Bit stream used: *DE2_70_TV_PIP.sof* or *DE2_70_TV_PIP.pof*
- Connect composite video output (yellow plug) of DVD player 1 and DVD player2 to the Video-in 1 and Video-in 2 RCA jack (J8 and J9) of the DE2-70 board respectively. Both DVD players must be configured to provide
 - 60 Hz refresh rate
 - 4:3 aspect ratio
 - Non-progressive video
- Connect the VGA output of the DE2-70 board to a VGA monitor (both LCD and CRT type of monitors should work)
- Connect the one audio output of the DVD player to the line-in port of the DE2-70 board and connect a speaker to the line-out port. If the audio output jacks from the DVD player are of

RCA type, then an adaptor will be needed to convert to the mini-stereo plug supported on the DE2-70 board; this is the same type of plug supported on most computers

- Load the bit stream into FPGA.
- The detailed configuration for switching video source of main and sub window are listed in Table 6.1.

Figure 6.4 illustrates the setup for this demonstration.

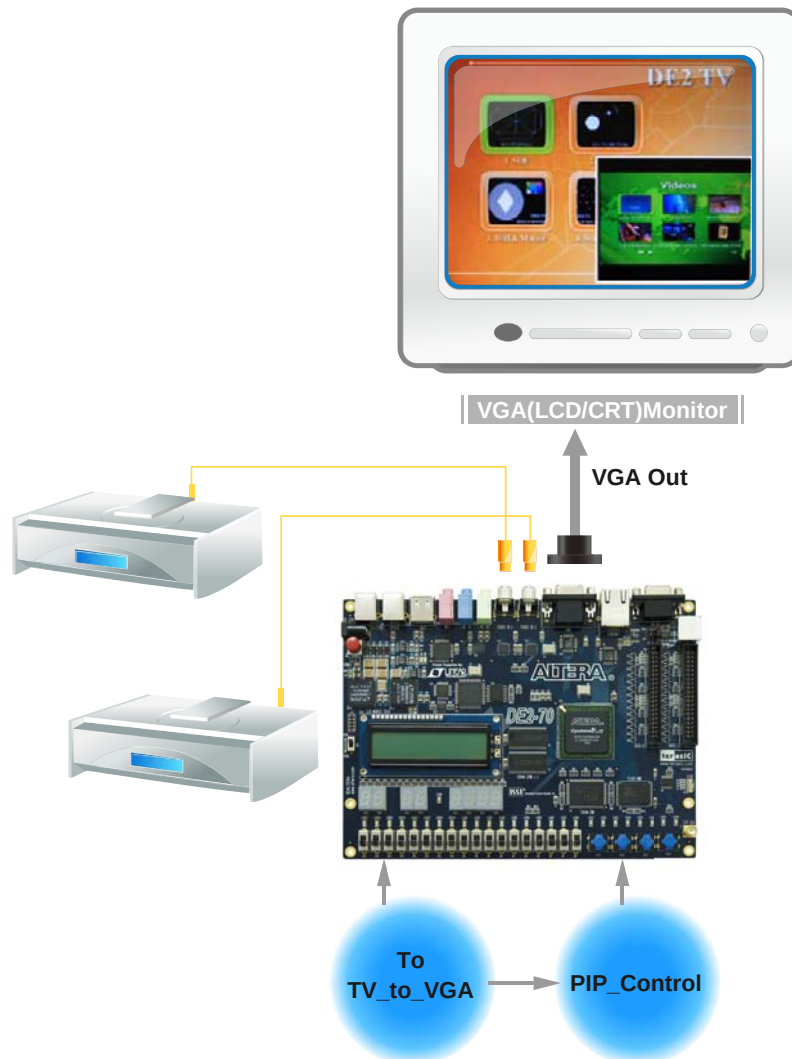


Figure 6.4. The setup for the TV box PIP demonstration.

Configuration	VGA Display Mode	Video source
SW[17] = OFF;	Signal display mode	Video in 2

SW[16] = OFF		
SW[17] = OFF; SW[16] = ON	Signal display mode	Video in 1
SW[17] = ON; SW[16] = OFF	PIP display mode	Main window: Video in 2 Sub window : Video in 1
SW[17] = ON; SW[16] = ON	PIP display mode	Main window: Video in 1 Sub window : Video in 2

Table 6.1. The setup for the TV box PIP demonstration

6.4 USB Paintbrush

USB is a popular communication method used in many multimedia products. The DE2-70 board provides a complete USB solution for both host and device applications. In this demonstration, we implement a Paintbrush application by using a USB mouse as the input device.

This demonstration uses the device port of the Philips ISP1362 chip and the Nios II processor to implement a USB mouse movement detector. We also implemented a video frame buffer with a VGA controller to perform the real-time image storage and display. Figure 6.5 shows the block diagram of the circuit, which allows the user to draw lines on the VGA display screen using the USB mouse. The *VGA Controller* block is integrated into the Altera Avalon bus so that it can be controlled by the Nios II processor.

Once the program running on the Nios II processor is started, it will detect the existence of the USB mouse connected to DE2-70 board. Once the mouse is moved, the Nios II processor is able to keep track of the movement and record it in a frame buffer memory. The *VGA Controller* will overlap the data stored in the frame buffer with a default image pattern and display the overlapped image on the VGA display.

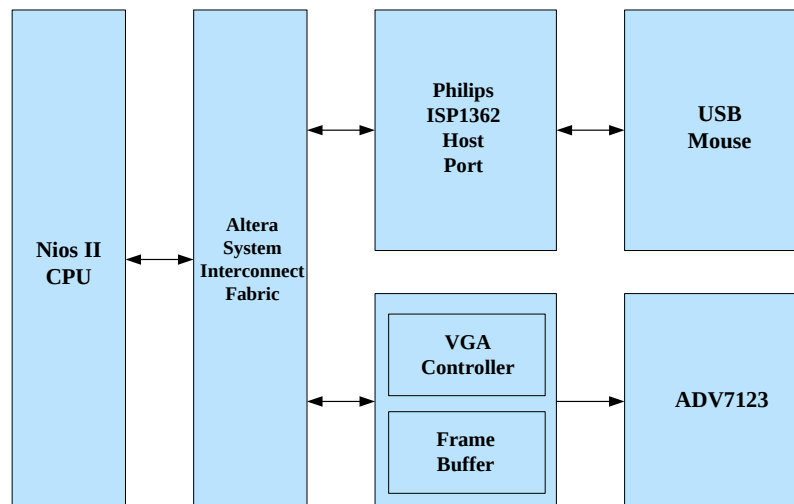


Figure 6.5. Block diagram of the USB paintbrush demonstration.

Demonstration Setup, File Locations, and Instructions

Project directory: *DE2_70_NIOS_HOST_MOUSE_VGA*

Bit stream used: *DE2_70_NIOS_HOST_MOUSE_VGA.sof*

Nios II Workspace: *DE2_70_NIOS_HOST_MOUSE_VGA\Software*

- Connect a USB Mouse to the USB Host Connector (type A) of the DE2-70 board
- Connect the VGA output of the DE2-70 board to a VGA monitor (both LCD and CRT type of monitors should work)
- Load the bit stream into FPGA
- Run the Nios II and choose *DE2_70_NIOS_HOST_MOUSE_VGA* as the workspace. Click on the **Compile** and **Run** button
- You should now be able to observe a blue background with an Altera logo on the VGA display
- Move the USB mouse and observe the corresponding movements of the cursor on the screen
- Left-click mouse to draw white dots/lines and right-click the mouse to draw blue dots/lines on the screen.

Figure 6.6 illustrates the setup for this demonstration.

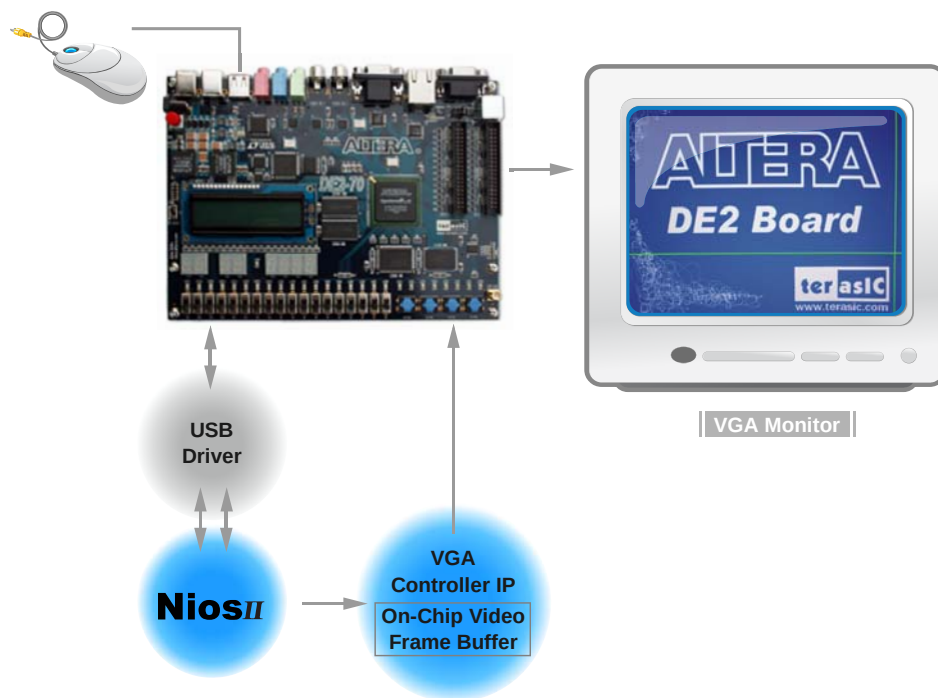


Figure 6.6. The setup for the USB paintbrush demonstration.

6.5 USB Device

Most USB applications and products operate as USB devices, rather than USB hosts. In this demonstration, we show how the DE2-70 board can operate as a USB device that can be connected to a host computer. As indicated in the block diagram in Figure 6.7, the Nios II processor is used to communicate with the host computer via the host port on the DE2-70 board's Philips ISP1362 device.

After connecting the DE2-70 board to a USB port on the host computer, a software program has to be executed on the Nios II processor to initialize the Philips ISP1362 chip. Once the software program is successfully executed, the host computer will identify the new device in its USB device list and ask for the associated driver; the device will be identified as a *Philips PDIUSB12 SMART Evaluation Board*. After completion of the driver installation on the host computer, the next step is to run a software program on the host computer called *ISP1362DcUsb.exe*; this program communicates with the DE2-70 board.

In the *ISP1362DcUsb* program, clicking on the **Add** button in the window panel of the software causes the host computer to send a particular USB packet to the DE2-70 board; the packet will be received by the Nios II processor and will increment the value of a hardware counter. The value of the counter is displayed on one of the board's 7-segment displays, and also on the green LEDs. If

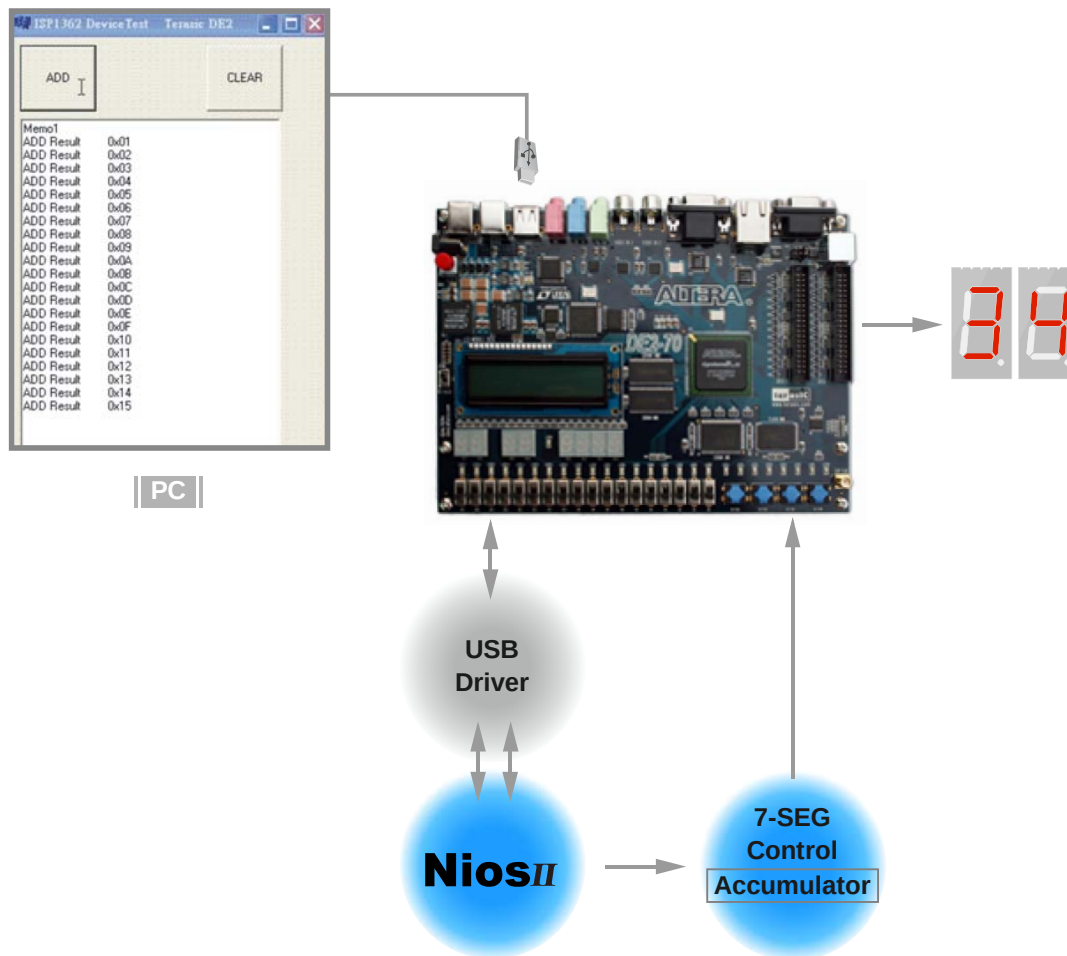


Figure 6.8. The setup for the USB device demonstration.

6.6 A Karaoke Machine

This demonstration uses the microphone-in, line-in, and line-out ports on the DE2-70 board to create a Karaoke Machine application. The Wolfson WM8731 audio CODEC is configured in the master mode, where the audio CODEC generates AD/DA serial bit clock (BCK) and the left/right channel clock (LRCK) automatically. As indicated in Figure 6.9, the I2C interface is used to configure the Audio CODEC. The sample rate and gain of the CODEC are set in this manner, and the data input from the line-in port is then mixed with the microphone-in port and the result is sent to the line-out port.

For this demonstration the sample rate is set to 48 kHz. Pressing the pushbutton *KEY0* reconfigures the gain of the audio CODEC via the I2C bus, cycling through one of the ten predefined gains (volume levels) provided by the device.

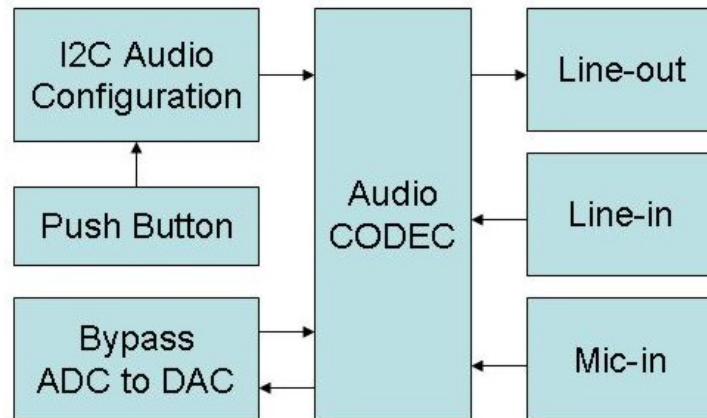


Figure 6.9. Block diagram of the Karaoke Machine demonstration.

Demonstration Setup, File Locations, and Instructions

- Project directory: *DE2-70_i2sound*
- Bit stream used: *DE2-70_i2sound.sof* or *DE2-70_i2sound.pof*
- Connect a microphone to the microphone-in port (pink color) on the DE2-70 board
- Connect the audio output of a music-player, such as an MP3 player or computer, to the line-in port (blue color) on the DE2-70 board
- Connect a headset/speaker to the line-out port (green color) on the DE2-70 board
- Load the bit stream into the FPGA
- You should be able to hear a mixture of the microphone sound and the sound from the music player
- Press *KEY0* to adjust the volume; it cycles between volume levels 0 to 9

Figure 6.10 illustrates the setup for this demonstration.

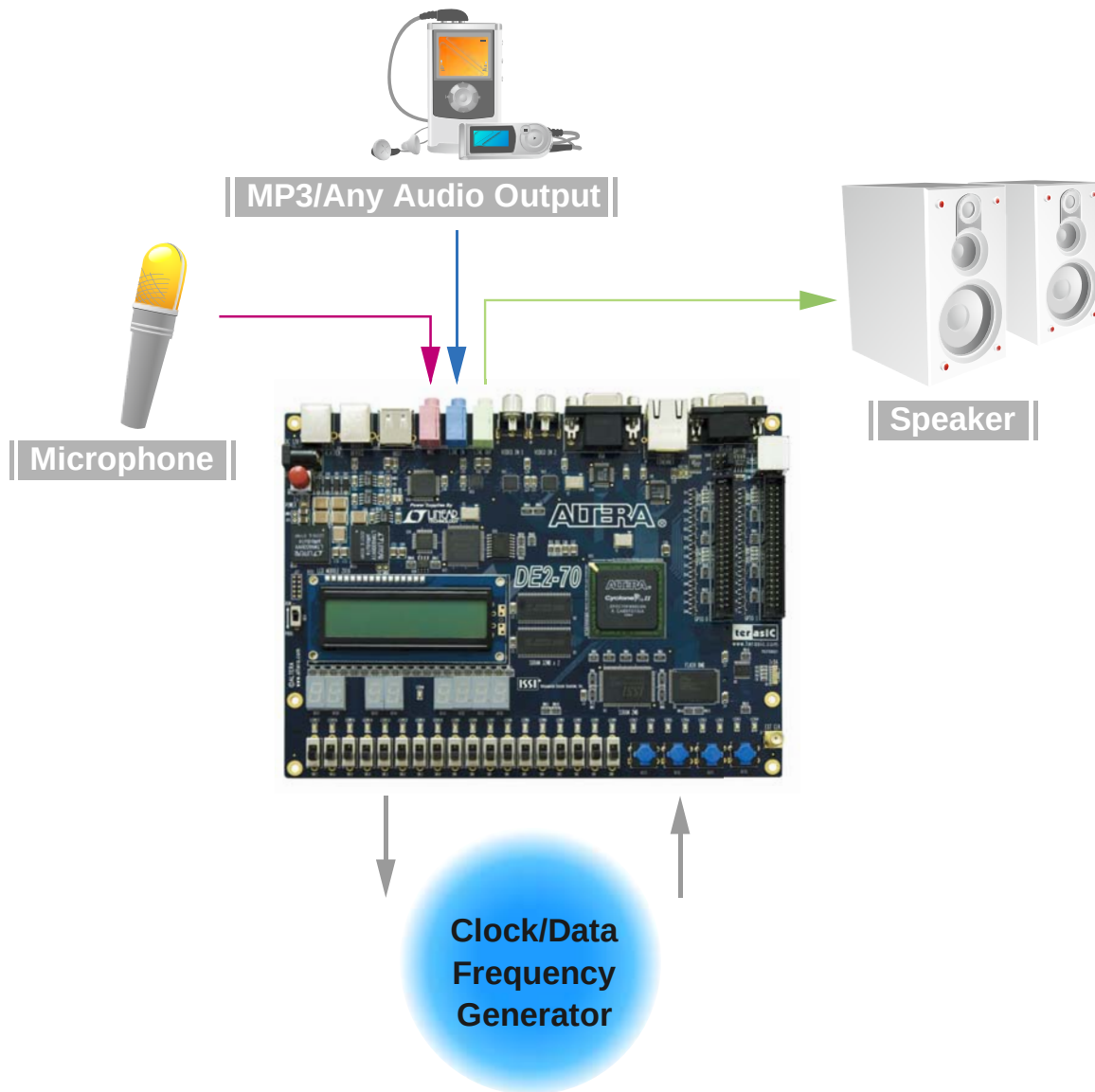


Figure 6.10. The setup for the Karaoke Machine.

6.7 Ethernet Packet Sending/Receiving

In this demonstration, we will show how to send and receive Ethernet packets using the Fast Ethernet controller on DE2-70 board. As illustrated in Figure 6.11, we use the Nios II processor to send and receive Ethernet packets using the DM9000A Ethernet PHY/MAC Controller. The demonstration can be set up to use either a loop-back connection from one board to itself, or two DE2-70 boards connected together.

On the transmitting side, the Nios II processor sends 64-byte packets every 0.5 seconds to the DM9000A. After receiving the packet, the DM9000A appends a four-byte checksum to the packet and sends it to the Ethernet port.

On the receiving side, the DM9000A checks every packet received to see if the destination MAC

address in the packet is identical to the MAC address of the DE2-70 board. If the packet received does have the same MAC address or is a broadcast packet, the DM9000A will accept the packet and send an interrupt to the Nios II processor. The processor will then display the packet contents in the Nios II IDE console window.

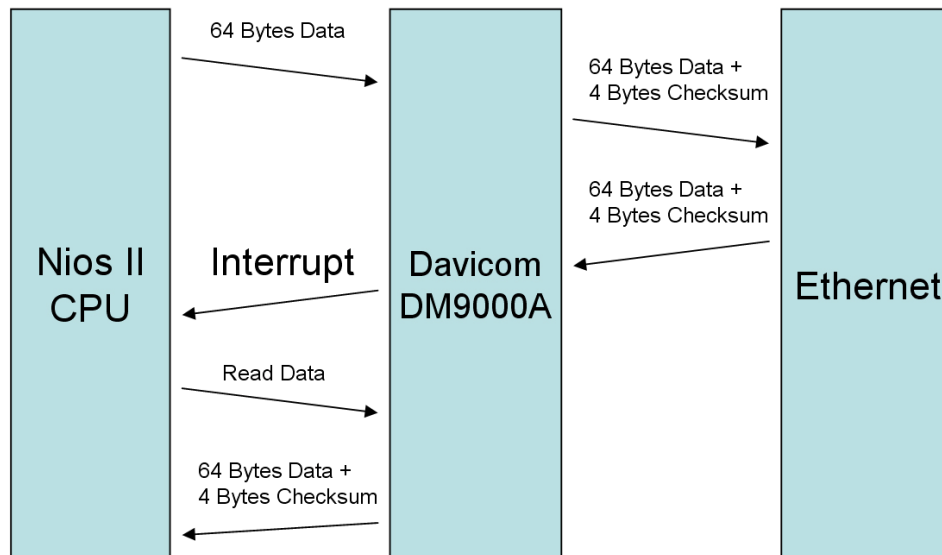


Figure 6.11. Packet sending and receiving using the Nios II processor.

Demonstration Setup, File Locations, and Instructions

- Project directory: *DE2_70_NET*
- Bit stream used: *DE2_70_NET.sof*
- Nios II Workspace: *DE2_70_NET\Software*
- Plug a CAT5 loop-back cable into the Ethernet connector of DE2-70
- Load the bit stream into the FPGA
- Run the Nios II IDE under the workspace *DE2_70_NET*
- Click on the **Compile** and **Run** button
- You should now be able to observe the contents of the packets received (64-byte packets sent, 68-byte packets received because of the extra checksum bytes)

Figure 6.12 illustrates the setup for this demonstration.

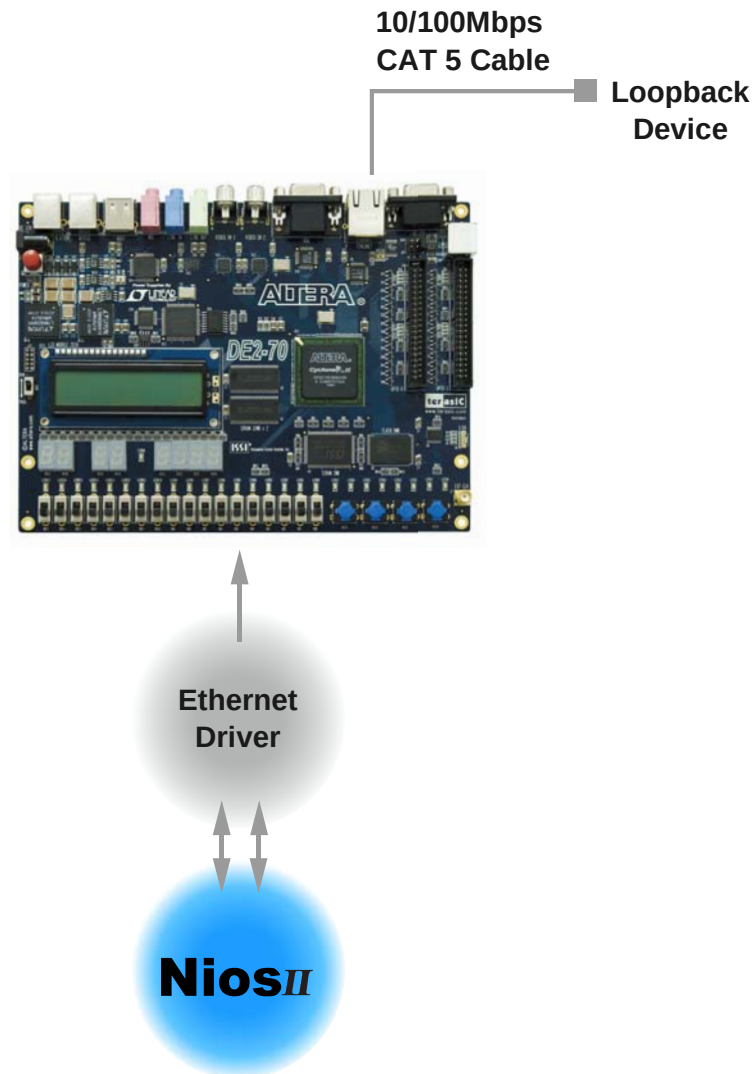


Figure 6.12. The setup for the Ethernet demonstration.

6.8 SD Card Music Player

Many commercial media/audio players use a large external storage device, such as an SD card or CF card, to store music or video files. Such players may also include high-quality DAC devices so that good audio quality can be produced. The DE2-70 board provides the hardware and software needed for SD card access and professional audio performance so that it is possible to design advanced multimedia products using the DE2-70 board.

In this demonstration we show how to implement an SD Card Music Player on the DE2-70 board, in which the music files are stored in an SD card and the board can play the music files via its CD-quality audio DAC circuits. We use the Nios II processor to read the music data stored in the SD Card and use the Wolfson WM8731 audio CODEC to play the music.

Figure 6.13 shows the hardware block diagram of this demonstration. The system requires a 50 MHz clock provided from the board. The PLL generates a 100-MHZ clock for NIOS II processor and the other controllers except for the audio controller. The audio chip is controlled by the Audio Controller which is a user-defined SOPC component. This audio controller needs an input clock running at 18.432 MHz. In this design, the clock is provided by the PLL block. The audio controller requires the audio chip working in master mode, so the serial bit (BCK) and the left/right channel clock (LRCK) are provided by the audio chip. The 7-segment display is controlled by the Seg-7 Controller which also is a user-defined SOPC component. Two PIO pins are connected to the I2C bus. The I2C protocol is implemented by software. Four PIO pins are connected to the SD CARD socket. SD 1-Bit Mode is used to access the SD card and is implemented by software. All of the other SOPC components in the block diagram are SOPC Builder built-in components.

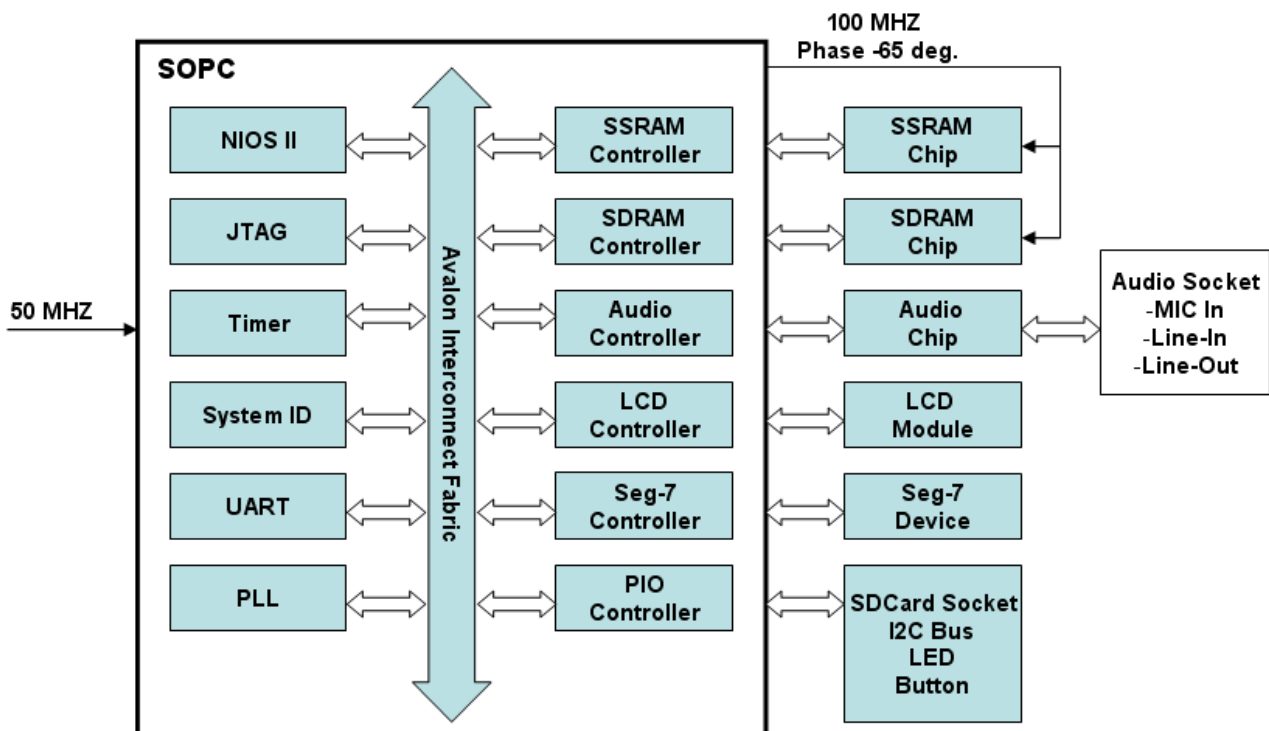


Figure 6.13. Block diagram of the SD music player demonstration.

Figure 6.14 shows the software stack of this demonstration. **SD 1-Bit Mod** block implements the SD 1-bit mode protocol for reading raw data from the SD card. The **FAT16** block implements FAT16 file system for reading wave files that stored in the SD card. In this block, only read function is implemented. The **WAVE Lib** block implements WAVE file decoding function for receiving audio signal from wave files. The **I2C** block implements I2C protocol for configuring audio chip. The **SEG7** block implements displaying function for display elapsed playing time. The **Audio** block implements audio FIFO checking function and audio signal sending/receiving function.

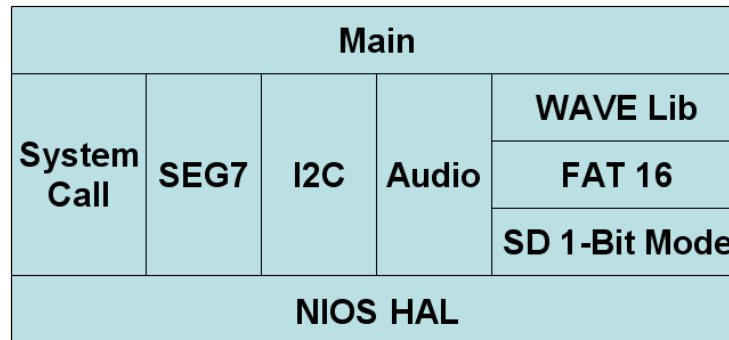


Figure 6.14. Software Stack of the SD music player demonstration.

The audio chip should be configured before sending audio signal to the audio chip. The main program uses I2C protocol to configure the audio chip working in master mode, the audio interface as I2S with 16-bits per channel, and sampling rate according to the wave file content. In audio playing loop, the main program reads 512-byte audio data from the SD card, and then writes the data to DAC FIFO in the Audio Controller. Before writing the data to the FIFO, the program have to make sure the FIFO is not full. The design also mixes the audio signal from the microphone-in and line-in for the Karaoke-style effects by enabling the BYPASS and SITETONE functions in the audio chip.

Finally, users can obtain the status of the SD music player from the 2x16-LCD module, the 7 segment display and the LEDs. The top and bottom row of the LCD module will display the file name of the music that is playing on the DE2-70 board and the value of music volume, respectively. The 7 segments display will show how long the music file has been played. The LED will indicate the audio signal strength.

Demonstration Setup, File Locations, and Instructions

- Project directory: *DE2_70_SD_Card_Audio_Player*
- Bit stream used: *DE2_70_SD_Card_Audio_Player.sof*
- Nios II Workspace: *DE2_70_SD_Card_Audio_Player\Software*
- Format your SD card into *FAT16* format
- Put the played wave files to the root directory of the SD card. The provided wave files must have a sample rate of either 96K, 48K, 44.1K, 32K, or 8K. Besides, the wave files must be stereo and 16 bits per channel. Also, the file name must be short filename.
- Load the bitstream into the FPGA on the DE2-70 board.
- Run the Nios II IDE under the workspace *DE2_70_SD_Card_Audio_Playe\Software*
- Connect a headset or speaker to the DE2-70 board and you should be able to hear the music played from the SD Card

- Press **KEY3** on the DE2-70 board can play the next music file stored in the SD card.
- Press **KEY2** and **KEY1** will increase and decrease the output music volume respectively.

Figure 6.16 illustrates the setup for this demonstration.

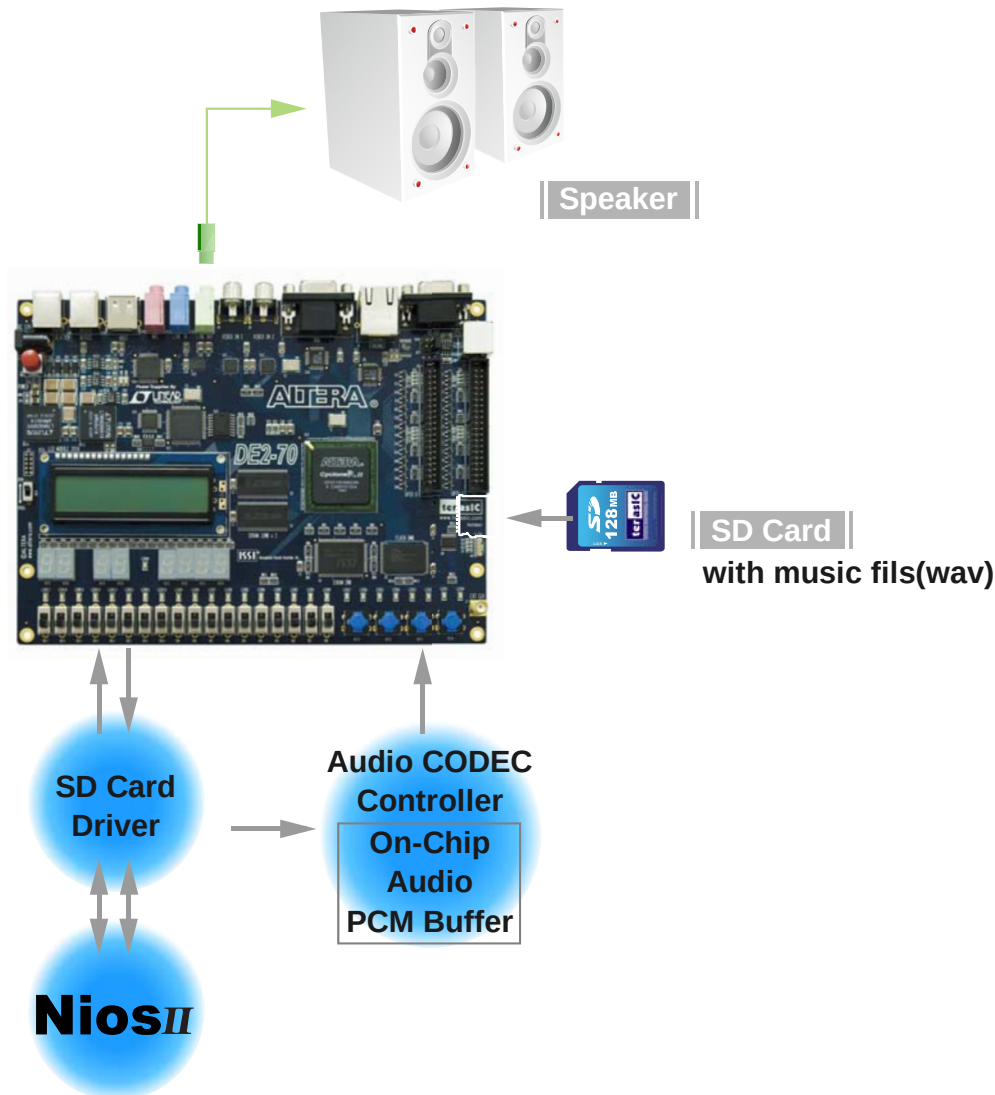


Figure 6.16. The setup for the SD music player demonstration.

6.9 Music Synthesizer Demonstration

This demonstration shows how to implement a Multi-tone Electronic Keyboard using DE2-70 board with a PS/2 Keyboard and a speaker.

PS/2 Keyboard is used as the piano keyboard for input. The Cyclone II FPGA on the DE2-70 board serves as the Music Synthesizer SOC to generate music and tones. The VGA connected to the DE2-70 board is used to show which key is pressed during the playing of the music.

Figure 6.15 shows the block diagram of the design of the Music Synthesizer. There are four major blocks in the circuit: *DEMO_SOUND*, *PS2_KEYBOARD*, *STAFF*, and *TONE_GENERATOR*. The *DEMO_SOUND* block stores a demo sound for user to play; *PS2_KEYBOARD* handles the users' input from PS/2 keyboard; The *STAFF* block draws the corresponding keyboard diagram on VGA monitor when key(s) are pressed. The *TONE_GENERATOR* is the core of music synthesizer SOC.

User can switch the music source either from *PS2_KEYBOARD* or the *DEMO_SOUND* block using SW9. To repeat the demo sound, users can press KEY1.

The *TONE_GENERATOR* has two tones: (1) String. (2) Brass, which can be controlled by SW0. The audio codec used on the DE2-70 board has two channels, which can be turned ON/OFF using SW1 and SW2.

Figure 6.17 illustrates the setup for this demonstration.

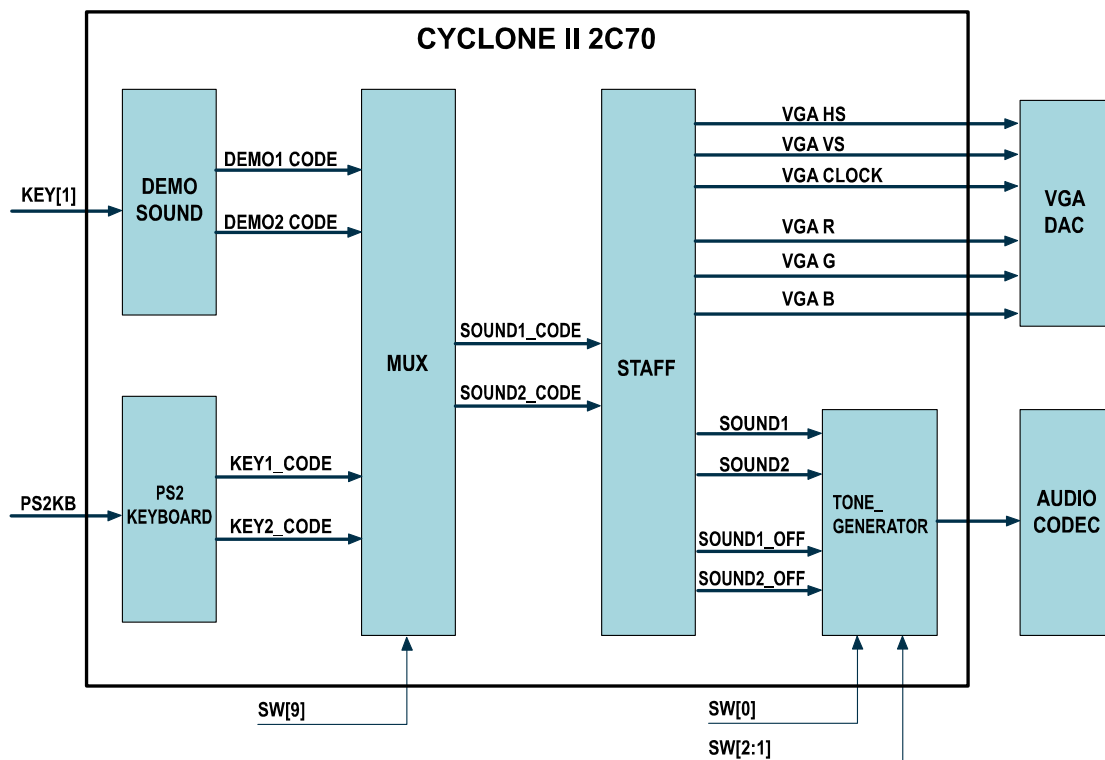


Figure 6.17. Block diagram of the Music Synthesizer design

Demonstration Setup, File Locations, and Instructions

- Project directory: *DE2_70_Synthesizer*

- Bit stream used: *DE2_70_Synthesizer.sof* or *DE2-70_Synthesizer.pof*
- Connect a PS/2 Keyboard to the DE2-70 board.
- Connect the VGA output of the DE2-70 board to a VGA monitor (both LCD and CRT type of monitors should work)
- Connect the Lineout of the DE2-70 board to a speaker.
- Load the bit stream into FPGA.
- Make sure all the switches (SW[9:0]) are set to 0 (Down Position)
- Press KEY1 on the DE2-70 board to start the music demo
- Press KEY0 on the DE2-70 board to reset the circuit

Table 6.2 and 6.3 illustrate the usage of the switches, pushbuttons (KEYs), PS/2 Keyboard.

- Switches and Pushbuttons

Signal Name	Description
KEY[0]	Reset Circuit
KEY[1]	Repeat the Demo Music
SW[0]	OFF: BRASS, ON: STRING
SW[9]	OFF: DEMO, ON: PS2 KEYBOARD
SW[1]	Channel-1 ON / OFF
SW[2]	Channel-2 ON / OFF

Table 6.2. Usage of the switches, pushbuttons (KEYs).

- PS/2 Keyboard

Signal Name	Description
Q	-#4
A	-5
W	-#5
S	-6
E	-#6
D	-7
F	1
T	#1
G	2
Y	#2
H	3
J	4

I	#4
K	5
O	#5
L	6
P	#6
:	7
"	+1

Table 6.3. Usage of the PS/2 Keyboard's keys.

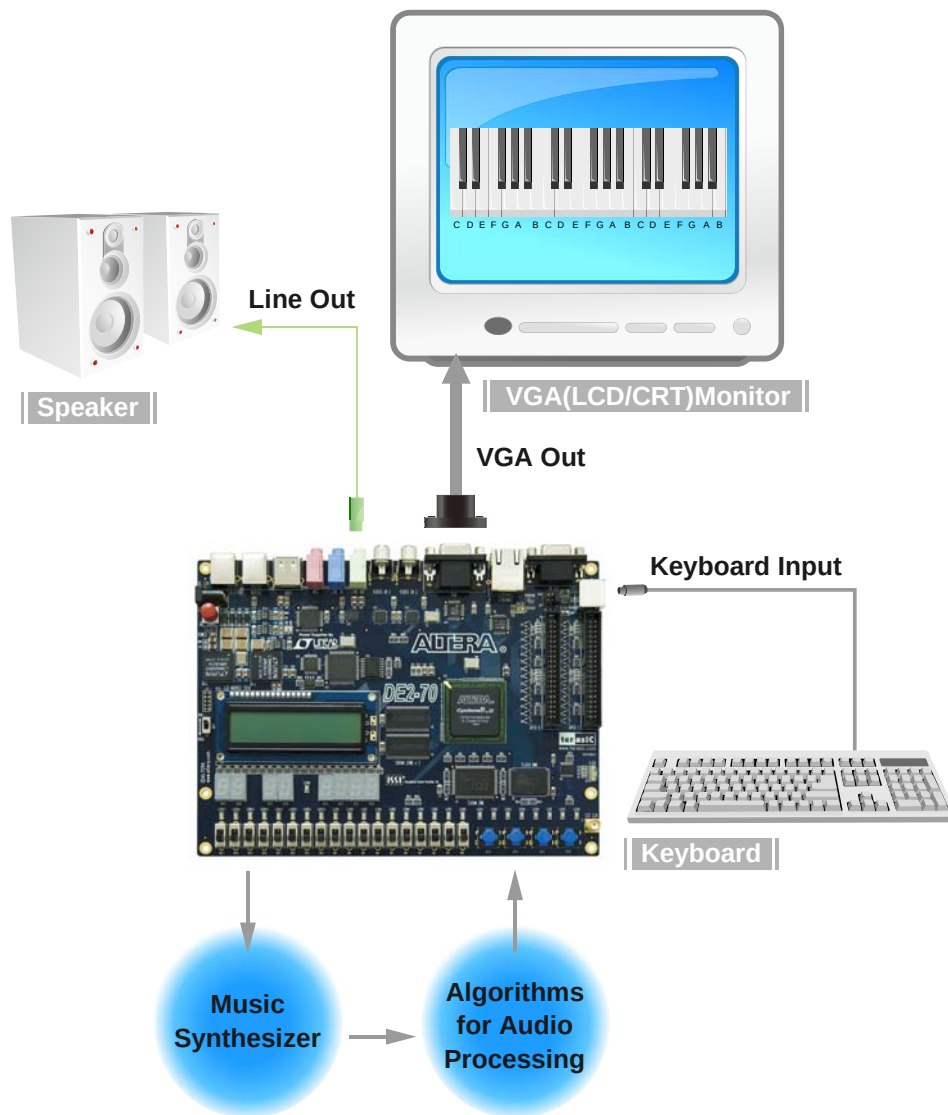


Figure 6.16. The Setup of the Music Synthesizer Demonstration.

6.10 Audio Recording and Playing

This demonstration shows how to implement an audio recorder and player using the DE2-70 board with the built-in Audio CODEC chip. This demonstration is developed based on SOPC Builder and NIOS II IDE.

Figure 6.18 shows the man-machine interface of this demonstration. Two push buttons and six toggle switches are used for users to configure this audio system: SW0 is used to specify recording source to be Line-in or MIC-In. SW1 is to enable/disable MIC Boost when the recording source is MIC-In. SW2 is used to enable/disable Zero-Cross Detection for audio playing. SW3, SW4 and SW5 are used to specify recording sample rate as 96K, 48K, 44.1K, 32K, or 8K. The 16x2 LCD is used to indicate the Recording/Playing status. The seg7 is used to display Recording/Playing duration with time unit in 1/100 second. The LED is used to indicate the audio signal strength. Table 6.4 summarizes the usage of toggle switches for configuring the audio recorder and player.

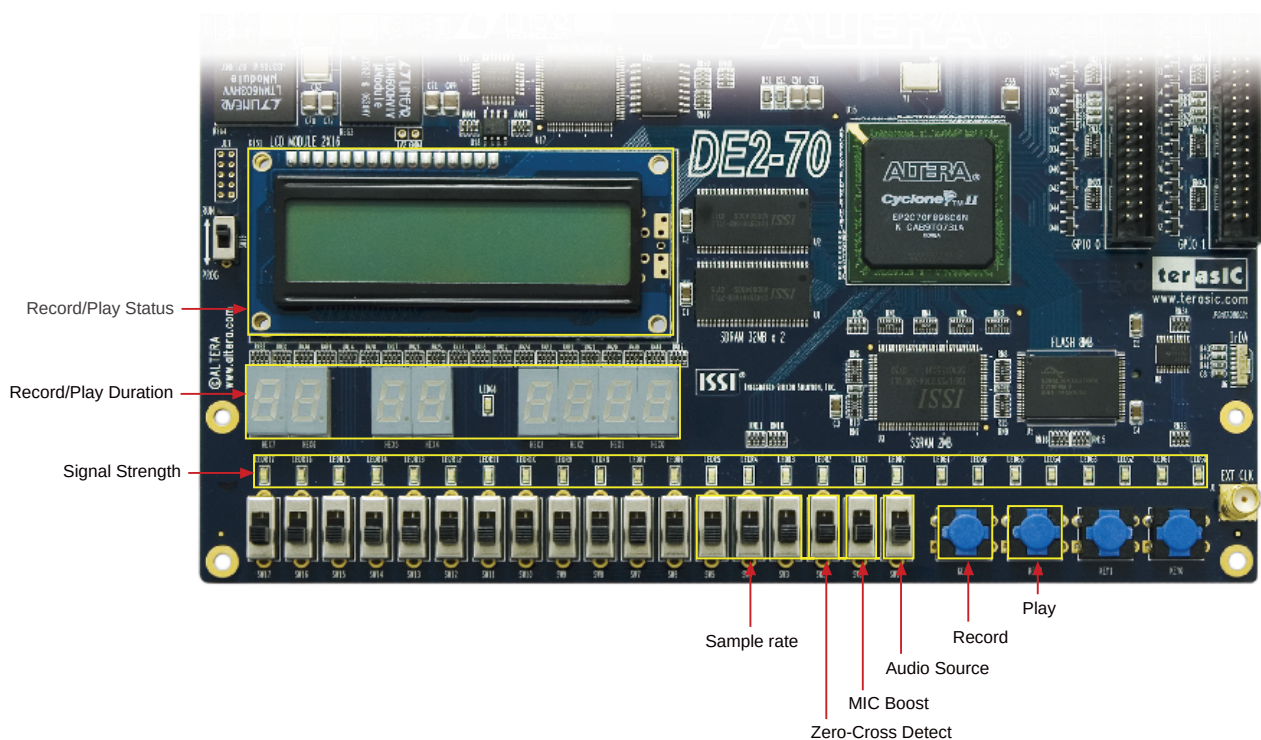


Figure 6.18. Man-Machine Interface of Audio Recorder and Player.

Figure 6.19 shows the block diagram of the design of the Audio Recorder and Player. There are hardware part and software part in the block diagram. The software part means the Nios II program that stored in SSRAM. The software part is built by Nios II IDE in C programming language. The

hardware part is built by SOPC Builder under Quartus II. The hardware part includes all the other blocks. The “AUDIO Controller” is a user-defined SOPC component. It is designed to send audio data to the audio chip or receive audio data from the audio chip.

The audio chip is programmed through I2C protocol which is implemented in C code. The I2C pin from audio chip is connected to SOPC System Interconnect Fabric through PIO controllers. In this example, the audio chip is configured in Master Mode. The audio interface is configured as I2S and 16-bit mode. A 18.432MHz clock generated by the PLL is connected to the XTI/MCLK pin of the audio chip through the AUDIO Controller.

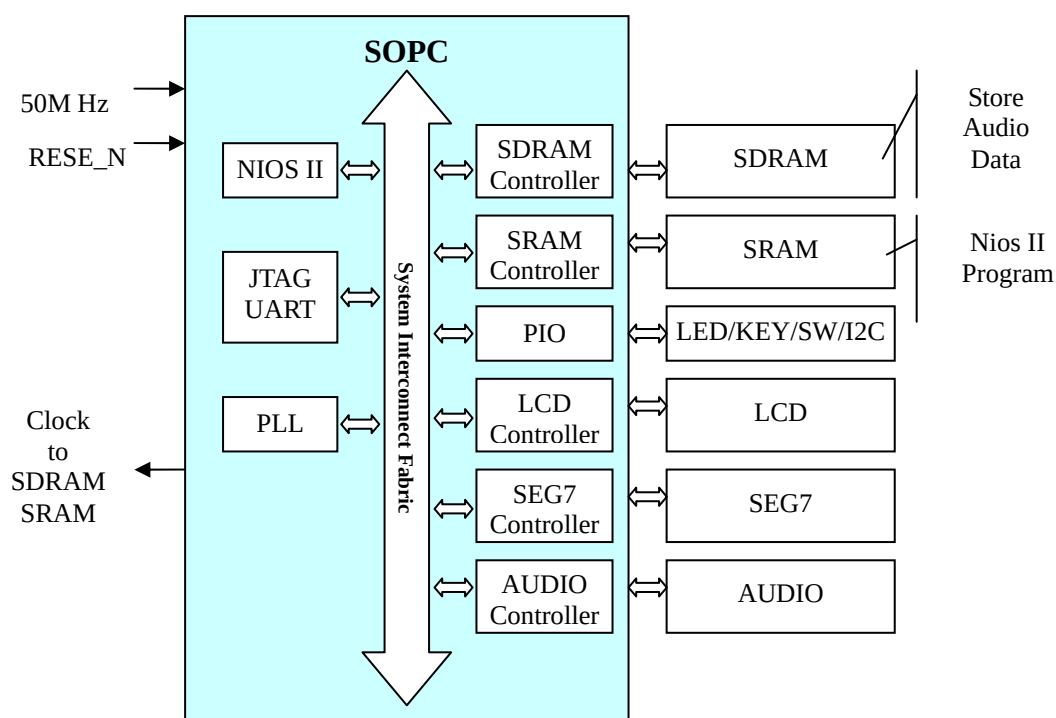


Figure 6.19. Block diagram of the audio recorder and player.

Demonstration Setup, File Locations, and Instructions

- Hardware Project directory: *DE2_70_AUDIO*
- Bit stream used: *DE2P_TOP.sof*
- Software Project directory: *DE2_70_AUDIO\software\project_audio*
- Software Execution File: *DE2_70_AUDIO\software\project_auido\audio\debug\audio.elf*
- Connect an Audio Source to the LINE-IN port of the DE2-70 board.
- Connect a Microphone to MIC-IN port on the DE2-70 board.
- Connect a speaker or headset to LINE-OUT port on the DE2-70 board.
- Load the bit stream into FPGA. (note *1)

- Load the Software Execution File into FPGA. (note *1)
- Configure audio with the toggle switches.
- Press KEY3 on the DE2-70 board to start/stop audio recoding (note *2)
- Press KEY2 on the DE2-70 board to start/stop audio playing (note *3)

Note:

- (1). Execute *DE2_70_AUDIO\demo batch\audio.bat* will download .sof and .elf files.
- (2). Recording process will stop if audio buffer is full.
- (3). Playing process will stop if audio data is played completely.

Toggle Switches	0 – DOWN Position	1 – UP Position
SW0	Audio is from MIC	Audio is from LINE-IN
SW1	Disable MIC Boost	Enable MIC Boost
SW2	Disable Zero-cross Detection	Enable Zero-cross Detection

SW5 (0 – DOWN; 1- UP)	SW4 (0 – DOWN; 1-UP)	SW3 (0 – DOWN; 1-UP)	Sample Rate
0	0	0	96K
0	0	1	48K
0	1	0	44.1K
0	1	1	32K
1	0	0	8K
Unlisted combination			96K

Table 6.4. Toggle switch setting for audio recorder and player.

Chapter 7

Appendix

7.1 Revision History

Version	Change Log
V1.0	Initial Version (Preliminary)
V1.01	1. Add appendix chapter. 2. Modify Chapter 2,3,4,5,6.
V1.02	Modify Figure 6.8
V1.03	Modify clock frequency of the VGA DAC
V1.04	Modify Section 5.4

7.2 Copyright Statement

Copyright © 2008 Terasic Technologies. All rights reserved.